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Internship Report

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Reinforcement Learning for Mean-Field Control

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Abstract

Mean-field control describes the optimal control of a large population of interacting agents through the law of a single representative agent, replacing a state space that grows with the population by a measure-valued one. This breaks the property that makes REINFORCE model-free: the population flow is generated by the policy, so the parameters act on the reward and the transition kernel a second time through the law, a channel the classical policy score omits. A recent construction restores a model-free estimator by perturbing the logits of the population vector, but is tied to a finite state space.

This report develops a perturbation framework that separates the policy-gradient construction from the representation of the population law. The population argument is replaced by the pushforward $M_t^{\lambda, \theta} = T_t^\lambda \# \mu_t^\theta$ of a random transport map, which reduces to the probability-preserving mixture $(1-\lambda)\mu_t^\theta + \lambda q_t$ on the simplex in finite state spaces, and to a random displacement read through a finite-dimensional representation in continuous ones. In both cases the gradient of the perturbed objective splits formally into the usual policy-score term and a population-score term, both estimable from simulated trajectories without differentiating the transition kernel, the reward, or the population recursion. On a finite state space the analysis is carried through: the perturbed law stays within total variation λ of the original flow, the perturbed objective and its gradient converge to their unperturbed counterparts as $\lambda \rightarrow 0$, the estimator is unbiased for the perturbed gradient under oracle sensitivities, and a mean-square-error decomposition quantifies the resulting trade-off. On a continuous state space we establish the construction and the convergence of the perturbed objective, but not their gradient-level counterparts, so that estimator is built by analogy and validated empirically; closing it is the main open item of the internship, which runs to the end of October. A forward recursion for the sensitivity flow makes the simulator cost linear rather than quadratic in the horizon.

On six benchmarks evaluated against REINFORCE and the logit-perturbed baseline under matched simulator budgets, no estimator dominates. The transport estimator improves on REINFORCE by an order of magnitude on the two-state problem and by wide margins on both continuous-state benchmarks; the logit-perturbed estimator does better on distribution planning and advertising; and on cybersecurity plain REINFORCE is best, at a fraction of the cost. An evaluation-only diagnostic accounts for the split: whether REINFORCE is directionally competitive is predicted not by the magnitude of the correction but by its alignment with the policy-score term, which separates the benchmarks into two groups with no intermediate case. The full code is available at <https://github.com/adonis107/RL-MFC>.

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Chapter 1

Introduction

1.1 Scientific context and motivation

Mean-field control (MFC) provides a framework for controlling large populations of interacting agents under a common objective, in problems where the evolution or the cost of each agent depends not only on its own state and action, but also on the distribution of the population. In a computer network, for instance, the probability that a machine becomes compromised depends on the proportion of compromised machines, while a central controller allocates protection resources. Modelling every agent individually leads to a state space whose dimension grows with the population size. In the mean-field limit, the empirical distribution is replaced by the law of a representative agent, yielding a control problem whose formulation no longer depends explicitly on the number of agents [22]. This reduction makes large systems tractable, at the price of a measure-valued state variable.

Most numerical methods for MFC assume that the transition dynamics and the cost functions are known. They rely on dynamic programming, on forward–backward systems of Hamilton–Jacobi–Bellman and Fokker–Planck equations, or on direct differentiation through the population dynamics [24, 2, 15]. In each case the value function and the population distribution evolve jointly, so knowledge of the model is central to both the analysis and the numerical solution.

Policy-gradient methods are attractive here because they optimize a parametrized policy directly and accommodate randomized policies, continuous actions, and neural-network parametrizations. In a classical Markov decision process, REINFORCE expresses the gradient of the expected cost using only sampled returns and the score of the policy [29]; the representation is model-free because, conditionally on the current state and action, the transition kernel does not depend on the policy parameters. This property no longer holds in mean-field control. The population law is generated by the policy itself, so changing the parameters changes the population flow, which then changes the policies, costs, and transition kernels evaluated at subsequent times. The classical policy score accounts for the direct effect of the parameters on the sampled actions, but not for their indirect effect through the population law, and REINFORCE applied without modification is therefore biased.

A recent method addresses this difficulty in discrete-time, finite-state MFC by perturbing the logits of the population vector and deriving an additional likelihood-ratio correction [19], establishing that fully model-free policy gradients are possible in this setting. The construction relies on one logit coordinate per state, however, and is therefore tied to finite state spaces: it does not extend to a continuous state space, where the population state is a probability measure with no canonical finite-dimensional parametrization.

The motivation of this work is to develop a perturbation framework that separates the policy-gradient construction from the representation of the population law. In finite state spaces, probability-preserving perturbations can be defined directly on the simplex; in continuous state spaces, they can instead be constructed through transport maps. The remaining question is then one of representation: a probability measure must be described through a finite collection of relevant functionals, such as moments, interaction fields, or learned features, and the score entering the gradient estimator is associated with the joint law of these functionals rather than with the population measure directly. This connects model-free MFC with representation learning on spaces of probability measures, and provides a common framework for discrete- and continuous-state problems.

1.2 Internship context

This internship is carried out at the Centre de Mathématiques Appliquées (CMAP) of École Polytechnique, in connection with the Institut Europlace de Finance, within the probability group and the MathsFi team. It is supervised by Prof. Huy  n Pham, with regular research meetings held together with Samy Mekkaoui and Yadh Hafsi, whose close supervision shaped the orientation of the project. On the academic side, the internship forms part of my final year at CentraleSup  lec and of the MVA master’s programme at   cole Normale Sup  rieure Paris-Saclay.

The initial topic concerned the numerical resolution of mean-field games and mean-field control problems using deep-learning methods. It was intentionally broad, and the first part of the work consisted in exploring several approaches at the interface of mean-field control and machine learning before progressively identifying a more specific research question around reinforcement-learning methods. The project therefore fits the broader agenda of the host team, which connects stochastic control, McKean–Vlasov dynamics, numerical probability, and machine learning. This report reflects the state of the project at the time of the MVA presentation, in late August, whereas the internship continues until the end of October; some theoretical developments and numerical conclusions should therefore be read as intermediate results.

The project draws on several MVA courses, and reflects their interdisciplinary character: *Reinforcement Learning* (D. Basu, E. Kaufmann, O. Maillard), which provides the most direct methodological background, *Optimal Transport for Machine Learning* (G. Peyr  , J. Delon) for the continuous-state extension, *Stochastic Calculus in Machine Learning* (A. Oliviero-Durmus), *Convex Optimization and Applications in Machine Learning* (A. d’Aspremont), *Interactions* (J. Randon-Furling), and *Deep Learning* (V. Lepetit, M. Vakalopoulou).

1.3 Objectives of the internship

The central objective of the internship was to investigate whether policy-gradient methods for discrete-time mean-field control could be made fully model-free without relying on the finite-state logit representation of [19], that is, using only simulated trajectories, policy evaluations and observed costs, without differentiating or explicitly knowing the transition dynamics.

The first objective was theoretical: to identify the additional dependence created by the population flow, to construct a perturbation of this flow for which a likelihood-ratio representation can be derived, to show that the perturbed gradient approximates the gradient of the original MFC objective as the perturbation vanishes, and to study the bias–variance trade-off it introduces. The second was to develop constructions adapted to both discrete and continuous state spaces: on the simplex, a probability-preserving perturbation replacing the logit one; in continuous state spaces, a perturbation through transport maps whose score is expressed through a finite-dimensional representation of the perturbed law. This second case raises a related representation-learning question, namely which functionals of the population distribution carry the information required by the dynamics, the costs and the policy. The third objective was algorithmic: converting these representations into practical Monte Carlo estimators, including an estimator of the sensitivity of the population flow, while reusing simulated trajectories across the horizon, controlling computational and memory costs, and selecting the perturbation scales, possibly adaptively.

Finally, the numerical objective was to evaluate the resulting methods against standard REINFORCE and the finite-state MF-REINFORCE method under matched simulation budgets, on four finite-state and two continuous-state benchmarks. Beyond final performance, the study examines gradient bias and variance, sensitivity to the perturbation scale, computational cost, and robustness to particle approximations of the population flow.

1.4 Contributions

The main contribution of this work is a perturbation-based framework for model-free policy gradients in discrete-time mean-field control. The central idea is to randomize the population argument so that its

dependence on the policy parameters is transferred to the distribution of the perturbation. The gradient of the perturbed objective then decomposes into the usual policy-score term and an additional population-score term, both of which admit likelihood-ratio representations and are therefore estimable from simulated trajectories without differentiating the transition kernel. The accompanying theory provides the assurances needed for the method to be meaningful: the perturbed population laws remain controlled perturbations of the original flow, the perturbed objectives and gradients converge back to their unperturbed counterparts as the perturbation scale vanishes, the Monte Carlo estimators are unbiased or consistent for the corresponding perturbed gradients, and bias–variance results quantify the finite-sample trade-off that the scale creates.

A transport view of perturbed population flows. We formulate perturbations directly at the level of probability measures. Given the population law $\mu_t^\theta = \mathbb{P}_{X_t^\theta}$, we introduce a random transport map T_t^λ and replace the population argument by $M_t^{\lambda,\theta} = T_t^\lambda \# \mu_t^\theta$, where the perturbation scale λ controls the distance between the original and perturbed population laws. In continuous state spaces this includes random maps of the form $T_t^\lambda(\omega, x) = x + \lambda f_t(\omega, x)$, whereas in finite state spaces the same idea becomes a perturbation on the simplex. For both discrete and continuous MFC models, the estimator is driven by a random, probability-preserving deformation of the population law rather than by a model-specific coordinate system.

A likelihood-ratio correction for the mean-field dependence. The perturbed population argument is random and its distribution depends on θ through μ_t^θ , so the gradient of the perturbed objective contains, besides the usual REINFORCE policy-score term, a score term associated with the generated population argument. This is the correction missing from the classical REINFORCE formula in MFC. The report derives the decomposition and identifies the conditions under which it can be estimated without differentiating the transition kernel, the reward, or the population recursion.

A finite-dimensional representation for continuous laws. In a general continuous state space, there is no canonical dominating measure on $\mathcal{P}(\mathcal{X})$, so the law of $M_t^{\lambda,\theta}$ need not have a usable density. We therefore introduce a representation map $\Psi(m) = (F_1(m), \dots, F_d(m))$ and work with the random vector $Z_t^{\lambda,\theta} = \Psi(M_t^{\lambda,\theta})$, writing the population-score correction through the density of $Z_t^{\lambda,\theta}$ in $\mathbb{R}^d$. This separates the policy-gradient construction from the choice of representation: Ψ may be given by exact sufficient statistics such as moments, or by interaction fields, Fourier or kernel features, or learned summaries of the population law.

A simplex perturbation for finite-state MFC. For $\mu \in \Delta_N$, we define the discrete transport perturbation $\Phi_\mu^\lambda(q) = (1 - \lambda)\mu + \lambda q$ with $q \in \Delta_N$, and use $M_t^{\lambda,\theta} = (1 - \lambda)\mu_t^\theta + \lambda q_t$ as the perturbed population argument, for which the simplex coordinates give an exact finite-dimensional representation. We derive the density and the score of $M_t^{\lambda,\theta}$ explicitly, obtaining a model-free gradient estimator that acts directly on probability vectors. Compared with the logit perturbation of [19], the construction does not require strictly positive population coordinates and is expressed in the geometry of the simplex itself; under regularity assumptions the perturbed gradient converges to the unperturbed one as $\lambda \rightarrow 0$, with a first-order error.

A recursive sensitivity estimator and practical algorithms. The population-score term requires the sensitivity of the population flow with respect to the policy parameters. We derive a forward recursion for this quantity, in which the sensitivity at time t depends only on earlier ones, so that the whole sensitivity flow can be estimated from shared simulated trajectories. With cumulative computations this gives a simulator cost linear in the horizon, against the quadratic cost of estimating each truncated sensitivity separately. The estimators are implemented as fixed-scale and adaptive Monte Carlo algorithms, the adaptive version updating the main and auxiliary perturbation scales by comparing gradient estimates at nearby scales, so as to balance perturbation bias and estimator variance without a full environment-specific grid search.

1.5 Status of the contributions

The contributions above are at different stages, and Table 1.1 states which. The finite-state construction is complete: every step from the perturbation estimate to the mean-square-error decomposition is proved. The continuous-state construction is complete as far as the objective, and open beyond it: the representation, the perturbation, the density and the score are established, and the perturbed objective is shown to converge, but the gradient-level counterparts are not yet proved. The continuous-state experiments are therefore run

with an estimator built by analogy with the finite-state case, and the reader should treat their guarantees as an extrapolation rather than a theorem. This is the main limitation of the report and is discussed again in Section 6.2.

Result	Setting	Where	Status
Perturbation estimate $d_{TV} \leq \lambda$	finite state	Thm. 1	proved
Density and score of the perturbation	finite state	Prop. 1	proved
Objective convergence $ J^\lambda - J \leq C_T \lambda$	finite state	Thm. 2	proved
Consistency of optimizers	finite state	Cor. 2	proved
Gradient-level convergence	finite state	Thm. 3	proved
Policy-gradient representation	finite state	Thm. 4	proved
Unbiasedness under oracle sensitivities	finite state	Prop. 3	proved
Auxiliary sensitivity error	finite state	Prop. 4	proved
Bias and mean-square-error decomposition	finite state	Thm. 5	proved
Representation and perturbation estimate	continuous	Thm. 6	proved
Density and score of the coordinates	continuous	Prop. 6	proved
Objective convergence $J_K^\lambda \rightarrow J$	continuous	Thm. 7	proved
Gradient-level convergence	continuous	—	<i>open</i>
Policy-gradient representation	continuous	—	<i>open</i>
Estimator of $D_{\theta} z_{K,t}^\theta$	continuous	—	implemented for exact moment charts; no general theorem
Bias and mean-square-error decomposition	continuous	—	<i>open</i>
Fixed-scale and adaptive algorithms	both	Ch. 4	implemented, 640 runs
Four theoretical predictions	both	Sec. 5.5	measured; three confirmed, unbiasedness only for $\lambda \leq 0.2$
Alignment predicts whether REINFORCE is competitive	both	Sec. 5.6	empirical, not a theorem

Table 1.1: Status of each contribution: what is proved, what is implemented and measured, and what remains open. The continuous-state analysis is established at the level of the objective and open at the level of the gradient; the empirical regularity of Section 5.6 is an observation over six benchmarks and one controlled ablation, not a result.

1.6 Organization of the report

Chapter 2 covers the background on Markov decision processes, policy-gradient methods and mean-field control, and positions the work relative to existing model-free approaches. Chapter 3 develops the perturbation framework and its convergence and bias–variance results, and Chapter 4 turns them into fixed-scale and adaptive Monte Carlo algorithms. Chapter 5 presents the numerical protocol and the benchmark suite, and Chapter 6 discusses what the results establish, the limits of the approach and the extensions it suggests.

Chapter 2

Background and Related Work

This chapter outlines the framework motivating this internship by contrasting classical policy gradients with the challenges of mean-field control. We review finite-horizon Markov decision processes and compare model-based, value-based and policy-based approaches. We then introduce likelihood-ratio policy gradients and REINFORCE, and highlight a defining structural assumption: environment transitions are completely independent of policy parameters given a state–action pair.

Mean-field control breaks this assumption. Because the environment relies on a population distribution driven by the policy, it creates a feedback loop that directly invalidates the classical likelihood-ratio trick. By identifying exactly where this breakdown occurs, we motivate the introduction of a perturbation framework designed to recover a model-free score representation.

2.1 Markov decision processes

Markov decision processes (MDPs) provide the standard mathematical framework for sequential decision-making under uncertainty and form the basis of classical reinforcement learning [25, 27]. An agent interacts with a stochastic environment over a sequence of time steps: at each time, it observes the current state, selects an action according to a policy, receives a reward, and the system evolves randomly to a new state.

Throughout this report, we consider a finite time horizon $\mathcal{T} = \{0, \dots, T\}$, although the same concepts could be extended to infinite-horizon and discounted settings. Let $\mathcal{X}$ denote the state space and $\mathcal{A}$ the action space. These spaces may be finite, as in the discrete-state experiments considered later, or more general measurable spaces, as required for continuous-state problems. The model is specified by an initial distribution $\mu_0 \in \mathcal{P}(\mathcal{X})$, transition kernels $P_t(dx' | x, a)$, running costs $r_t : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}$, and a terminal cost $g : \mathcal{X} \rightarrow \mathbb{R}$.

A randomized Markov policy is a sequence $\pi = (\pi_t)_{t=0}^{T-1}$ of kernels from $\mathcal{X}$ to $\mathcal{A}$. Under π , the controlled process satisfies

$$\begin{cases} X_0 & \sim \mu_0, \\ \alpha_t & \sim \pi_t(\cdot | X_t), \\ X_{t+1} & \sim P_t(\cdot | X_t, \alpha_t), \end{cases} \quad t = 0, \dots, T-1.$$

A deterministic feedback rule is recovered when $\pi_t(\cdot | x)$ is a Dirac measure. Randomized policies are used here because their action probabilities can be differentiated, which is the basis of the likelihood-ratio estimator introduced below. They also provide the exploration needed to compare different actions from simulated trajectories.

The last relation encodes the Markov property: conditionally on (X_t, α_t) , the law of X_{t+1} is independent of the preceding trajectory. Consequently, the law of $\tau = (X_0, \alpha_0, \dots, X_{T-1}, \alpha_{T-1}, X_T)$ factorizes as

$$\mathbb{P}^\pi(d\tau) = \mu_0(dx_0) \prod_{t=0}^{T-1} \pi_t(da_t | x_t) P_t(dx_{t+1} | x_t, a_t). \quad (2.1)$$

The performance criterion is the expected cumulative cost

$$J(\pi) = \mathbb{E}^\pi \left[\sum_{t=0}^{T-1} r_t(X_t, \alpha_t) + g(X_T) \right], \quad \inf_{\pi} J(\pi).$$

When the model is known, this problem can be approached by dynamic programming. Reinforcement learning instead assumes access to sampled transitions and costs, without requiring the kernels to be known or tractable in closed form. For this work, the important consequence of (2.1) is that the environment kernels do not change when the parameters of a policy are varied. This separation makes the classical likelihood-ratio policy gradient model-free.

Reinforcement learning methods differ in what they attempt to learn from these samples. Value-based methods first estimate a value or state-action value function and then derive a policy. Policy-gradient methods instead parameterize the policy and optimize its performance directly. The latter choice is natural for this work because the policy is already the object being parametrized, whereas the transition mechanism is treated as a simulator. It also avoids differentiating the dynamics, at the cost of a noisier Monte Carlo gradient.

2.2 Policy-gradient methods

Policy-gradient methods optimize a differentiable family of randomized policies directly. They include the REINFORCE algorithm of [29] and the broader policy-gradient framework of [28]. Let $\Theta \subseteq \mathbb{R}^D$ and suppose that π_t^θ admits a positive differentiable density with respect to a fixed reference measure $\nu_{\mathcal{A}}$: $\pi_t^\theta(da | x) = p_t^\theta(a | x)\nu_{\mathcal{A}}(da)$. We write $J(\theta) = J(\pi^\theta)$ and seek a trajectory-based estimator of $\nabla_\theta J(\theta)$.

2.2.1 Likelihood-ratio representation

The score-function, or likelihood-ratio, identity states that, under conditions justifying differentiation under the integral sign,

$$\nabla_\theta \mathbb{E}_{Z \sim q_\theta} [F(Z)] = \mathbb{E}_{Z \sim q_\theta} [F(Z) \nabla_\theta \log q_\theta(Z)] \quad (2.2)$$

whenever F has no explicit dependence on θ [14]. Applied to the trajectory law (2.1), the terms involving μ_0 and P_t disappear from the score because they do not depend on θ . Hence

$$\nabla_\theta \log q_\theta(\tau) = \sum_{t=0}^{T-1} \nabla_\theta \log p_t^\theta(a_t | x_t).$$

The transition kernels still determine which trajectories are observed, but their values and derivatives are not needed in the estimator.

Let $C_t = \sum_{s=t}^{T-1} r_s(X_s, \alpha_s) + g(X_T)$ denote the cumulative cost-to-go from time t . The policy score has zero conditional mean,

$$\mathbb{E} \left[\nabla_\theta \log p_t^\theta(\alpha_t | X_t) \mid X_t \right] = \int_{\mathcal{A}} \nabla_\theta p_t^\theta(a | X_t) \nu_{\mathcal{A}}(da) = 0. \quad (2.3)$$

Costs incurred before time t are measurable before α_t is sampled and therefore have zero expectation when multiplied by the score. Combining this causality argument with (2.2) gives the finite-horizon policy-gradient formula

$$\nabla_\theta J(\theta) = \mathbb{E}^{\pi^\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log p_t^\theta(\alpha_t | X_t) C_t \right]. \quad (2.4)$$

Thus credit is assigned using future costs only, rather than the full trajectory cost.

2.2.2 Monte Carlo estimator and baselines

Given N independent trajectories generated under π^θ , let $C_t^{(k)}$ be the cost-to-go along trajectory k . Equation (2.4) gives the REINFORCE estimator

$$\widehat{\nabla_\theta J(\theta)} = \frac{1}{N} \sum_{k=1}^N \sum_{t=0}^{T-1} \nabla_\theta \log p_t^\theta(\alpha_t^{(k)} | X_t^{(k)}) C_t^{(k)}.$$

In this sense, REINFORCE replaces the unknown conditional expected cost-to-go by its realized Monte Carlo value on each trajectory. This avoids learning a separate value function, but is also the main source of variance. Under the preceding regularity assumptions this estimator is unbiased, and a cost-minimization update takes the form $\theta_{n+1} = \theta_n - \eta_n \widehat{\nabla_{\theta} J}(\theta_n)$ for a suitable step size $\eta_n > 0$. It is model-free in the relevant sense: it uses sampled costs and the policy score, but neither the transition kernels nor its derivatives.

The estimator may nevertheless have high variance. If $b_t(X_t)$ is any integrable baseline independent of A_t conditionally on X_t , then (2.3) implies

$$\nabla_{\theta} J(\theta) = \mathbb{E}^{\pi^{\theta}} \left[\sum_{t=0}^{T-1} \nabla_{\theta} \log p_t^{\theta}(\alpha_t | X_t) (C_t - b_t(X_t)) \right].$$

Subtracting the baseline therefore preserves unbiasedness while potentially reducing variance. A common choice is the conditional expected cost-to-go $b_t(x) = \mathbb{E}^{\pi^{\theta}}[C_t | X_t = x]$, although it is not in general the variance-minimizing baseline [9]. Estimating this quantity jointly with the policy leads to actor-critic methods. The present work instead builds on the score-function structure of REINFORCE.

2.3 Mean-field control

Mean-field control (MFC) describes the cooperative control of a large population of interacting agents. In a system of N agents, the interaction typically occurs through the empirical distribution $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_t^{i,N}}$. The dynamics and cost of one agent may depend on its own state and action, but also on μ_t^N . Directly treating the joint state $(X_t^{1,N}, \dots, X_t^{N,N})$ quickly becomes intractable as N grows. The mean-field limit replaces the empirical distribution by the law $\mu_t = \mathcal{L}(X_t)$ of a representative agent. Under suitable assumptions, this yields a limiting control problem expressed in terms of the representative state and the population law, rather than the joint state of all N agents [22].

Mean-field control should be distinguished from mean-field games. In a mean-field game, each agent optimizes its own objective while treating the population flow as fixed, and an equilibrium condition subsequently requires the flow generated by the optimal response to coincide with the prescribed flow. In mean-field control, a central planner chooses a policy for the whole population and internalizes its effect on the population flow. The problems studied in this report are of the latter, cooperative type.

2.3.1 Representative-agent formulation

In discrete time, a randomized feedback policy may depend on both the individual state and the current population law:

$$\begin{cases} X_0 & \sim \mu_0, \\ \alpha_t & \sim \pi_t(\cdot | X_t, \mu_t), \\ X_{t+1} & \sim P_t(\cdot | X_t, \mu_t, \alpha_t), \\ \mu_t & = \mathcal{L}(X_t). \end{cases} \quad t = 0, \dots, T-1,$$

The corresponding objective is

$$J(\pi) = \mathbb{E}_{\mu_0, \pi} \left[\sum_{t=0}^{T-1} r_t(X_t, \mu_t, \alpha_t) + g(X_T, \mu_T) \right], \quad \inf_{\pi} J(\pi).$$

Unlike an exogenous state variable, the population law is not sampled independently of the policy. Once μ_t and π_t are fixed, its next value is determined by

$$\mu_{t+1}(B) = \int_{\mathcal{X} \times \mathcal{A}} P_t(B | x, a, \mu_t) \pi_t(da | x, \mu_t) \mu_t(dx)$$

for every measurable set $B \subseteq \mathcal{X}$. Thus, changing the policy changes the whole sequence $(\mu_t)_{t=0}^T$, and this sequence in turn changes the policy, dynamics and costs evaluated at later times. This feedback is the defining feature of the problem.

2.3.2 Lifted control problem

At the population level, the state of the control problem is μ_t rather than X_t alone. A control action at this level is a rule assigning an action distribution to each individual state. This gives a lifted MDP on $\mathcal{P}(\mathcal{X})$, for which dynamic programming can be formulated when the model is known [22]. The lifted viewpoint also makes clear why the population law cannot simply be omitted from the state: two populations may contain an agent in the same state x while inducing different transitions, costs and optimal decisions.

The difficulty is that the lifted state space is large. If $\mathcal{X}$ contains d states, μ_t belongs to a $(d-1)$ -dimensional simplex. For a continuous state space, $\mathcal{P}(\mathcal{X})$ is infinite-dimensional.

2.3.3 Representing the population law

For a finite state space, the population law is represented exactly by a vector of state probabilities. There is no comparable canonical vector representation when the state space is continuous. In computations, one instead describes a measure m through functions

$$\Psi(m) = (F_1(m), \dots, F_d(m)).$$

such as moments, Fourier coefficients, interaction fields or kernel features. For example, one may take $F_j(m) = \langle \varphi_j, m \rangle = \int_{\mathcal{X}} \varphi_j(x) m(dx)$. A sufficiently rich separating family of test functions characterizes the measure. A finite collection is exact when the problem has a corresponding finite-dimensional structure, and otherwise provides a task-dependent approximation.

A related approach is based on cylindrical representations $\Phi_J(m) = (\langle \varphi_1, m \rangle, \dots, \langle \varphi_J, m \rangle)$, which reduce an operator on probability measures to a function of finitely many measure functionals. Mekkaoui, Pham and Warin establish a universal approximation result of this form for operators on labelled conditional distributions and polynomial moment maps in their numerical implementation [18].

This representation issue also enters the policy-gradient construction. After the population law is randomized, the mean-field argument $M_t^{\lambda, \theta}$ is itself a random measure. The finite-dimensional object whose density is studied is therefore

$$Z_t^{\lambda, \theta} = \Psi(M_t^{\lambda, \theta}), \quad \mathcal{L}(Z_t^{\lambda, \theta}) \in \mathcal{P}(\mathbb{R}^d). \quad (2.5)$$

Thus the problem is not only to represent the population law, but to determine the joint law of functionals that characterize the relevant part of it. This law may admit an ordinary density and score even though there is no canonical Lebesgue measure on $\mathcal{P}(\mathcal{X})$. The choice, or learning, of Ψ is therefore part of the method: it must retain the distributional information used by the dynamics, costs and policy while remaining estimable from population samples.

2.3.4 Model-free reinforcement learning for MFC

MFC is closely related to cooperative multi-agent reinforcement learning (MARL). In a finite-agent formulation, learning a joint policy becomes difficult as the joint state and action spaces grow with population. Mean-field methods replace the individual interactions by aggregate population quantities, as in mean-field MARL [30]. For cooperative systems, the connection is made more precise in [20], which studies how finite-agent MARL problems can be approximated by MFC problems. Mean-field MARL and MFC are not interchangeable, however the latter is the limiting control problem of a cooperative population under a common objective, which is the setting considered here.

When the transition and cost functions are unknown, the policy or value function must be learned directly from simulated or observed trajectories. Most model-free methods for MFC are value-based. A two-timescale mean-field Q -learning algorithm in discrete state and action spaces is proposed in [1]. In [3], MFC is formulated as a lifted mean-field MDP and tabular and deep Q -learning methods are developed on this space. These methods avoid using the transition model explicitly, but the value function still takes a population distribution as part of its input.

Policy-based methods optimize a parametrized policy directly and are useful when the policy has a compact representation or the action space is continuous. Related actor–critic methods have been developed for continuous-time MFC with randomized policies [8], and more recently for extended MFC with deterministic policies [5]. These approaches use continuous-time gradient representations and therefore differ from the discrete-time likelihood-ratio setting of this report. In discrete time, standard REINFORCE is not directly applicable: changing the policy parameters also changes the population flow μ_t^θ , hence the costs and transition kernels evaluated along that flow. The usual policy-score term does not account for this dependence.

The closest work is [19], which introduces the first fully model-free policy-gradient method for discrete-time MFC with a finite state space. A strictly positive population vector is represented by its logits, Gaussian noise is added in logit space, and the perturbed vector is mapped back to the simplex by a softmax. A change of variables then replaces the unknown measure derivatives by an additional score term. The resulting MF-REINFORCE algorithm estimates the sensitivity of the population logits through an auxiliary simulation procedure. The authors prove convergence of the perturbed gradient as the perturbation vanishes and give explicit bias and mean-square error bounds for the estimator.

This construction is the starting point of the present work, but its logit representation is tied to finite state spaces: it requires one coordinate per state and has no direct counterpart on $\mathcal{P}(\mathcal{X})$ when $\mathcal{X}$ is continuous. The method developed here separates the perturbation from the representation of the law. In the discrete case, the perturbation is applied directly on the simplex through a convex mixture, leading to an explicit simplex-density score and a recursive sensitivity estimator that reuses trajectories across the horizon. In the continuous case, the law is perturbed by probability-preserving push-forward maps, while the score is studied through the distribution features in (2.5). The extension is therefore not only a change of state space: it requires selecting or learning a representation of the population law and characterizing the law of that representation under the perturbation.

Chapter 3

Methodology

This chapter introduces the mathematical construction used to turn a mean-field control problem into a model-free policy-gradient objective. We first fix the canonical controlled process and its policy parametrization, then define the randomized population perturbation whose limiting behavior justifies the transport estimators developed in the next chapter.

To keep the chapter focused on the construction and on the consequences of the main results, the more technical proofs are collected in Appendix A. The main text keeps the statements and proof ideas needed to follow how the method fits together.

3.1 Canonical construction of the control problem

Fix a horizon $T \in \mathbb{N}^*$ and denote by $\mathcal{T} = \{0, 1, \dots, T\}$ the set of time indices. Let $\mathcal{X}$ be the state space and let $\mathcal{A}$ be the one-step action space, both assumed to be Polish. Given a random variable Y on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, we denote by $\mathbb{P}_Y$ or $\mathcal{L}(Y)$ its law under $\mathbb{P}$.

Individual actions take values in $\mathcal{A}$, whereas $\mathbb{A}$ denotes the set of admissible control processes, so that the notation agrees with Chapter 2. Given $\alpha = (\alpha_t)_{t=0}^{T-1} \in \mathbb{A}$ and an initial condition $\xi \sim \mu_0$, the controlled state process satisfies

$$\begin{cases} X_0 = \xi, \\ X_{t+1} = F_t(X_t, \mathbb{P}_{X_t}, \alpha_t, \epsilon_{t+1}), \end{cases} \quad t = 0, \dots, T-1,$$

or equivalently $\mathcal{L}(X_{t+1} \mid X_t, \mathbb{P}_{X_t}, \alpha_t) = P_t(\cdot \mid X_t, \mathbb{P}_{X_t}, \alpha_t)$. The reward functional is

$$\sup_{\alpha \in \mathbb{A}} J(\alpha) := \mathbb{E} \left[\sum_{t=0}^{T-1} r_t(X_t, \mathbb{P}_{X_t}, \alpha_t) + g(X_T, \mathbb{P}_{X_T}) \right].$$

In the model-free setting, controls are sampled from randomized Markov policies $\pi = (\pi_t)_{t=0}^{T-1}$ in the class Π , with $\pi_t : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{A})$ and $\alpha_t \sim \pi_t(\cdot \mid X_t, \mathbb{P}_{X_t})$, which gives the equivalent problem

$$\sup_{\pi \in \Pi} J(\pi) := \mathbb{E}^\pi \left[\sum_{t=0}^{T-1} r_t(X_t, \mathbb{P}_{X_t}, \alpha_t) + g(X_T, \mathbb{P}_{X_T}) \right], \quad (3.1)$$

where $\mathbb{E}^\pi$ is the expectation under the trajectory law induced by π .

To solve (3.1) by policy gradient methods, we restrict to a parametrized family $\theta \mapsto \pi^\theta = (\pi_t^\theta)_{t=0}^{T-1}$, with $\theta \in \Theta \subset \mathbb{R}^k$. We write X^θ and α^θ for the corresponding state and action processes, and the finite-dimensional objective becomes $\sup_{\theta \in \Theta} J(\theta) = J(\pi^\theta)$.

Throughout, *model-free* means that the estimator uses only simulated transitions, evaluations of the policy density and its score, and observed rewards: it never differentiates, and never evaluates in closed form, the transition kernel P_t , the reward r_t , or the population recursion. Exact population recursions and closed-form optima do appear in Chapter 5, but only outside the training loop, to evaluate a frozen policy or to supply a reference; no such quantity enters a gradient step.

The difficulty is that the policy and the reward depend on the population law $(\mathbb{P}_{X_t^\theta})_{t \in \mathcal{T}}$, whose sensitivity with respect to θ is not directly available in a model-free setting. The construction below introduces an

exogenous randomization of these laws. The perturbed objective will converge to the original one as the perturbation scale λ tends to zero, while providing a likelihood-ratio handle on the missing population derivative.

All the processes below are realized on a single canonical space carrying an initial variable $\xi \sim \mu_0$, transition noises $(\epsilon_t)_{t=1}^T$, policy noises $(\tilde{\epsilon}_t)_{t=0}^{T-1}$ and perturbation variables $(f_t)_{t=0}^T$, all independent. By the representation theorem for probability kernels [11], the transition kernel and the randomized policy can then be written as measurable maps F_t and $\tilde{F}_t^\theta$ of a state, a population argument and a noise, so that the unperturbed process $(X_t^\theta, \alpha_t^\theta)$ is a deterministic function of θ and of the noises. This is what allows the perturbed and unperturbed processes to be compared pathwise, under common random numbers, throughout the chapter. Appendix A.1 records the construction.

3.2 Perturbed process and value function

We now randomize the population argument while keeping the reference population flow $(\mathbb{P}_{X_t^\theta})_{t \in \mathcal{T}}$. For the abstract construction, assume that each perturbation variable $f \in \mathbf{F}$ induces a measurable map $T_f^\lambda : \mathcal{X} \rightarrow \mathcal{X}$. In the vector-space case used by the continuous benchmarks, this is written as $T_f^\lambda = I_d + \lambda f$. For any $\theta \in \Theta$ and $\lambda \geq 0$, define

$$\Phi_t^{\lambda, \theta}(f) := T_f^\lambda \# \mathbb{P}_{X_t^\theta}, \quad \mathbb{P}_t^{\lambda, \theta}(\omega) := \Phi_t^{\lambda, \theta}(f_t(\omega)).$$

The perturbed trajectory associated with θ is then

$$\begin{cases} X_0^{\lambda, \theta}(\omega) = x_0, \\ \alpha_t^{\lambda, \theta}(\omega) = \tilde{F}_t^\theta \left(X_t^{\lambda, \theta}(\omega), \mathbb{P}_t^{\lambda, \theta}(\omega), \tilde{\epsilon}_t(\omega) \right), \\ X_{t+1}^{\lambda, \theta}(\omega) = F_t \left(X_t^{\lambda, \theta}(\omega), \mathbb{P}_t^{\lambda, \theta}(\omega), \alpha_t^{\lambda, \theta}(\omega), \epsilon_{t+1}(\omega) \right), \end{cases} \quad t = 0, \dots, T-1.$$

Conditionally on the realized perturbation, this process satisfies

$$\begin{cases} X_0^{\lambda, \theta} \sim \mu_0, \\ \mathcal{L} \left(\alpha_t^{\lambda, \theta} \mid X_t^{\lambda, \theta}, \mathbb{P}_t^{\lambda, \theta} \right) = \pi_t^\theta(\cdot \mid X_t^{\lambda, \theta}, \mathbb{P}_t^{\lambda, \theta}), \\ \mathcal{L} \left(X_{t+1}^{\lambda, \theta} \mid X_t^{\lambda, \theta}, \mathbb{P}_t^{\lambda, \theta}, \alpha_t^{\lambda, \theta} \right) = P_t(\cdot \mid X_t^{\lambda, \theta}, \mathbb{P}_t^{\lambda, \theta}, \alpha_t^{\lambda, \theta}). \end{cases}$$

The perturbed reward functional is

$$J^\lambda(\theta) := \mathbb{E} \left[\sum_{t=0}^{T-1} r_t(X_t^{\lambda, \theta}, \mathbb{P}_t^{\lambda, \theta}, \alpha_t^{\lambda, \theta}) + g(X_T^{\lambda, \theta}, \mathbb{P}_T^{\lambda, \theta}) \right], \quad \sup_{\theta \in \Theta} J^\lambda(\theta).$$

Equivalently, write

$$\begin{aligned} s &= (x_0, f_0, a_0, \dots, x_{T-1}, f_{T-1}, a_{T-1}, x_T, f_T), \\ \tilde{s} &= (x_0, m_0, a_0, \dots, x_{T-1}, m_{T-1}, a_{T-1}, x_T, m_T). \end{aligned}$$

The trajectory law can be expressed either in terms of the perturbation variables f_t ,

$$p_\theta^\lambda(ds) = \mu_0(dx_0) \prod_{\tau=0}^{T-1} \nu(df_\tau) \pi_\tau^\theta(da_\tau \mid x_\tau, \Phi_\tau^{\lambda, \theta}(f_\tau)) P_\tau(dx_{\tau+1} \mid x_\tau, \Phi_\tau^{\lambda, \theta}(f_\tau), a_\tau) \cdot \nu(df_T),$$

or in terms of the induced random population arguments $m_t = \Phi_t^{\lambda, \theta}(f_t)$,

$$\tilde{p}_\theta^\lambda(d\tilde{s}) = \mu_0(dx_0) \prod_{\tau=0}^{T-1} \left(\Phi_\tau^{\lambda, \theta} \# \nu \right) (dm_\tau) \pi_\tau^\theta(da_\tau \mid x_\tau, m_\tau) P_\tau(dx_{\tau+1} \mid x_\tau, m_\tau, a_\tau) \cdot \left(\Phi_T^{\lambda, \theta} \# \nu \right) (dm_T).$$

Therefore

$$\begin{aligned} J^\lambda(\theta) &= \int \left[\sum_{t=0}^{T-1} r_t(x_t, \Phi_t^{\lambda, \theta}(f_t), a_t) + g(x_T, \Phi_T^{\lambda, \theta}(f_T)) \right] p_\theta^\lambda(ds) \\ &= \int \left[\sum_{t=0}^{T-1} r_t(x_t, m_t, a_t) + g(x_T, m_T) \right] \tilde{p}_\theta^\lambda(d\tilde{s}). \end{aligned}$$

3.3 Discrete-state space

Let $\mathcal{X} = \{1, \dots, N\}$. Define the probability simplex on $\mathcal{X}$ as $\Delta_N := \{p \in \mathbb{R}_+^N : \sum_{i=1}^N p_i = 1\}$ and denote Δ_N° its relative interior. For a policy parameter $\theta \in \Theta$, let $\mu_t^\theta(i) := \mathbb{P}(X_t^\theta = i)$ be the law of the unperturbed controlled state process, for $i \in \{1, \dots, N\}$.

We introduce a random perturbation of the population law μ_t^θ while keeping the perturbed object inside the simplex. Consider a stochastic matrix $Q_t = (Q_t^{i,j})_{1 \leq i,j \leq N}$, and focus on the rank-one perturbation $Q_t^{j,i} = q_t^i$ where q_t is a random element of Δ_N° . The perturbed population law is then $M_t^{\lambda,\theta}(\omega) := (1 - \lambda)\mu_t^\theta + \lambda q_t(\omega)$. Notice that $M_t^{\lambda,\theta}$ is a random probability vector, but it should not be identified with the law of the perturbed state process defined below.

We now define the perturbed controlled process

$$\begin{cases} X_0^{\lambda,\theta} & \sim \mu_0, \\ \alpha_t^{\lambda,\theta} & \sim \pi_t^\theta(\cdot \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta}), \\ X_{t+1}^{\lambda,\theta} & \sim P_t(\cdot \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta}, \alpha_t^{\lambda,\theta}), \end{cases} \quad t = 0, \dots, T-1.$$

The perturbed objective is

$$J^\lambda(\theta) = \mathbb{E} \left[\sum_{t=0}^{T-1} r_t(X_t^{\lambda,\theta}, M_t^{\lambda,\theta}, \alpha_t^{\lambda,\theta}) + g(X_T^{\lambda,\theta}, M_T^{\lambda,\theta}) \right],$$

where the expectation is taken over the perturbations, the state process and the randomized actions.

3.3.1 Density and score of the randomized perturbation

For each t , let U_t be an $\mathbb{R}^{N-1}$ -valued random vector with density ρ with respect to the Lebesgue measure on $\mathbb{R}^{N-1}$. Define $q_t = \varphi(U_t) \in \Delta_N^\circ$, with the map $\varphi : \mathbb{R}^{N-1} \rightarrow \Delta_N^\circ$ given by

$$\varphi(u) = \left(\frac{\exp(u_1)}{1 + \sum_{j=1}^{N-1} \exp(u_j)}, \dots, \frac{\exp(u_{N-1})}{1 + \sum_{j=1}^{N-1} \exp(u_j)}, \frac{1}{1 + \sum_{j=1}^{N-1} \exp(u_j)} \right).$$

Its inverse is $\varphi^{-1}(p) = \left(\log \frac{p_1}{p_N}, \dots, \log \frac{p_{N-1}}{p_N} \right)$, $p \in \Delta_N^\circ$.

For fixed (t, λ, θ) , define the affine map $\phi_t^{\lambda,\theta} : \Delta_N^\circ \rightarrow \Delta_N^\circ$ such that $\phi_t^{\lambda,\theta}(p) := (1 - \lambda)\mu_t^\theta + \lambda p$. Then

$$M_t^{\lambda,\theta}(\omega) = \phi_t^{\lambda,\theta}(\varphi(U_t(\omega))), \quad \mathcal{L}(M_t^{\lambda,\theta}) = (\phi_t^{\lambda,\theta} \circ \varphi) \# \mathcal{L}(U_t).$$

Theorem 1 (Perturbation estimate). *Let $\lambda \in (0, 1)$ and let $q_t \in \Delta_N$. For the perturbed population law $M_t^{\lambda,\theta}(\omega) = (1 - \lambda)\mu_t^\theta + \lambda q_t(\omega)$, we have*

$$d_{\text{TV}}(M_t^{\lambda,\theta}(\omega), \mu_t^\theta) \leq \lambda, \quad \text{for every } \omega.$$

Moreover, if $\mathcal{X}$ is endowed with a metric $d_{\mathcal{X}}$ of diameter $\text{diam}(\mathcal{X}) := \max_{i,j \in \mathcal{X}} d_{\mathcal{X}}(i, j)$, then $W_1(M_t^{\lambda,\theta}(\omega), \mu_t^\theta) \leq \text{diam}(\mathcal{X})\lambda$, for every ω . Consequently, for any of the distances above, there exists a function $h(\lambda)$ such that

$$\mathbb{E} \left[d(M_t^{\lambda,\theta}, \mu_t^\theta) \right] \leq h(\lambda), \quad h(\lambda) \xrightarrow{\lambda \downarrow 0} 0.$$

The two estimates follow from $M_t^{\lambda,\theta} - \mu_t^\theta = \lambda(q_t - \mu_t^\theta)$ and from the domination of W_1 by the total-variation distance; see Appendix A.2.1.

We now compute the density of $M_t^{\lambda,\theta}$ on the simplex and its score.

Proposition 1 (Density of the logistic perturbation). *Assume that U_t has density ρ on $\mathbb{R}^{N-1}$. Then $q_t = \varphi(U_t)$ admits the density*

$$f_q(p) = \rho \left(\log \frac{p_1}{p_N}, \dots, \log \frac{p_{N-1}}{p_N} \right) \frac{1}{\prod_{i=1}^N p_i}, \quad p \in \Delta_N^\circ.$$

The proof is a change-of-variables calculation on the simplex coordinates and is given in Appendix A.2.2.

Proposition 2 (Density of the perturbed population law). *Let $\Lambda_t^{\lambda, \theta} := \{(1 - \lambda)\mu_t^\theta + \lambda m, m \in \Delta_N^\circ\}$. For $m \in \Lambda_t^{\lambda, \theta}$, define $p_i^{\lambda, \theta}(m) := (m_i - (1 - \lambda)\mu_t^\theta(i))/\lambda$ for $i = 1, \dots, N$; equivalently, $p^{\lambda, \theta}(m) = (\phi_t^{\lambda, \theta})^{-1}(m)$. Then $M_t^{\lambda, \theta}$ admits the density*

$$f_t^{\lambda, \theta}(m) = \frac{1}{\lambda^{N-1}} \rho \left(\log \frac{p_1^{\lambda, \theta}(m)}{p_N^{\lambda, \theta}(m)}, \dots, \log \frac{p_{N-1}^{\lambda, \theta}(m)}{p_N^{\lambda, \theta}(m)} \right) \frac{1}{\prod_{i=1}^N p_i^{\lambda, \theta}(m)} \mathbf{1}_{\Lambda_t^{\lambda, \theta}}(m),$$

and $\mathcal{L}(M_t^{\lambda, \theta})(dm) = f_t^{\lambda, \theta}(m)dm$, where dm denotes the Lebesgue measure on the simplex in the coordinates $(m_1, \dots, m_{N-1})$.

The proof follows by applying the affine change of variables after the logistic transformation; see Appendix A.2.2.

Corollary 1 (Score of the perturbation density). *Assume that ρ is strictly positive and continuously differentiable on $\mathbb{R}^{N-1}$. Let $m \in \Lambda_t^{\lambda, \theta}$. Write $p_i := p_i^{\lambda, \theta}(m)$, and set $z_i := \log \frac{p_i}{p_N}$ and $a_i(z) := \partial_{z_i} \log \rho(z)$ for $i = 1, \dots, N - 1$. Then for fixed m in the interior of the support,*

$$\nabla_\theta \log f_t^{\lambda, \theta}(m) = -\frac{1 - \lambda}{\lambda} \sum_{k=1}^{N-1} H_k(p) \nabla_\theta \mu_t^\theta(k),$$

where $H_k(p) = \frac{a_k(z)}{p_k} + \frac{\sum_{i=1}^{N-1} a_i(z)}{p_N} - \frac{1}{p_k} + \frac{1}{p_N}$ for $k = 1, \dots, N - 1$.

The proof differentiates the previous density at fixed m and collects the coefficients of the independent coordinates of $\nabla_\theta \mu_t^\theta$; see Appendix A.2.2.

3.3.2 Convergence of the perturbed objective

We now show that the perturbed control problem is an approximation of the original one when λ is small. The law of the perturbed state process, $\nu_t^{\lambda, \theta} := \mathcal{L}(X_t^{\lambda, \theta})$, needs to be shown to remain close to μ_t^θ .

For $\theta \in \Theta$, define the averaged transition kernel, a probability measure on $\mathcal{X}$ and the conditional law of the next state X_{t+1} given $X_t = i$, after averaging over the randomized action $\alpha_t \sim \pi_t^\theta(\cdot | i, m)$.

$$K_t^\theta(j | i, m) := \int_{\mathcal{A}} P_t(j | i, m, a) \pi_t^\theta(da | i, m), \quad i, j \in \mathcal{X}, \quad m \in \Delta_N,$$

and the averaged running reward

$$R_t^\theta(i, m) := \int_{\mathcal{A}} r_t(i, m, a) \pi_t^\theta(da | i, m), \quad i \in \mathcal{X}, \quad m \in \Delta_N.$$

The unperturbed law and the perturbed dynamics each then satisfy

$$\mu_{t+1}^\theta(j) = \sum_{i=1}^N \mu_t^\theta(i) K_t^\theta(j | i, \mu_t^\theta), \quad \nu_{t+1}^{\lambda, \theta}(j) = \mathbb{E} \left[K_t^\theta(j | X_t^{\lambda, \theta}, M_t^{\lambda, \theta}) \right], \quad j \in \mathcal{X}.$$

Indeed, by the tower property,

$$\begin{aligned} \nu_{t+1}^{\lambda, \theta}(j) &= \mathbb{P}(X_{t+1}^{\lambda, \theta} = j) = \mathbb{E} \left[\mathbf{1}_{\{X_{t+1}^{\lambda, \theta} = j\}} \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\mathbf{1}_{\{X_{t+1}^{\lambda, \theta} = j\}} \mid X_t^{\lambda, \theta}, M_t^{\lambda, \theta}, \alpha_t^{\lambda, \theta} \right] \right] \\ &= \mathbb{E} \left[P_t(j | X_t^{\lambda, \theta}, M_t^{\lambda, \theta}, \alpha_t^{\lambda, \theta}) \right]. \end{aligned}$$

Conditioning now on $(X_t^{\lambda,\theta}, M_t^{\lambda,\theta})$, and using $\alpha_t^{\lambda,\theta} \sim \pi_t^\theta(\cdot \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta})$, we obtain

$$\begin{aligned}\nu_{t+1}^{\lambda,\theta} &= \mathbb{E} \left[\int_A P_t(j \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta}, a) \pi_t^\theta(da \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta}) \right] \\ &= \mathbb{E} \left[K_t^\theta(j \mid X_t^{\lambda,\theta}, M_t^{\lambda,\theta}) \right].\end{aligned}$$

Assumption 1 (Uniform regularity). *There exists constants $L_K, L_R, L_g \geq 0$ and $\bar{R}, \bar{g} < \infty$ such that, for every $\theta \in \Theta$, $t = 0, \dots, T-1$, $i \in \mathcal{X}$, and $m, m' \in \Delta_N$,*

$$\begin{aligned}d_{\text{TV}} \left(K_t^\theta(\cdot \mid i, m), K_t^\theta(\cdot \mid i, m') \right) &\leq L_K d_{\text{TV}}(m, m'), \\ |R_t^\theta(i, m) - R_t^\theta(i, m')| &\leq L_R d_{\text{TV}}(m, m'), \\ |g(i, m) - g(i, m')| &\leq L_g d_{\text{TV}}(m, m'), \\ |R_t^\theta(i, m)| &\leq \bar{R}, \quad |g(i, m)| \leq \bar{g}.\end{aligned}$$

Lemma 1 (Stability of the state marginal). *Under Assumption 1, for every $\theta \in \Theta$ and every $t = 0, \dots, T$,*

$$d_{\text{TV}}(\nu_t^{\lambda,\theta}, \mu_t^\theta) \leq L_K \lambda t.$$

The proof is a finite-horizon induction based on the dual representation of total variation; see Appendix A.2.3.

Theorem 2 (Convergence of the perturbed objective). *Under Assumption 1, for every $\theta \in \Theta$,*

$$|J^\lambda(\theta) - J(\theta)| \leq C_T \lambda.$$

If the constants in the assumption are uniform in θ , then the convergence is uniform.

The proof decomposes running and terminal reward errors into marginal and population-argument errors, then uses Lemma 1; see Appendix A.2.3.

Corollary 2 (Consistency of optimizers). *Assume that Θ is compact, that J is continuous on Θ , and that $\sup_{\theta \in \Theta} |J^\lambda(\theta) - J(\theta)| \xrightarrow{\lambda \downarrow 0} 0$. Let $\theta_\lambda^* \in \operatorname{argmax}_{\theta \in \Theta} J^\lambda(\theta)$ and suppose that, along a sequence $\lambda_n \downarrow 0$, $\theta_{\lambda_n}^* \rightarrow \bar{\theta}$.*

Then $\bar{\theta} \in \operatorname{argmax}_{\theta \in \Theta} J(\theta)$. In particular, if J admits a unique maximizer θ^ , then $\theta_\lambda^* \rightarrow \theta^*$ as $\lambda \downarrow 0$.*

This is the standard argmax-continuity argument under uniform convergence; see Appendix A.2.4.

3.3.3 Gradient-level convergence

Theorem 2 proves the convergence of the perturbed analysis, but a stronger gradient-level version is needed for the statistical analysis of the policy-gradient estimator. This requires additional smoothness assumptions. We use the notation $D_t^\theta[h] := \partial_h \mu_t^\theta$ and $E_t^{\lambda,\theta}[h] := \partial_h \nu_t^{\lambda,\theta}$ for the directional derivatives in the direction $h \in \mathbb{R}^{d_\theta}$, where d_θ is the dimension of the parameter space. We assume $|h| = 1$.

Assumption 2 (Smoothness of the averaged dynamics). *The maps $(\theta, m) \mapsto K_t^\theta(\cdot \mid i, m)$ and $(\theta, m) \mapsto R_t^\theta(i, m)$ are continuously differentiable with respect to θ and m , uniformly in t and i . The terminal reward $m \mapsto g(i, m)$ is continuously differentiable.*

Moreover, there exists a constant $C_1 > 0$ such that, for every unit vector h , every vector $v \in \mathbb{R}^N$ satisfying $\sum_i v_i = 0$, and every $m, m' \in \Delta_N$,

$$\begin{aligned}d_{\text{TV}} \left(\partial_h K_t^\theta(\cdot \mid i, m), \partial_h K_t^\theta(\cdot \mid i, m') \right) &\leq C_1 d_{\text{TV}}(m, m'), \\ d_{\text{TV}} \left(D_m K_t^\theta(\cdot \mid i, m)[v], D_m K_t^\theta(\cdot \mid i, m')[v] \right) &\leq C_1 \|v\|_1 d_{\text{TV}}(m, m'), \\ d_{\text{TV}} \left(D_m K_t^\theta(\cdot \mid i, m)[v], D_m K_t^\theta(\cdot \mid i, m)[v'] \right) &\leq C_1 \|v - v'\|_1.\end{aligned}$$

Similarly,

$$\begin{aligned} \left| \partial_h R_t^\theta(i, m) - \partial_h R_t^\theta(i, m') \right| &\leq C_1 d_{\text{TV}}(m, m'), \\ \left| D_m R_t^\theta(i, m)[v] - D_m R_t^\theta(i, m')[v] \right| &\leq C_1 \|v\|_1 d_{\text{TV}}(m, m'), \\ \left| D_m R_t^\theta(i, m)[v] - D_m R_t^\theta(i, m)[v'] \right| &\leq C_1 \|v - v'\|_1. \end{aligned}$$

The same type of bounds hold for g , namely

$$\begin{aligned} \left| D_m g(i, m)[v] - D_m g(i, m')[v] \right| &\leq C_1 \|v\|_1 d_{\text{TV}}(m, m'), \\ \left| D_m g(i, m)[v] - D_m g(i, m)[v'] \right| &\leq C_1 \|v - v'\|_1. \end{aligned}$$

Finally, all first-order derivatives appearing above are uniformly bounded.

Lemma 2 (Stability of the differentiated state marginal). *Under Assumption 2, there exists a constant $C_T^D > 0$ such that, for every $\theta \in \Theta$, every $\lambda \in (0, 1)$, every unit direction h , and every $t = 0, \dots, T$, $\|E_t^{\lambda, \theta}[h] - D_t^\theta[h]\|_1 \leq C_T^D \lambda$.*

The proof differentiates the two marginal recursions, compares the resulting triangular systems, and applies a finite-horizon induction; see Appendix A.2.5.

Theorem 3 (Gradient-level convergence). *Under Assumption 2, there exists a constant $C_T^\nabla > 0$ such that, for every $\theta \in \Theta$ and every $\lambda \in (0, 1)$,*

$$\|\nabla_\theta J^\lambda(\theta) - \nabla_\theta J(\theta)\| \leq C_T^\nabla \lambda.$$

If the constants in the assumption are uniform in θ , then the convergence is uniform in θ .

The proof compares directional derivatives of J^λ and J term by term and uses the differentiated marginal stability estimate; see Appendix A.2.5.

3.3.4 Policy-gradient formula for the perturbed problem

We now derive a likelihood-ratio representation for the gradient of the perturbed objective J^λ . The perturbation introduced transfers the dependence on θ through the population law to the density $f_t^{\lambda, \theta}$ of the random variable $M_t^{\lambda, \theta}$. Differentiating J^λ therefore produces a standard policy-score term and an additional score term coming from the density of the perturbation.

Define the perturbed return

$$\mathcal{R}_\theta^\lambda := \sum_{t=0}^{T-1} r_t(X_t^{\lambda, \theta}, M_t^{\lambda, \theta}, \alpha_t^{\lambda, \theta}) + g(X_T^{\lambda, \theta}, M_T^{\lambda, \theta}).$$

The complete trajectory of the perturbed system is denoted by

$$\Gamma^\lambda := (M_0^{\lambda, \theta}, X_0^{\lambda, \theta}, \alpha_0^{\lambda, \theta}, \dots, M_{T-1}^{\lambda, \theta}, X_{T-1}^{\lambda, \theta}, \alpha_{T-1}^{\lambda, \theta}, M_T^{\lambda, \theta}, X_T^{\lambda, \theta}).$$

The law of this trajectory has density, with respect to the product reference measure, proportional to

$$\mu_0(x_0) \prod_{t=0}^{T-1} \left[f_t^{\lambda, \theta}(m_t) p_t^\theta(a_t \mid x_t, m_t) P_t(x_{t+1} \mid x_t, a_t, m_t) \right] f_T^{\lambda, \theta}(m_T).$$

Here, the transition probabilities P_t do not depend directly on θ . Their dependence on θ has been moved to the random population argument $M_t^{\lambda, \theta}$ through its density $f_t^{\lambda, \theta}$. Because the support $\Lambda_t^{\lambda, \theta}$ may move with θ , the following likelihood-ratio statement is understood under a common-support or vanishing-boundary-term condition; equivalently, differentiation of the induced trajectory law is justified without extra boundary contributions.

Assumption 3 (Differentiability and integrability). *For every t , the randomized policy admits a density $p_t^\theta(a \mid i, m)$ with respect to a reference measure $\nu_{\mathcal{A}}$ on $\mathcal{A}$, strictly positive and continuously differentiable in θ ; the perturbation density $f_t^{\lambda, \theta}$ is strictly positive on the relative interior of Δ_N and continuously differentiable in θ ; and differentiation under the integral sign is justified, equivalently the trajectory score S_θ^λ of (3.2) satisfies $\mathbb{E}[\|\mathcal{R}_\theta^\lambda S_\theta^\lambda\|] < \infty$.*

Theorem 4 (Policy-gradient formula for the perturbed objective). *Under the previous assumptions, $\nabla_\theta J^\lambda(\theta) = \mathbb{E}[\mathcal{R}_\theta^\lambda S_\theta^\lambda]$, where, using Corollary 1, the score of the perturbed trajectory is*

$$\begin{aligned} S_\theta^\lambda &:= \sum_{t=0}^{T-1} \nabla_\theta \log p_t^\theta(\alpha_t^{\lambda, \theta} \mid X_t^{\lambda, \theta}, M_t^{\lambda, \theta}) + \sum_{t=0}^T \nabla_\theta \log f_t^{\lambda, \theta}(M_t^{\lambda, \theta}) \\ &= \sum_{t=0}^{T-1} \nabla_\theta \log p_t^\theta(\alpha_t^{\lambda, \theta} \mid X_t^{\lambda, \theta}, M_t^{\lambda, \theta}) - \frac{1-\lambda}{\lambda} \sum_{t=0}^T \sum_{k=1}^{N-1} H_k(q_t) \nabla_\theta \mu_t^\theta(k). \end{aligned} \quad (3.2)$$

The proof differentiates the density of the perturbed trajectory under the integral sign; the transition kernels carry no direct θ -dependence once the sampled population arguments are fixed, so only the policy and perturbation densities contribute. See Appendix A.2.6.

Corollary 3 (Baseline version). *Let $b \in \mathbb{R}$ be any deterministic baseline. Then $\nabla_\theta J^\lambda(\theta) = \mathbb{E}[(\mathcal{R}_\theta^\lambda - b) S_\theta^\lambda]$. More generally, one may use any baseline b such that $\mathbb{E}[b S_\theta^\lambda] = 0$.*

This follows from $\mathbb{E}[S_\theta^\lambda] = 0$; see Appendix A.2.6.

3.3.5 Estimating the sensitivity and assembling the estimator

The gradient formula of Theorem 4 still contains the population-flow sensitivity $D_t^\theta(k) := \nabla_\theta \mu_t^\theta(k)$, which is not observed along a single trajectory and for which the plain likelihood-ratio identity gives no exact model-free estimator, because the kernel and the policy both read the whole law μ_t^θ . We estimate it by applying the same perturbation idea one level down. An auxiliary batch is simulated under the population argument $\bar{M}_s^{\eta, \theta} = (1 - \eta)\mu_s^\theta + \eta \bar{q}_s$, at an auxiliary scale η that need not equal λ , and differentiating the auxiliary marginal $\mu_{t,k}^\eta(\theta) = \mathbb{P}(\bar{X}_t^{\eta, \theta} = k)$ by the likelihood-ratio identity gives, in the limit $\eta \downarrow 0$, a representation of $D_t^\theta(k)$ as an expectation of an indicator against the accumulated policy and perturbation scores up to time $t - 1$. Two features of that representation matter. It is triangular in time, since D_t^θ involves only $D_0^\theta, \dots, D_{t-1}^\theta$, and $D_0^\theta = 0$ because μ_0 does not depend on θ ; the whole sensitivity flow can therefore be estimated by a single forward recursion, which is what makes the simulator cost linear rather than quadratic in the horizon. Averaging over n auxiliary trajectories yields the plug-in estimator $\hat{D}_t^{\eta, n, \theta}(k)$, with $\hat{D}_t^{\eta, n, \theta}(N) = -\sum_{k < N} \hat{D}_t^{\eta, n, \theta}(k)$.

Combining it with the main batch gives the estimator used throughout. With B perturbed trajectories, perturbed returns $\mathcal{R}_\theta^{\lambda, (b)}$, a deterministic baseline b_0 , policy score $\mathcal{S}_{\text{pol}}^{\lambda, \theta, (b)}$ and estimated perturbation score

$$\hat{\mathcal{S}}_{\text{pert}}^{\lambda, \eta, n, \theta, (b)} := -\frac{1-\lambda}{\lambda} \sum_{t=0}^T \sum_{k=1}^{N-1} H_k(q_t^{(b)}) \hat{D}_t^{\eta, n, \theta}(k),$$

the transport policy-gradient estimator is

$$\hat{G}_{\lambda, \eta, B, n}(\theta) := \frac{1}{B} \sum_{b=1}^B \left(\mathcal{R}_\theta^{\lambda, (b)} - b_0 \right) \left(\mathcal{S}_{\text{pol}}^{\lambda, \theta, (b)} + \hat{\mathcal{S}}_{\text{pert}}^{\lambda, \eta, n, \theta, (b)} \right). \quad (3.3)$$

Replacing $\hat{D}^{\eta, n, \theta}$ by the exact sensitivities defines the oracle estimator $G_{\lambda, B}(\theta)$, which is not implementable but is exactly unbiased for $\nabla_\theta J^\lambda(\theta)$ and separates the effect of the transport perturbation from the error of estimating the mean-field derivative. The baseline plays no role beyond variance reduction. Appendix A.2.7 gives the auxiliary process, the derivation of the sensitivity identity, and the full definitions; Chapter 4 gives the resulting algorithms.

The oracle estimator $G_{\lambda,B}$ only contains the perturbation bias and the ordinary Monte Carlo variance. The practical estimator $\hat{G}_{\lambda,\eta,B,n}$ adds a third source of error: the auxiliary error made when D_t^θ is replaced by $\hat{D}_t^{\eta,n,\theta}$.

Proposition 3 (Unbiasedness and error of the oracle estimator). *Under the assumptions of Theorem 3, assume in addition that the perturbed returns and scores have bounded second moments, with the perturbation-score contribution scaling as $O(\lambda^{-1})$. Then*

$$\begin{aligned}\mathbb{E}[G_{\lambda,B}(\theta)] &= \nabla_\theta J^\lambda(\theta), \\ \mathbb{E}\left[\|G_{\lambda,B}(\theta) - \nabla_\theta J^\lambda(\theta)\|^2\right] &\leq C_T \frac{1 + \lambda^{-2}}{B}.\end{aligned}$$

Consequently, with respect to the original objective,

$$\begin{aligned}\|\mathbb{E}[G_{\lambda,B}(\theta)] - \nabla_\theta J(\theta)\| &\leq C_T \lambda, \\ \mathbb{E}\left[\|G_{\lambda,B}(\theta) - \nabla_\theta J(\theta)\|^2\right] &\leq C_T \left(\lambda^2 + \frac{1 + \lambda^{-2}}{B}\right).\end{aligned}$$

Unbiasedness is Theorem 4 together with the baseline identity, and the error bound combines the variance of a B -sample average with the gradient convergence estimate; see Appendix A.2.8.

Proposition 4 (Auxiliary sensitivity error). *Assume the gradient-level convergence estimate of Theorem 3 and suppose that the auxiliary likelihood-ratio scores have uniformly bounded second moments, with the perturbation-density score scaling as $O(\eta^{-1})$. Then, for every $t = 0, \dots, T$ and $k = 1, \dots, N-1$, there exists a constant $C_T > 0$ such that*

$$\mathbb{E}\left[\|\hat{D}_t^{\eta,n,\theta}(k) - D_t^\theta(k)\|^2\right] \leq C_T \left(\eta^2 + \frac{1}{n\eta^2}\right).$$

The two terms are the $O(\eta)$ bias of the η -perturbed target and the $O((n\eta^2)^{-1})$ variance of an n -sample likelihood-ratio average, propagated through the recursion by induction on t ; see Appendix A.2.8.

The full decomposition is stated through the main-batch coefficients $A_{t,k}^{\lambda,B}(\theta)$, the empirical average of the perturbed return against the perturbation score at coordinate k and time t , and their population counterparts $a_{t,k}^\lambda(\theta)$; both are defined in Appendix A.2.8.

Theorem 5 (Bias and mean-square error). *Assume the hypotheses of Proposition 4. Assume also that the perturbed returns, policy scores and perturbation scores have bounded second moments uniformly in θ , with the perturbation-score contribution scaling as $O(\lambda^{-1})$. Then*

$$\begin{aligned}\|\mathbb{E}[\hat{G}_{\lambda,\eta,B,n}(\theta)] - \nabla_\theta J(\theta)\| &\leq C_T \left(\lambda + \frac{1}{\lambda} \sqrt{\eta^2 + \frac{1}{n\eta^2}}\right), \\ \mathbb{E}\left[\|\hat{G}_{\lambda,\eta,B,n}(\theta) - \nabla_\theta J(\theta)\|^2\right] &\leq C_T \left(\lambda^2 + \frac{1 + \lambda^{-2}}{B} + \frac{1}{\lambda^2} \left(\eta^2 + \frac{1}{n\eta^2}\right)\right).\end{aligned}$$

The proof conditions on the auxiliary batch, which isolates the plug-in error of the estimated sensitivities, and then applies the conditional variance decomposition; see Appendix A.2.8.

The bound displays the usual tuning tradeoff. Decreasing λ reduces the deterministic perturbation bias, but increases both the variance of the perturbation score and the amplification of the auxiliary sensitivity error. The auxiliary radius η has the same structure: small η reduces the bias of the sensitivity target, while the likelihood-ratio variance grows like η^{-2} .

3.4 Continuous-state space

3.4.1 Gaussian-mixture representation and perturbation of the population law

We now consider the continuous-state setting $\mathcal{X} = \mathbb{R}^d$, and denote by $\mathcal{P}_1(\mathbb{R}^d)$ the set of probability measures with finite first moment. For a policy parameter $\theta \in \Theta$, let $\mu_t^\theta := \mathcal{L}(X_t^\theta) \in \mathcal{P}_1(\mathbb{R}^d)$, for $t \in \mathcal{T}$, be the population law of the unperturbed controlled process.

The main difference with the finite-state setting is that μ_t^θ is now an infinite-dimensional object. In the discrete case, the population law is exactly represented by a point of a finite-dimensional simplex. No analogous exact representation exists for a general probability measure on $\mathbb{R}^d$. We therefore introduce a finite-dimensional approximation of the population law before applying the randomization argument.

Fix an integer $K \geq 1$. We consider the family of K -component Gaussian mixtures

$$\mathcal{G}_K := \left\{ \sum_{k=1}^K w_k \mathcal{N}(m_k, \Sigma_k) : w \in \Delta_K^\circ, m_k \in \mathbb{R}^d, \Sigma_k \in \mathbb{S}_{++}^d \right\},$$

where Δ_K° denotes the relative interior of the K -dimensional probability simplex and $\mathbb{S}_{++}^d$ the set of symmetric positive-definite $d \times d$ matrices.

To perturb elements of $\mathcal{G}_K$ while automatically preserving the constraints on the weights and covariance matrices, we work in unconstrained Euclidean coordinates. Let $q_K := (K-1) + Kd + K\frac{d(d+1)}{2}$. For $z = (\beta, m_1, \dots, m_K, s_1, \dots, s_K) \in \mathbb{R}^{q_K}$, with $\beta \in \mathbb{R}^{K-1}$, define the mixture weights by the logistic parametrization

$$w_k(\beta) = \frac{\exp(\beta_k)}{1 + \sum_{j=1}^{K-1} \exp(\beta_j)}, \quad k = 1, \dots, K-1,$$

$$w_K(\beta) = \frac{1}{1 + \sum_{j=1}^{K-1} \exp(\beta_j)}.$$

For each k , the vector $s_k \in \mathbb{R}^{d(d+1)/2}$ parametrizes a lower-triangular matrix $L(s_k)$ whose diagonal entries are exponentials of the corresponding diagonal coordinates of s_k . We then set $\Sigma(s_k) := L(s_k)L(s_k)^\top \in \mathbb{S}_{++}^d$. This defines a smooth decoder

$$\Gamma_K : \mathbb{R}^{q_K} \longrightarrow \mathcal{G}_K, \quad \Gamma_K(z) := \sum_{k=1}^K w_k(\beta) \mathcal{N}(m_k, \Sigma(s_k)).$$

The map Γ_K transforms arbitrary Euclidean parameters into a valid probability measure. This plays the same role as the softmax map in the finite-state construction: perturbations can be performed in an unconstrained Euclidean space and subsequently mapped back to the space of admissible probability measures.

Finite-dimensional representation of the population flow. Let $\mathcal{R}_K : \mathcal{P}_1(\mathbb{R}^d) \rightarrow \mathbb{R}^{q_K}$ be a measurable representation map, and set $z_{K,t}^\theta := \mathcal{R}_K(\mu_t^\theta)$ and $\bar{\mu}_{K,t}^\theta := \Gamma_K(z_{K,t}^\theta)$, the K -component mixture representation of the true population law. The map $\mathcal{R}_K$ is kept abstract; it may be a population-level mixture fitting procedure, and in the implementation it is replaced by an estimator built from particles sampled from μ_t^θ . The statistical procedure that constructs it is conceptually separate from the policy-gradient argument. For a metric $d_{\mathcal{P}}$ on $\mathcal{P}_1(\mathbb{R}^d)$, the representation error is

$$\delta_{K,t}^\theta := d_{\mathcal{P}}(\mu_t^\theta, \bar{\mu}_{K,t}^\theta), \quad \delta_K^\theta := \max_{t \in \mathcal{T}} \delta_{K,t}^\theta.$$

Unlike in the finite-state parametrization this error is generally non-zero at fixed K , and it is independent of the perturbation scale introduced below. It vanishes when $\mu_t^\theta \in \mathcal{G}_K$ and $\mathcal{R}_K$ is exact; in particular the Gaussian setting is the case $K = 1$, so the linear-quadratic model of Chapter 5 is an exact finite-dimensional subcase of the construction.

Two conventions are in force throughout. Since Γ_K is not injective under permutations of the components, we assume that $\mathcal{R}_K$ selects a deterministic representative, for instance through a fixed component ordering; and for the differentiation arguments below we restrict attention to non-degenerate representations, with mixture weights bounded away from zero and covariance matrices uniformly positive definite, which prevents label collisions and singular components along the population flow.

Random perturbation of the Gaussian-mixture coordinates. Let $(U_t)_{t \in \mathcal{T}}$ be i.i.d. $\mathbb{R}^{q_K}$ -valued random variables, independent of the state and policy noises, with common distribution ν admitting a density ρ with respect to Lebesgue measure. For $\lambda > 0$, define the randomized mixture parameter

$$Z_{K,t}^{\lambda,\theta} := z_{K,t}^\theta + \lambda U_t,$$

and the corresponding randomized population law

$$M_{K,t}^{\lambda,\theta} := \Gamma_K \left(Z_{K,t}^{\lambda,\theta} \right) = \Gamma_K \left(z_{K,t}^\theta + \lambda U_t \right).$$

By construction, $M_{K,t}^{\lambda,\theta} \in \mathcal{G}_K \subset \mathcal{P}_1(\mathbb{R}^d)$ almost surely. Moreover, $M_{K,t}^{0,\theta} = \bar{\mu}_{K,t}^\theta$.

The three objects must be kept apart: μ_t^θ is the true population law, $\bar{\mu}_{K,t}^\theta$ its deterministic finite-dimensional representation, and $M_{K,t}^{\lambda,\theta}$ the randomized argument used to build the perturbed problem; the last is not the law of the perturbed state process defined next. That process replaces the population argument in the policy and the transition kernel by $M_{K,t}^{\lambda,\theta}$,

$$\begin{cases} X_0^{K,\lambda,\theta} \sim \mu_0, \\ \alpha_t^{K,\lambda,\theta} \sim \pi_t^\theta \left(\cdot \mid X_t^{K,\lambda,\theta}, M_t^{K,\lambda,\theta} \right), \\ X_{t+1}^{K,\lambda,\theta} \sim P_t \left(\cdot \mid X_t^{K,\lambda,\theta}, M_{K,t}^{\lambda,\theta}, \alpha_t^{K,\lambda,\theta} \right), \end{cases} \quad t = 0, \dots, T-1,$$

and its objective $J_K^\lambda(\theta)$ is (3.1) evaluated along that process, the expectation running over the mixture perturbations, the state process and the randomized actions.

There are therefore two distinct approximation scales, $\mu_t^\theta \rightarrow \bar{\mu}_{K,t}^\theta$ as $K \rightarrow \infty$ and $M_{K,t}^{\lambda,\theta} \rightarrow \bar{\mu}_{K,t}^\theta$ as $\lambda \downarrow 0$. At fixed K , sending λ to zero removes the randomization but not the representation error, so recovering the original objective requires both limits. The following estimate makes the decomposition explicit.

Assumption 4 (Regularity of the Gaussian-mixture decoder). *Let $d_{\mathcal{P}} = W_1$. There exist $\lambda_0 > 0$ and $C_{\Gamma,K} < \infty$ such that, for all $\theta \in \Theta$, $t \in \mathcal{T}$, and $\lambda \in [0, \lambda_0]$,*

$$\mathbb{E} \left[W_1 \left(\Gamma_K(z_{K,t}^\theta + \lambda U_t), \Gamma_K(z_{K,t}^\theta) \right) \right] \leq C_{\Gamma,K} \lambda.$$

This assumption is a local regularity condition on the finite-dimensional chart. It is satisfied under uniform non-degeneracy and moment bounds on the mixture parameters together with suitable moment conditions on the perturbation distribution.

Theorem 6 (Representation and perturbation estimate). *Under Assumption 4, for every $\theta \in \Theta$, $t \in \mathcal{T}$, and $\lambda \in [0, \lambda_0]$,*

$$\mathbb{E} \left[W_1 \left(M_{K,t}^{\lambda,\theta}, \mu_t^\theta \right) \right] \leq \delta_{K,t}^\theta + C_{\Gamma,K} \lambda. \quad (3.4)$$

Consequently, if $\sup_{\theta \in \Theta} \max_{t \in \mathcal{T}} \delta_{K,t}^\theta \xrightarrow{K \rightarrow \infty} 0$, then

$$\sup_{\theta \in \Theta} \max_{t \in \mathcal{T}} \mathbb{E} \left[W_1 \left(M_{K,t}^{\lambda,\theta}, \mu_t^\theta \right) \right] \xrightarrow[\lambda \downarrow 0]{K \rightarrow \infty} 0.$$

The bound is the triangle inequality applied through the decoder, followed by Assumption 4; see Appendix A.3.1.

The estimate (3.4) is the continuous-state analogue of the simplex perturbation estimate obtained in the finite-state setting. There is, however, one additional term: $\delta_{K,t}^\theta$ measures the structural error caused by replacing an infinite-dimensional population law with a finite-dimensional Gaussian mixture. The subsequent convergence and policy-gradient analysis must therefore keep the limits in K and λ separate.

Finally, the use of the Euclidean coordinates $Z_{K,t}^{\lambda,\theta}$ has an additional advantage. We do not need to construct a dominating measure or a density directly on the infinite-dimensional space $\mathcal{P}(\mathbb{R}^d)$. The likelihood-ratio correction can instead be derived from the ordinary finite-dimensional density of $Z_{K,t}^{\lambda,\theta} = z_{K,t}^\theta + \lambda U_t$.

3.4.2 Density and score of the randomized coordinates

Because the randomization acts on the Euclidean coordinates rather than on $\mathcal{P}(\mathbb{R}^d)$, the likelihood-ratio correction is an ordinary finite-dimensional computation. Since $Z_{K,t}^{\lambda,\theta} = z_{K,t}^\theta + \lambda U_t$ with U_t of density ρ , a change of variables gives the density $\varrho_{K,t}^{\lambda,\theta}(z) = \lambda^{-q_K} \rho((z - z_{K,t}^\theta)/\lambda)$, whose dependence on θ is carried entirely by the translation $z_{K,t}^\theta$. Differentiating it yields the score

$$\nabla_\theta \log \varrho_{K,t}^{\lambda,\theta}(Z_{K,t}^{\lambda,\theta}) = -\frac{1}{\lambda} \left(D_\theta z_{K,t}^\theta \right)^\top \nabla \log \rho(U_t), \quad (3.5)$$

which for $U_t \sim \mathcal{N}(0, \Xi)$ becomes $\lambda^{-1} (D_\theta z_{K,t}^\theta)^\top \Xi^{-1} U_t$, and $\lambda^{-1} (D_\theta z_{K,t}^\theta)^\top U_t$ in the isotropic case. This is the direct continuous-state analogue of the perturbation-score term of the finite-state construction, and it depends on the unknown mean-field dynamics only through the population-flow sensitivity $D_\theta z_{K,t}^\theta = D_\theta \mathcal{R}_K(\mu_t^\theta)$, the counterpart of the gradient of the logits. The singular factor λ^{-1} is again the source of the bias-variance trade-off. In practice the coordinates of the mixture representation have different scales, so a block-diagonal Ξ over weight, mean and covariance coordinates is preferable, and the score formula stays explicit.

The construction never requires the random measure $M_{K,t}^{\lambda,\theta} = \Gamma_K(Z_{K,t}^{\lambda,\theta})$ to admit a density on $\mathcal{P}(\mathbb{R}^d)$ or on $\mathcal{G}_K$, which would be awkward since Γ_K is not injective under permutations; the Euclidean variable is retained as part of the augmented trajectory and every likelihood-ratio computation runs against Lebesgue measure on $\mathbb{R}^{q_K}$. Augmenting the trajectory by $(Z_{K,t}^{\lambda,\theta})_{t \in \mathcal{T}}$ makes the structural point explicit: the trajectory density depends on θ only through the policy and the perturbation density, so the transition kernel is never differentiated. Appendix A.3.2 gives the two propositions, their proofs and the augmented formulation.

3.4.3 Convergence of the perturbed objective

We now show that the Gaussian-mixture perturbed control problem approximates the original mean-field control problem. There are two distinct sources of approximation. First, for a fixed number K of Gaussian components, the random perturbation $M_{K,t}^{\lambda,\theta} = \Gamma_K(z_{K,t}^\theta + \lambda U_t)$ approaches the deterministic Gaussian-mixture representation $\bar{\mu}_{K,t}^\theta$ as $\lambda \downarrow 0$. Second, even when $\lambda = 0$, the represented law $\bar{\mu}_{K,t}^\theta$ need not coincide with the true population law μ_t^θ .

Accordingly, we decompose

$$J_K^\lambda(\theta) - J(\theta) = \left(J_K^\lambda(\theta) - J_K^0(\theta) \right) + \left(J_K^0(\theta) - J(\theta) \right), \quad (3.6)$$

where J_K^0 denotes the control problem obtained by replacing the population argument by the deterministic Gaussian-mixture representation $\bar{\mu}_{K,t}^\theta$.

The first term in (3.6) is the perturbation error, whereas the second is the finite-dimensional representation error.

Averaged coefficients and the two error terms. Both terms of (3.6) are controlled by the same stability argument, carried out on the averaged controlled kernel $\mathcal{K}_t^\theta(dx' | x, m) = \int_{\mathcal{A}} P_t(dx' | x, a, m) \pi_t^\theta(da | x, m)$ and the averaged running reward $\mathcal{R}_t^\theta(x, m) = \int_{\mathcal{A}} r_t(x, a, m) \pi_t^\theta(da | x, m)$, which are the coefficients that drive the population recursion. Writing $\nu_{K,t}^{\lambda,\theta} = \mathcal{L}(X_t^{K,\lambda,\theta})$ for the law of the perturbed state process, and $\nu_{K,t}^{0,\theta}$ for

its counterpart at $\lambda = 0$, the independence of U_t from the state and action noises up to time t gives the recursion $\nu_{K,t+1}^{\lambda,\theta} = \mathbb{E}[\mathcal{K}_t^\theta(\cdot \mid X_t^{K,\lambda,\theta}, M_{K,t}^{\lambda,\theta})]$. Note that $\nu_{K,t}^{0,\theta} \neq \bar{\mu}_{K,t}^\theta$ in general: the second is an approximation of the population law, the first is the law of the state process obtained by using that approximation as an exogenous population argument.

Assumption 5 (Wasserstein stability of the controlled dynamics). *There exist constants $L_{\mathcal{K},x}, L_{\mathcal{K},\mu}, L_{\mathcal{R},x}, L_{\mathcal{R},\mu}, L_{g,x}, L_{g,\mu} \geq 0$ such that, uniformly in $\theta \in \Theta$ and $t = 0, \dots, T-1$,*

$$W_1(\mathcal{K}_t^\theta(\cdot \mid x, m), \mathcal{K}_t^\theta(\cdot \mid x', m')) \leq L_{\mathcal{K},x}|x - x'| + L_{\mathcal{K},\mu}W_1(m, m'), \quad (3.7)$$

$$|\mathcal{R}_t^\theta(x, m) - \mathcal{R}_t^\theta(x', m')| \leq L_{\mathcal{R},x}|x - x'| + L_{\mathcal{R},\mu}W_1(m, m'), \quad (3.8)$$

$$|g(x, m) - g(x', m')| \leq L_{g,x}|x - x'| + L_{g,\mu}W_1(m, m').$$

We further assume the moment conditions required for all the Wasserstein distances and expectations below to be finite.

It is stated directly in terms of the averaged coefficients because these are the quantities entering the population recursion; it follows from Wasserstein regularity of $(x, m) \mapsto P_t(\cdot \mid x, a, m)$ uniformly in a together with regularity of the randomized policy, and working with $\mathcal{K}_t^\theta$ and $\mathcal{R}_t^\theta$ keeps the argument independent of the action-space parametrization.

Under this assumption, a single Wasserstein stability estimate for the state marginal, applied twice, bounds the two terms of (3.6) separately: the perturbation error by $C_T C_{\Gamma,K} \lambda$ and the representation error by $C_T \delta_K^\theta$, with

$$C_T := \sum_{t=0}^{T-1} [L_{\mathcal{R},x} L_{\mathcal{K},\mu} H_t(L_{\mathcal{K},x}) + L_{\mathcal{R},\mu}] + L_{g,x} L_{\mathcal{K},\mu} H_T(L_{\mathcal{K},x}) + L_{g,\mu}, \quad (3.9)$$

where $H_t(L) = \sum_{j=0}^{t-1} L^j$ for $t \geq 1$ and $H_0(L) = 0$. Appendix A.3.4 states the stability lemma and the two intermediate propositions, and proves them. Adding the two estimates gives the main result of this subsection.

Combining the two preceding propositions gives the main result of this subsection.

Theorem 7 (Convergence of the Gaussian-mixture perturbed objective). *Under Assumptions 4 and 5, for every $\theta \in \Theta$,*

$$|J_K^\lambda(\theta) - J(\theta)| \leq C_T (C_{\Gamma,K} \lambda + \delta_K^\theta).$$

In particular, for fixed K ,

$$\limsup_{\lambda \downarrow 0} |J_K^\lambda(\theta) - J(\theta)| \leq C_T \delta_K^\theta.$$

Furthermore, if (K_n, λ_n) is a sequence satisfying $\delta_{K_n}^\theta \rightarrow 0$ and $C_{\Gamma,K_n} \lambda_n \rightarrow 0$, then

$$J_{K_n}^{\lambda_n}(\theta) \rightarrow J(\theta).$$

The order of the two limits matters. At fixed K , sending $\lambda \downarrow 0$ only removes the artificial randomization and recovers J_K^0 , not J ; recovering the original problem requires consistency of the representation, and the correct joint requirement is $C_{\Gamma,K} \lambda_K \rightarrow 0$ rather than $\lambda_K \rightarrow 0$, unless the decoder constants $C_{\Gamma,K}$ are bounded uniformly in K .

If the population flow belongs exactly to $\mathcal{G}_K$, so that $\mu_t^\theta = \bar{\mu}_{K,t}^\theta$ for all t , then $\delta_K^\theta = 0$ and the bound reduces to $|J_K^\lambda(\theta) - J(\theta)| \leq C_T C_{\Gamma,K} \lambda$. This includes the Gaussian linear-quadratic setting with $K = 1$, which is the case studied in Chapter 5.

For policy optimization, pointwise convergence in θ is not sufficient by itself. Uniform convergence over the policy class allows us to transfer optimality properties from the perturbed problems to the original MFC problem.

Define $\Delta_K := \sup_{\theta \in \Theta} \delta_K^\theta$.

Corollary 4 (Uniform convergence). *Suppose that the constants in Assumptions 4 and 5 are uniform in θ . Then*

$$\sup_{\theta \in \Theta} |J_K^\lambda(\theta) - J(\theta)| \leq C_T (C_{\Gamma, K} \lambda + \Delta_K).$$

Consequently, if $\Delta_{K_n} \rightarrow 0$ and $C_{\Gamma, K_n} \lambda_n \rightarrow 0$, then

$$\sup_{\theta \in \Theta} |J_{K_n}^{\lambda_n}(\theta) - J(\theta)| \rightarrow 0.$$

As a direct consequence, the optimal values converge.

Corollary 5 (Convergence of optimal values and policies). *Assume the hypotheses of Corollary 4 and suppose that Θ is compact and that $J : \Theta \rightarrow \mathbb{R}$ is continuous. Let $V := \sup_{\theta \in \Theta} J(\theta)$, and $V_K^\lambda := \sup_{\theta \in \Theta} J_K^\lambda(\theta)$. Then*

$$|V_K^\lambda - V| \leq C_T (C_{\Gamma, K} \lambda + \Delta_K).$$

Hence, for any sequence (K_n, λ_n) satisfying $\Delta_{K_n} \rightarrow 0$ and $C_{\Gamma, K_n} \lambda_n \rightarrow 0$, we have $V_{K_n}^{\lambda_n} \rightarrow V$.

Moreover, let $\theta_n^* \in \operatorname{argmax}_{\theta \in \Theta} J_{K_n}^{\lambda_n}(\theta)$. Every accumulation point of $(\theta_n^*)_{n \geq 1}$ belongs to $\operatorname{argmax}_{\theta \in \Theta} J(\theta)$. In particular, if J admits a unique maximizer θ^* , then $\theta_n^* \rightarrow \theta^*$.

The value estimate follows from $|\sup_{\theta} f - \sup_{\theta} g| \leq \sup_{\theta} |f - g|$, and the statement on maximizers is the standard argmax -continuity argument under uniform convergence; see Appendix A.3.5.

Theorem 7 establishes convergence at the objective level. For policy-gradient learning, however, we require the stronger property $\nabla_{\theta} J_K^\lambda(\theta) \rightarrow \nabla_{\theta} J(\theta)$. Objective convergence alone does not imply convergence of gradients.

From mixtures to moment charts. One point matters before the experiments. The construction above is stated for the Gaussian-mixture chart Γ_K because that is the case in which a general population law on $\mathbb{R}^d$ admits a finite-dimensional representation with an explicit decoder, and it is the case in which the representation error δ_K^θ is genuinely present. Nothing in the argument uses the mixture structure itself: what it uses is a finite-dimensional chart $z \mapsto \Gamma(z)$, a randomization $Z^{\lambda, \theta} = z^\theta + \lambda U$ within it, and a sensitivity $D_\theta z^\theta$. Any such chart gives the same density, the same score (3.5) and the same objective-convergence estimate. The continuous-state benchmarks of Chapter 5 are instances of this kind with an exact low-dimensional chart rather than a fitted mixture. On both, the perturbed population argument is one-dimensional, given by the mean, because that is the only functional of the law through which the policy, the transition and the reward depend on the population; the chart is exact and $\delta_K^\theta = 0$ by construction. The remaining moments do not disappear: the variance is propagated alongside the mean by the closed recursion of each benchmark, but only to evaluate the exact objective and the closed-form optimum at a frozen policy, never as part of the perturbed argument or of a gradient step. They should therefore be read as exact-chart instances of the continuous construction, testing the perturbation and the sensitivity recursion in isolation, and not as a test of Gaussian-mixture fitting, which is not exercised anywhere in this report.

We have so far introduced the Gaussian-mixture representation and perturbation of the population law, derived the density and score of the randomized GMM parameters, and proved the convergence of the perturbed objective.

The continuous-state theory is therefore still incomplete at this stage. The remaining work includes proving the gradient-level convergence, obtaining a policy-gradient formula for the perturbed problem, estimating the population-flow sensitivity, providing a Monte Carlo estimator of the perturbed policy gradient, and finally obtaining a bias and mean-square error decomposition. Additional benchmarks, algorithmic variants and numerical methods are natural directions for the remaining part of the internship.

Chapter 4

Algorithms

The previous chapter derived policy-gradient identities for perturbed mean-field control problems. This chapter describes how these identities are turned into practical algorithms. The emphasis is algorithmic rather than benchmark-specific: choices such as horizons, perturbation grids, validation initial laws, and simulator budgets are deferred to Chapter 5.

All methods follow the same outer loop. At iteration m , a Monte Carlo estimator $\hat{G}_m(\theta_m)$ of the relevant policy gradient is computed, the policy parameter is updated with Adam [12], and diagnostic quantities such as the objective estimate, gradient norm, and validation objective are recorded. Most benchmarks are written in reward-maximization form; the linear-quadratic benchmark is implemented as a cost minimization and reported with the corresponding sign convention in Chapter 5. When a loss-minimization interface is used for a reward objective, the optimizer is applied to the negative estimated gradient.

4.1 Baseline policy-gradient estimators

REINFORCE. The first baseline is the standard likelihood-ratio estimator. Given B trajectories generated by the current policy, let $\mathcal{R}_\theta^{(b)}$ be the return of trajectory b and let

$$\mathcal{S}_{\text{pol}}^{\theta,(b)} = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_t^\theta \left(\alpha_t^{\theta,(b)} \mid X_t^{\theta,(b)}, \mu_t \right)$$

be its policy score. With a deterministic or batch-mean baseline b_0 , REINFORCE uses

$$\hat{G}_{\text{RF}}(\theta) = \frac{1}{B} \sum_{b=1}^B \left(\mathcal{R}_\theta^{(b)} - b_0 \right) \mathcal{S}_{\text{pol}}^{\theta,(b)}.$$

This estimator differentiates the sampled policy but treats the supplied population argument as fixed. In a mean-field control problem it therefore misses the derivative of the population flow with respect to θ .

MF-REINFORCE. The second baseline is the finite-state MF-REINFORCE estimator of [19]. It perturbs the population law through a logit transformation and uses likelihood-ratio identities to estimate the missing population-flow derivative. This provides a model-free reference estimator for discrete mean-field control, but its auxiliary sampling cost grows quadratically with the time horizon because separate perturbations are used across intermediate time indices.

4.2 Transport policy-gradient

The transport estimator is the practical version of the estimator constructed in Chapter 3. It uses two perturbation scales. The main scale λ defines the perturbed objective and appears in the final policy-gradient estimate. The auxiliary scale η is used to estimate the population-flow sensitivity $D_t^\theta = \nabla_\theta \mu_t^\theta$. When the exact sensitivity is available, the resulting oracle estimator is unbiased for $\nabla_\theta J^\lambda(\theta)$ by Proposition 3. In the model-free implementation, D_t^θ is replaced by the recursive estimator $\hat{D}_t^{\eta,n,\theta}$.

Algorithm 1 Discrete fixed-scale transport policy gradient

Require: policy parameter θ , scales λ, η , main batch size B , auxiliary batch size n , horizon T

- 1: Compute or estimate the population flow $(\mu_t^\theta)_{t=0}^T$
- 2: Initialize $\hat{D}_0^{\eta, n, \theta} = 0$
- 3: Simulate n auxiliary trajectories with perturbed laws $(1 - \eta)\mu_s^\theta + \eta q_s$
- 4: **for** $t = 1, \dots, T$ **do**
- 5: Estimate $\hat{D}_t^{\eta, n, \theta}$ from the shared auxiliary trajectories using the triangular likelihood-ratio recursion
- 6: **end for**
- 7: **for** $b = 1, \dots, B$ **do**
- 8: Simulate one perturbed trajectory with laws $M_t^{\lambda, \theta, (b)} = (1 - \lambda)\mu_t^\theta + \lambda q_t^{(b)}$
- 9: Compute the return $\mathcal{R}_\theta^{\lambda, (b)}$ and the policy score $\mathcal{S}_{\text{pol}}^{\lambda, \theta, (b)}$
- 10: Compute the perturbation score with $\hat{D}_t^{\eta, n, \theta}$:

$$\hat{\mathcal{S}}_{\text{pert}}^{\lambda, \eta, n, \theta, (b)} = -\frac{1 - \lambda}{\lambda} \sum_{t=0}^T \sum_{k=1}^{N-1} H_k(q_t^{(b)}) \hat{D}_t^{\eta, n, \theta}(k)$$

11: **end for**

12: Form

$$\hat{G}_{\lambda, \eta, B, n}(\theta) = \frac{1}{B} \sum_{b=1}^B \left(\mathcal{R}_\theta^{\lambda, (b)} - b_0 \right) \left(\mathcal{S}_{\text{pol}}^{\lambda, \theta, (b)} + \hat{\mathcal{S}}_{\text{pert}}^{\lambda, \eta, n, \theta, (b)} \right)$$

13: Update θ with Adam in the direction $\hat{G}_{\lambda, \eta, B, n}(\theta)$

The continuous-state transport estimator has the same structure. The difference is that the population law is represented through finite-dimensional features or moments, and the simplex perturbation is replaced by an environment-specific perturbation map. If Y_t^θ denotes the chosen law feature, the perturbed feature has the generic affine form $Y_t^{\lambda, \theta} = \Phi_\lambda(Y_t^\theta, \xi_t)$, where ξ_t is exogenous noise. The environment supplies both Φ_λ and the corresponding transport score. The auxiliary stage estimates $\nabla_\theta Y_t^\theta$, and the main stage combines the policy score with the transport score exactly as in Algorithm 1.

The statement is abstract because the feature Y_t^θ is whatever finite-dimensional chart the environment exposes. In the two continuous-state benchmarks of Chapter 5 it is the population mean, the only functional of the law that the coefficients read: $\Phi_\lambda(y, \xi) = (1 + \lambda\xi)y + \lambda\beta$ with ζ, β i.i.d. Gaussian, the score is the corresponding Gaussian score of Corollary 6, and $\nabla_\theta Y_t^\theta$ is estimated by the auxiliary recursion applied to the closed moment recursion of the benchmark. The exact moment recursion is used to propagate the population during training in the exact-flow runs and is replaced by a particle estimate in the particle-flow runs; in neither case is it differentiated, which is what keeps the estimator model-free in the sense of Section 3.1.

4.3 Adaptive transport

Fixed-scale transport requires choosing λ and η before training. The adaptive version keeps the same gradient estimator but periodically updates these scales from local bias-variance diagnostics. The idea is to compare gradient estimates at the current scales and at contracted scales. If reducing a scale changes the estimated gradient more than expected from Monte Carlo noise, the current scale is interpreted as too biased and is decreased; otherwise it can remain large enough to avoid excessive score variance.

Let $c_\lambda, c_\eta \in (0, 1)$ be contraction factors. At an adaptive checkpoint, four gradient estimates are computed using the scale pairs (λ, η) , $(c_\lambda \lambda, \eta)$, $(\lambda, c_\eta \eta)$, $(c_\lambda \lambda, c_\eta \eta)$. For each replication $r = 1, \dots, R$, denote the corresponding gradient estimates by $g_{++}^{(r)}$, $g_{-+}^{(r)}$, $g_{+-}^{(r)}$, $g_{--}^{(r)}$, where the first sign refers to the main scale λ and the second to the auxiliary scale η ; the sign $+$ means current scale and the sign $-$ means contracted scale. The effect of contracting only λ is estimated by averaging over the two auxiliary

scales,

$$\Delta_\lambda^{(r)} = \frac{1}{2} \left[\left(g_{++}^{(r)} - g_{-+}^{(r)} \right) + \left(g_{+-}^{(r)} - g_{--}^{(r)} \right) \right],$$

and similarly the effect of contracting only η is

$$\Delta_\eta^{(r)} = \frac{1}{2} \left[\left(g_{++}^{(r)} - g_{+-}^{(r)} \right) + \left(g_{-+}^{(r)} - g_{--}^{(r)} \right) \right].$$

For a collection of vectors $(U^{(r)})_{r=1}^R$, write

$$\bar{U} = \frac{1}{R} \sum_{r=1}^R U^{(r)}, \quad \widehat{\text{trCov}}(U) = \frac{1}{R-1} \sum_{r=1}^R \|U^{(r)} - \bar{U}\|^2,$$

with $\widehat{\text{trCov}}(U) = 0$ if $R = 1$. The raw squared discrepancies are corrected for Monte Carlo noise by subtracting the variance of the empirical mean:

$$\hat{d}_\lambda = \left(\|\bar{\Delta}_\lambda\|^2 - \frac{1}{R} \widehat{\text{trCov}}(\Delta_\lambda) \right)_+, \quad \hat{d}_\eta = \left(\|\bar{\Delta}_\eta\|^2 - \frac{1}{R} \widehat{\text{trCov}}(\Delta_\eta) \right)_+.$$

If the leading perturbation errors scale like λ^{p_λ} and η^{p_η} , the bias proxies used by the controller are

$$\hat{b}_\lambda = \frac{\hat{d}_\lambda}{(1 - c_\lambda^{p_\lambda})^2}, \quad \hat{b}_\eta = \frac{\hat{d}_\eta}{(1 - c_\eta^{p_\eta})^2}.$$

The variance proxy is the empirical trace covariance of the current-scale estimator, $\hat{v} = \widehat{\text{trCov}}(g_{++})$. The controller then uses the log-ratio signals

$$z_\lambda = \log \left(\frac{\hat{b}_\lambda + \delta}{\tau_\lambda \hat{v} + \delta} \right), \quad z_\eta = \log \left(\frac{\hat{b}_\eta + \delta}{\tau_\eta \hat{v} + \delta} \right),$$

where $\tau_\lambda, \tau_\eta > 0$ are target bias-to-variance ratios and $\delta > 0$ prevents numerical division by zero. Positive values indicate that the estimated perturbation bias is too large relative to variance.

Algorithm 2 Adaptive transport

Require: initial policy θ , initial scales λ, η , contraction factors c_λ, c_η , checkpoint interval

- 1: **for** training iterations $m = 0, 1, \dots$ **do**
 - 2: Compute a fixed-scale transport gradient estimate using the current λ, η
 - 3: Update θ with Adam
 - 4: **if** m is an adaptive checkpoint **then**
 - 5: Estimate g_{++} , g_{-+} , g_{+-} , and g_{--} over independent replications
 - 6: Form $\Delta_\lambda, \Delta_\eta$, the corrected discrepancies $\hat{d}_\lambda, \hat{d}_\eta$, and the variance proxy $\hat{v}$
 - 7: Set $\hat{b}_\lambda = \hat{d}_\lambda / (1 - c_\lambda^{p_\lambda})^2$ and $\hat{b}_\eta = \hat{d}_\eta / (1 - c_\eta^{p_\eta})^2$
 - 8: Compute z_λ and z_η from the bias-to-variance ratios
 - 9: Update transformed scale parameters with an Adam-style controller
 - 10: Clamp λ and η to their admissible intervals
 - 11: **end if**
 - 12: **end for**
-

The adaptive controller is deliberately separated from the policy optimizer. The policy update uses the current transport estimator, while the controller only changes future perturbation scales. Directional checks can also be included: if the current-scale and contracted-scale gradients point in incompatible directions, the corresponding scale is forced downward even when the scalar bias proxy is small.

4.4 Computational cost

The relevant cost unit is one simulated transition. For a horizon T and a main batch of size B , REINFORCE has per-update cost $C_{\text{RF}} = BT$. For finite-state MF-REINFORCE with n auxiliary trajectories and shared state-gradient reuse, the auxiliary procedure requires two perturbed forward simulations for each pair of time indices, yielding $C_{\text{MF}} = BT + nT(T + 1)$. Algorithm 1 of [19] instead repeats that auxiliary pass inside the loop over the main batch, at a cost of order nBT^2 ; Section 5.1 explains why the two differ and which is used to match budgets. The recursive transport estimator uses one auxiliary forward pass and one main forward pass, hence $C_{\text{tr}} = T(B + n)$. Adaptive transport adds the amortized cost of its checkpoints. If a checkpoint is performed every K iterations and uses R independent replications of the contracted-scale diagnostic, its average per-update cost is written abstractly as $C_{\text{ad}} = C_{\text{tr}} + \frac{1}{K}C_{\text{diag}}(R, T)$.

These expressions are used only to account for simulator effort. They do not include common optimizer overhead, tensor operations, logging, or deterministic validation. Their main implication is structural: MF-REINFORCE has quadratic horizon dependence in its auxiliary estimator, while transport keeps linear horizon dependence through the recursive sensitivity construction.

Chapter 5

Numerical Experiments

This chapter describes the numerical benchmark suite used to evaluate the proposed transport policy-gradient estimators. The goal is to compare REINFORCE, MF-REINFORCE, fixed-scale transport and adaptive transport under matched simulator budgets, while separating optimization performance from perturbation bias, estimator variance, and population-flow approximation error.

5.1 Experimentation protocol

The study has three goals: to measure whether the transport estimator improves the optimization performance of model-free policy gradients in mean-field control; to separate the effect of the perturbation scale from that of the estimator geometry, by varying the transport radii and comparing exact with particle population flows; and to test whether the adaptive controller of Algorithm 2 can select useful scales without a per-environment grid search.

A configuration is defined by the benchmark, the horizon, the algorithm, the population-flow mode, the perturbation parameters and the simulator budget, and each is repeated over five independent seeds. In a paired comparison the same seed fixes the policy initialization, the sequence of training initial laws, the Monte Carlo samples and the validation schedule, and reported curves show the mean and dispersion across seeds.

Algorithms compared. On the finite-state benchmarks we compare four methods: standard REINFORCE, which ignores the derivative of the population flow and serves as a lower-complexity baseline; the logit-perturbed MF-REINFORCE estimator of [19]; the discrete-transport estimator of Algorithm 1; and the adaptive method of Algorithm 2, initialized from the same simplex estimator but allowed to update its two perturbation scales during training. On the continuous-state benchmarks, REINFORCE is compared with the continuous transport estimator and its adaptive version.

The logit perturbation level of MF-REINFORCE is not retuned here. On the two-state, cybersecurity and distribution benchmarks we use the best-performing value reported in the reference study, $\varepsilon = 0.2, 1.0$ and 2.0 respectively; advertising is not part of that grid and we use $\varepsilon = 1.0$, a value that performs reasonably on the other finite-state benchmarks. This keeps the comparison focused on the proposed estimator rather than on a new hyperparameter search for the baseline.

Transport grids. The two-state benchmark serves as the calibration experiment, over the broad grid $\lambda \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$ against $\eta \in \{0.4, 0.6, 0.85, 0.95\}$. The λ -range is deliberately wide enough to expose both small-perturbation failures, where the score correction becomes unstable, and large-perturbation failures, where the optimized objective is too biased; the η -range tests the observation that the auxiliary estimator benefits from a comparatively large radius. After this calibration the auxiliary scale is fixed at $\eta = 0.85$ everywhere else and the main grid is reduced, as listed in each subsection.

Adaptive transport is initialized at $\lambda_0 = 0.2$, $\eta_0 = 0.8$, with the diagnostics of Section 4.3 computed every 100 training iterations from 4 replications. At four replications the debiased bias estimate is also frequently non-positive, which means the bias is unresolved rather than zero; the controller therefore updates a scale only on a resolved estimate, or when the direction test detects a sign change.

Horizons and flow modes. The two-state benchmark is run at $T = 2$, reproducing the reference toy problem, and at $T = 5$, which tests whether the estimators degrade when sensitivity information is propagated through more steps; both MF-REINFORCE and transport are run there with exact population

propagation and with a particle approximation, isolating the effect of replacing the deterministic recursion by an empirical law. The remaining benchmarks use exact propagation in the main comparison, since their finite-dimensional recursions are available exactly, and the linear–quadratic benchmark is additionally run with a particle flow.

Equal-budget comparison. All headline comparisons are made under an equal simulator-budget constraint, counted in simulated transitions per policy update and implemented by adjusting the number of main and auxiliary trajectories before training. The reference is the finite-state MF-REINFORCE configuration. Estimating the logit gradient at time t requires simulating a mean-field and a perturbed trajectory over t steps, for each of n auxiliary samples, so one pass over the whole horizon costs $2n \sum_{t=1}^T t = nT(T+1)$ transitions. Algorithm 1 of [19] performs that pass inside the loop over the B main trajectories, giving a per-update cost of $BT + BnT(T+1)$, of order nBT^2 , which is the complexity reported there. The logit gradient is however a property of the policy and of the population flow, not of an individual trajectory, so our implementation estimates it once per update and shares it across the main batch. The reference cost is therefore

$$C_{\text{MF}} = BT + nT(T+1), \quad (5.1)$$

the smaller of the two, and the conservative choice here since it is the budget every other estimator must match. The transport estimator has linear horizon dependence,

$$C_{\text{tr}} = T(B_{\text{tr}} + n_{\text{tr}}), \quad (5.2)$$

up to the common cost of particle-flow propagation when enabled. REINFORCE is given the same budget by increasing its number of trajectories, and adaptive transport the same nominal budget after its amortized checkpoint overhead. The realized budgets agree exactly across estimators on every benchmark; Appendix C.1 reports them together with the wall-clock cost, which the rule does not equalize.

Training and validation. Unless a subsection states otherwise, policies are optimized with Adam, a deterministic validation is performed every ten training iterations, and the validation policy is evaluated without exploration noise whenever the environment admits an exact population recursion. The training initial law is randomized, to test whether a policy generalizes across the relevant part of the simplex, while validation uses fixed initial laws or fixed grids; the exact objective, dynamic-programming reference or closed-form optimizer is used only after freezing the parameters. Each benchmark produces the same family of outputs: the mean validation reward against training iteration, the final objective and runtime, the policy error where a reference policy exists, the learned state or moment flow, and benchmark-specific diagnostics.

Estimator diagnostics and reproducibility. Where a benchmark provides an exact or automatic-differentiation gradient oracle, evaluation-only diagnostics are computed at saved policies: independent gradient replications at a fixed budget estimate bias, dispersion, mean-square error, cosine alignment with the oracle, and the size of the transport correction. These never enter a training update and are reported in Section 5.5. Every run writes its metadata, training history, final policy, runtime summary and budget estimate, and every figure and table below is regenerated from those artifacts, so the results are reproducible from the recorded seed and configuration. The suite comprises 640 runs and roughly 94 hours of single-core compute.

Hardware. All experiments were run on a single machine equipped with an NVIDIA RTX A4000 GPU. The policies are small, two hidden layers of width 32 on the finite-state benchmarks and a directly parametrized affine policy on the continuous-state ones, so a training step is dominated by trajectory simulation rather than by parameter updates; the runs were accordingly executed on the CPU device of that machine, with several configurations trained concurrently. All wall-clock timings were collected under these conditions and are therefore comparable across estimators.

Table 5.1 collects the final validation objective of every estimator on every benchmark, at the headline horizon and population-flow mode of the corresponding subsection. It is the reference point for the benchmark sections that follow, each of which discusses one problem in isolation; Section 6.1 reads the table as a whole.

Benchmark	Optimum	REINFORCE	MF-REINFORCE	Transport	(λ, η)	Adaptive
Two-state ($T = 5$)	-2.6400	-31.5104 ± 0.0001	-4.2053 ± 0.3324	-3.1332 ± 0.0423	(0.1, 0.95)	-3.3960 ± 0.1141
Cybersecurity ($T = 3$)	—	-0.1517 ± 0.0000	-0.1524 ± 0.0013	-0.1544 ± 0.0030	(0.4, 0.85)	-0.1602 ± 0.0032
Distribution ($T = 5$)	—	-0.2616 ± 0.0000	-0.1017 ± 0.0161	-0.2421 ± 0.0459	(0.4, 0.85)	-0.5596 ± 0.1225
Advertising ($T = 5$)	—	0.9688 ± 0.0000	0.9946 ± 0.0091	0.9795 ± 0.0089	(0.1, 0.85)	0.9727 ± 0.0078
Linear-quadratic ($T = 20$)	-7.2238	-7.5157 ± 0.0010	—	-7.2952 ± 0.0023	(0.1, 0.85)	-7.3021 ± 0.0030
Portfolio ($T = 10$)	-13.133	-20.535 ± 0.004	—	-17.980 ± 1.079	(0.4, 0.85)	-17.193 ± 0.594

Table 5.1: Final validation objective on every benchmark, as mean and standard deviation over five seeds, at the headline horizon and population-flow mode of each subsection. Higher is better throughout. The transport column reports the best entry of the fixed-scale grid and the scales (λ, η) that attain it; MF-REINFORCE is not defined on the continuous-state benchmarks. The optimum column gives the exact recursion evaluated at the closed-form optimal policy where one is available.

5.2 Discrete-state space benchmarks

5.2.1 Two-State Two-Action

We first consider the two-state two-action mean-field control problem from [19, 10]. Besides providing a simple sanity check, this benchmark admits an explicit optimal stationary policy and an exact recursion for the population law, enabling direct evaluation of the proposed gradient estimator against an oracle.

Model. The state and action spaces are $\mathcal{X} = \{0, 1\}$ and $\mathcal{A} = \{\text{ST}, \text{MV}\}$ respectively. ST leaves the current state unchanged, whereas MV attempts to move the agent to the other state. For $x \in \mathcal{X}$ and $a \in \mathcal{A}$, the transition kernel is given by

$$P(x' | x, a) = \begin{cases} \lambda_x \mathbf{1}_{\{a=\text{MV}\}}, & x' \neq x, \\ 1 - \lambda_x \mathbf{1}_{\{a=\text{MV}\}}, & x' = x, \end{cases}$$

where $\lambda_0, \lambda_1 \in (0, 1)$. The transition mechanism itself does not depend on the population distribution; the mean-field interaction enters through the reward. Writing $\bar{\mu} = p\delta_0 + (1-p)\delta_1$ the target population distribution, the running and terminal rewards are $r(x, a, \mu) = r(x, \mu) = \mathbf{1}_{\{x=1\}} - \mu(1)^2 - \kappa W_1(\mu, \bar{\mu})$, $g(x, \mu) = r(x, \mu)$, where the Wasserstein term reduces to $W_1(\mu, \bar{\mu}) = |\mu(1) - (1-p)|$ on the binary state space endowed with the distance $d(0, 1) = 1$.

The optimal stationary policy sends any initial population distribution to $\bar{\mu}$ after one time step and preserves it thereafter. It is available in closed form:

$$\begin{aligned} \pi^*(a | 0) &= \frac{\lambda_0 - 1 + p}{\lambda_0} \mathbf{1}_{\{a=\text{ST}\}} + \frac{1-p}{\lambda_0} \mathbf{1}_{\{a=\text{MV}\}}, \\ \pi^*(a | 1) &= \frac{\lambda_1 - p}{\lambda_1} \mathbf{1}_{\{a=\text{ST}\}} + \frac{p}{\lambda_1} \mathbf{1}_{\{a=\text{MV}\}}. \end{aligned}$$

Setup. As in the reference experiment the policy class is restricted to two Bernoulli logits $\theta = (\theta_0, \theta_1)$ and does not read the population argument, so the flow reduces to the scalar recursion (B.1) and both the objective (B.2) and its maximizer θ^* are available in closed form, with $\pi^*(\text{ST} | 0) = 0.2$ and $\pi^*(\text{ST} | 1) = 0.25$. We run $T = 2$ and $T = 5$, each with exact and with particle population flow, MF-REINFORCE at $\varepsilon = 0.2$, the broad calibration grid $\lambda \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$ against $\eta \in \{0.4, 0.6, 0.85, 0.95\}$, and adaptive transport from $(\lambda_0, \eta_0) = (0.2, 0.8)$, all at the equal budget derived from the MF-REINFORCE reference configuration $B = 200$, $n = 10$. Every ten iterations the policy is frozen and evaluated from the fixed law $\mu_0^{\text{val}} = (0.2, 0.8)$, without trajectory noise. Policy recovery is measured by the two state-wise errors

$$e_0(\theta) = \left| \pi^\theta(\text{ST} | 0) - 0.2 \right|, \quad e_1(\theta) = \left| \pi^\theta(\text{ST} | 1) - 0.25 \right|.$$

Appendix B.1 gives the model, the parameter values, the closed-form optimal policy and the full protocol.

Results. The exact recursion at the closed-form optimum gives $J(\theta^*) = -3.360$ at $T = 2$ and -2.640 at $T = 5$, against -5.000 and -4.532 at the initialization $\theta = (0, 0)$. Figure 5.1 shows that REINFORCE moves away from the optimum from the first iterations and ends at -14.118 and -31.510 , far below its own starting point. Estimators that differentiate the population flow improve instead, provided the perturbation is not too coarse: at $T = 2$ every transport configuration with $\lambda \leq 0.4$ ends above the initialization and only the four runs at $\lambda = 0.8$ fall below it, and at $T = 5$ the threshold moves to $\lambda \leq 0.2$. This is the sharpest illustration in the suite of the obstruction of Chapter 3: the population channel carries the sign of the useful direction, and an estimator that ignores it ascends the wrong function.

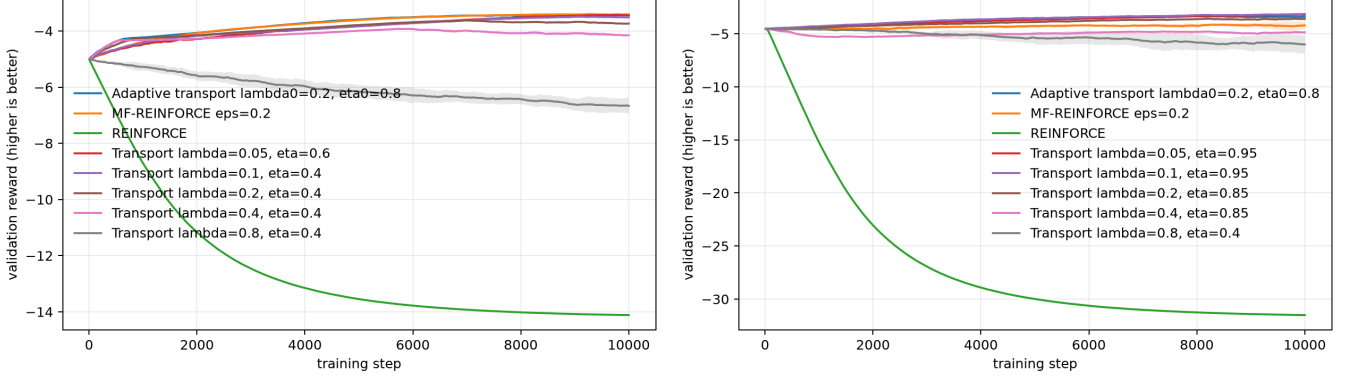


Figure 5.1: Two-state benchmark with exact population flow, validation reward against training iteration at $T = 2$ (left) and $T = 5$ (right), as mean over five seeds with one standard deviation shaded. For readability only the best η of each λ is drawn. REINFORCE descends away from its initialization at both horizons; the perturbed estimators ascend, except at the largest perturbation scales, where the bias of the perturbed objective dominates.

The horizon is what separates transport from MF-REINFORCE. Table 5.2 reports the objective gap and the policy-recovery error at both horizons. At $T = 2$ MF-REINFORCE is the more accurate estimator, recovering π^* to within 0.006 against 0.014 for the best transport configuration; at $T = 5$ the ordering reverses by a wide margin, the logit policy error growing by a factor 35 against 2.5 for transport, with an objective gap three times smaller. This is what the per-update costs (5.1) and (5.2) predict: at a fixed budget the quadratic horizon dependence of the logit estimator is paid for by a reduction in the number of trajectories, and the longer horizon exposes the resulting variance. Transport recovers the optimal policy to within 3.5% at $T = 5$.

Estimator	Scales	$J(\theta^*) - J(\hat{\theta})$		$\frac{1}{2}(e_0 + e_1)$	
		$T = 2$	$T = 5$	$T = 2$	$T = 5$
REINFORCE	—	10.758 ± 0.000	28.870 ± 0.000	0.4694 ± 0.0000	0.4691 ± 0.0000
MF-REINFORCE	$\varepsilon = 0.2$	0.047 ± 0.022	1.565 ± 0.332	0.0062 ± 0.0019	0.2196 ± 0.0167
Transport	$(0.05, 0.6) / (0.1, 0.95)$	0.084 ± 0.015	0.493 ± 0.042	0.0137 ± 0.0040	0.0347 ± 0.0057
Adaptive transport	$(0.2, 0.8)$	0.069 ± 0.017	0.756 ± 0.114	0.0100 ± 0.0027	0.1160 ± 0.0161

Table 5.2: Two-state benchmark with exact population flow, at matched simulator budgets. Both columns are evaluated at the frozen learned parameter $\hat{\theta}$ and reported as mean and standard deviation over the five seeds. The objective column is the exact unperturbed objective of the deterministic recursion (B.1) started from the fixed validation law $\mu_0^{\text{val}} = (0.2, 0.8)$, evaluated at the closed-form optimum θ^* of (B.2) and at $\hat{\theta}$, and differenced: $J(\theta^*; \mu_0^{\text{val}}) = -3.360$ at $T = 2$ and -2.640 at $T = 5$. Neither term carries Monte Carlo error, so the dispersion is entirely across seeds of $\hat{\theta}$. The policy column is the average of the two state-wise recovery errors e_0 and e_1 defined above. For transport the best entry of the (λ, η) grid is reported at each horizon, and the two winning pairs (λ, η) are listed in the scales column as $T = 2 / T = 5$; that column carries ε for MF-REINFORCE and (λ_0, η_0) for the adaptive controller. MF-REINFORCE is the more accurate estimator at $T = 2$ and loses to transport at $T = 5$, on both measures.

This benchmark also carries the calibration experiment announced in Section 5.1, reported in full in Appendix C.4, which fixes the two design choices used on the remaining benchmarks. In the main scale the optimum is interior, at $\lambda = 0.1$: below it the score of the perturbation density degrades, above it the perturbed objective is too far from the true one, and at $\lambda = 0.8$ the run ends below its initialization. In the auxiliary scale the objective improves over the whole grid whenever $\lambda \leq 0.1$ and peaks at $\eta = 0.85$, so at this horizon the auxiliary estimator prefers a radius near the top of the range. This is the regime in which the $(n\eta^2)^{-1}$ variance term of Proposition 4 dominates its $O(\eta)$ bias, which is what the small auxiliary sample sizes of the equal-budget allocation should produce. The preference is nonetheless horizon-dependent, which limits how far the calibration transfers: at $T = 2$ the ordering reverses, $\eta = 0.4$ being the best column at every $\lambda \geq 0.1$. Since the remaining benchmarks all have $T \geq 3$, $\eta = 0.85$ is retained from the longer-horizon evidence.

Replacing the exact recursion by a particle approximation costs 0.32 at $T = 5$, from -3.133 to -3.455 at the best scales, about two thirds of the remaining optimality gap. The substitution is therefore not free on the simplex, in contrast with the linear-quadratic benchmark; the ordering of the estimators and the calibration in η are nonetheless unchanged, and particle-flow transport still improves on exact-flow MF-REINFORCE.

Learned population flows. The objective and the policy error both summarize the learned policy by a single number, and Figure C.5 shows what the four estimators actually do to the population. The optimal policy holds the population at the target proportions $(p, 1 - p) = (0.6, 0.4)$ from the second step onwards, and the perturbed estimators reproduce this: MF-REINFORCE, transport and adaptive transport all drive the state-0 mass from its initial value 0.2 to within a few percent of 0.6 and hold it there. REINFORCE does the opposite. Its learned policy drives the population to the corner rather than to the target, which is what the objective value -31.5 and the policy error 0.469 measure; the flow makes clear that this is not slow convergence but convergence to a qualitatively different flow.

5.2.2 Cybersecurity

We next consider the cybersecurity benchmark of [19], inspired by the mean-field game model of [13] and its mean-field control formulation in [3]. A central planner controls the protection status of a large population of computers. Compared with the two-state toy problem, this example has a larger state space, nonlinear dependence on the population law through infection rates, and a neural-network policy depending explicitly on the population distribution.

Model. The state space is $\mathcal{X} = \{\text{DI}, \text{DS}, \text{UI}, \text{US}\}$, where the first letter indicates whether the computer is defended or undefended and the second whether it is infected or susceptible. The action space is $\mathcal{A} = \{0, 1\}$: action 0 keeps the current protection status, while action 1 switches between defended and undefended. To avoid confusion with the transport scale, the switching intensity is denoted by λ_{sw} .

For a population law $\mu \in \Delta_4$, the infection intensities are

$$\begin{aligned}\iota_D(\mu) &= v_H q_{\text{inf}}^D + \beta_{DD}\mu(\text{DI}) + \beta_{UD}\mu(\text{UI}), \\ \iota_U(\mu) &= v_H q_{\text{inf}}^U + \beta_{UU}\mu(\text{UI}) + \beta_{DU}\mu(\text{DI}).\end{aligned}$$

In the ordered basis (DI, DS, UI, US), the infinitesimal generator associated with action a is

$$Q^{\mu,a} = \begin{pmatrix} -q_{\text{rec}}^D - \lambda_{\text{sw}}a & q_{\text{rec}}^D & \lambda_{\text{sw}}a & 0 \\ \iota_D(\mu) & -\iota_D(\mu) - \lambda_{\text{sw}}a & 0 & \lambda_{\text{sw}}a \\ \lambda_{\text{sw}}a & 0 & -q_{\text{rec}}^U - \lambda_{\text{sw}}a & q_{\text{rec}}^U \\ 0 & \lambda_{\text{sw}}a & \iota_U(\mu) & -\iota_U(\mu) - \lambda_{\text{sw}}a \end{pmatrix}.$$

The discrete-time transition matrix is $P_{\Delta t}^{\mu,a} = \exp(\Delta t Q^{\mu,a})$. The cost of a state is $f(x) = k_D \mathbf{1}_{\{x \in \{\text{DI}, \text{DS}\}\}} + k_I \mathbf{1}_{\{x \in \{\text{UI}, \text{US}\}\}}$, and the running and terminal rewards are $r_{\Delta t}(x, a, \mu) = -\Delta t f(x)$ and $g_{\Delta t}(x, \mu) = -\Delta t f(x)$. With discount factor γ , the objective is

$$J_T(\theta; \mu_0) = \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t r_{\Delta t}(X_t, A_t, \mu_t^\theta) + \gamma^T g_{\Delta t}(X_T, \mu_T^\theta) \right].$$

Setup. The policy is a two-hidden-layer network of width 32 reading (t, μ) . We use the parameters and the horizon $T = 3$ of [19], MF-REINFORCE at $\varepsilon = 1$, the reduced transport grid $\lambda \in \{0.1, 0.2, 0.4\}$ at $\eta = 0.85$, and adaptive transport from $(\lambda_0, \eta_0) = (0.2, 0.8)$, at the equal budget derived from $B = 200$, $n = 1$. Training randomizes the initial law over Dirichlet(1, 1, 1, 1) for 20,000 iterations; validation propagates the frozen policy deterministically from the uniform law over $T_{\text{val}} = 50$ steps. Appendix B.2 gives the full specification.

Results. Cybersecurity is the first benchmark on which the population-flow derivative buys nothing; Table 5.3 collects the comparison.

Estimator	Final objective	Wall clock (s)
REINFORCE	-0.1517 ± 0.0000	532
MF-REINFORCE $\varepsilon = 1$	-0.1524 ± 0.0013	1 199
Transport $\lambda = 0.1, \eta = 0.85$	-0.1579 ± 0.0038	1 101
Transport $\lambda = 0.2, \eta = 0.85$	-0.1552 ± 0.0038	1 105
Transport $\lambda = 0.4, \eta = 0.85$	-0.1544 ± 0.0030	1 122
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	-0.1602 ± 0.0032	1 026

Table 5.3: Cybersecurity benchmark at $T = 3$ with exact population flow: final validation objective as mean and standard deviation over five seeds, and mean wall-clock cost per run, at a matched simulator budget. Higher is better.

REINFORCE reaches -0.1517 ± 0.0000 within roughly 500 iterations and stays there, while MF-REINFORCE needs the full 20,000 iterations to approach the same value from below and the transport runs end between -0.1544 and -0.1579 . Figure C.6 shows the whole comparison confined to a band of width 0.01, about 6% of the objective, so the benchmark is best read as a negative result with a small effect size rather than as a failure of the transport estimator. The correction-alignment diagnostic of Section 5.6 explains it directly: on this benchmark the mean-field correction accounts for one part in a thousand of the gradient and is exactly parallel to the policy-score term, so there is nothing for a population-aware estimator to recover and the extra variance of its auxiliary estimate is pure cost.

Learned state distributions. The reason the objectives are so close is visible in the learned flows of Figure C.3. REINFORCE, MF-REINFORCE and fixed-scale transport converge to the same policy: they stop defending altogether, and the population settles on the undefended equilibrium with terminal law $(\text{DI}, \text{DS}, \text{UI}, \text{US}) \approx (0.000, 0.000, 0.438, 0.562)$ for REINFORCE and within 0.02 of it for the other two. On this configuration the defence cost is not worth paying, and three of the four estimators find that out.

The adaptive controller does not. It ends with 79% of the population defended and a terminal law $(0.331, 0.462, 0.136, 0.071)$, so it pays the defence cost throughout and still ends with a slightly larger infected fraction, 0.467 against 0.438. This is the same failure recorded in Section 5.4: on this benchmark the controller contracts the main scale to $\lambda = 0.026$, an order of magnitude below the smallest calibrated value, and the resulting variance moves the policy to a qualitatively different flow rather than slowing it down. It is the clearest picture in the suite of what a badly scaled perturbation costs.

5.2.3 Distribution Planning

The distribution-planning benchmark is based on the finite-state control problem used in [19, 3]. The goal is to move a population on a discrete torus towards a prescribed target distribution while penalizing individual movement. This example tests the estimators on a higher-dimensional simplex: the population law has ten coordinates and the policy must coordinate local actions across all sites.

Model. The state space is the discrete torus $\mathcal{X} = \mathbb{Z}/10\mathbb{Z} = \{0, \dots, 9\}$ and the action space is $\mathcal{A} = \{-1, 0, +1\}$. Actions move an agent one site left, keep it in place, or move it one site right. The transition is deterministic: $P(x' | x, a, \mu) = \mathbf{1}_{\{x' = x + a \pmod{10}\}}$. The target distribution used in the experiments is $\mu_{\text{target}} = (0.02, 0.04, 0.09, 0.16, 0.19, 0.19, 0.16, 0.09, 0.04, 0.02)$. With movement penalty $c_{\text{mov}} = 0.01$, the

running and terminal rewards are

$$\begin{aligned} r(x, a, \mu) &= -c_{\text{mov}}|a| - \|\mu - \mu_{\text{target}}\|_2^2, \\ g(x, \mu) &= -\|\mu - \mu_{\text{target}}\|_2^2. \end{aligned}$$

The horizon is $T = 5$ for both training and validation.

Setup. The policy is a two-hidden-layer network of width 256 reading (t, μ) . The horizon is $T = 5$. MF-REINFORCE uses $\varepsilon = 2$, transport the grid $\lambda \in \{0.1, 0.2, 0.4\}$ at $\eta = 0.85$, and adaptive transport starts from $(\lambda_0, \eta_0) = (0.2, 0.8)$, at the equal budget derived from $B = 500$, $n = 10$. Training randomizes the initial law over $\text{Dirichlet}(\mathbf{1}_{10})$ for 30,000 iterations; validation starts from the uniform law. With nine free coordinates this is the largest population argument in the suite. Appendix B.3 gives the full specification.

Results. Distribution planning separates the two population-aware estimators more sharply than any other benchmark; Table 5.4 collects the comparison.

Estimator	Final objective	Wall clock (s)
REINFORCE	-0.2616 ± 0.0000	263
MF-REINFORCE $\varepsilon = 2$	-0.1017 ± 0.0161	6 622
Transport $\lambda = 0.1, \eta = 0.85$	-0.2455 ± 0.0375	6 802
Transport $\lambda = 0.2, \eta = 0.85$	-0.2715 ± 0.0611	6 708
Transport $\lambda = 0.4, \eta = 0.85$	-0.2421 ± 0.0459	6 497
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	-0.5596 ± 0.1225	5 975

Table 5.4: Distribution planning benchmark at $T = 5$ with exact population flow: final validation objective as mean and standard deviation over five seeds, and mean wall-clock cost per run, at a matched simulator budget. Higher is better.

MF-REINFORCE reaches -0.1017 ± 0.0161 within the first few hundred iterations and improves slowly thereafter, whereas REINFORCE is frozen at -0.2616 with zero seed dispersion, never leaving the neighbourhood of the uniform law. The transport runs sit between the two but much closer to REINFORCE, at -0.2421 for $\lambda = 0.4$, and Figure C.4 shows why the average is not the whole story: the transport curves oscillate over tens of thousands of iterations without settling, and their seed dispersion is comparable to their distance from REINFORCE. At $\lambda = 0.4$ the five seeds end between -0.1986 and -0.2963 , straddling the REINFORCE value of -0.2616 ; no transport seed at any scale comes within 0.09 of the MF-REINFORCE result.

The qualitative check is nevertheless passed. Figure 5.2 shows that the transport policy moves mass coherently from the uniform law towards the target mode, concentrating it slightly too much on the central sites, rather than producing the local oscillation that a badly scaled sensitivity estimate would cause. The failure is one of accuracy, not of qualitative behaviour.

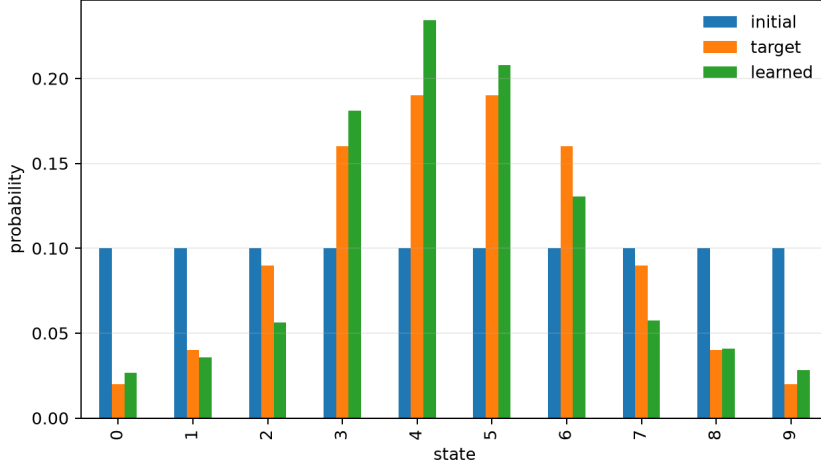


Figure 5.2: Distribution-planning benchmark at $T = 5$: terminal population law learned by fixed-scale transport at $\lambda = 0.4$, against the uniform initial law and the target. The learned flow is coherent transport towards the target mode rather than local oscillation, but it overshoots the centre and undershoots the shoulders.

5.2.4 Targeted Advertising

The targeted-advertising benchmark is based on the model introduced by Motte and Pham in [21]. A planner repeatedly decides whether to advertise to a large population in order to increase the proportion of customers while paying an advertising cost. The example is useful for the present study because the mean-field interaction enters through the transition kernel, while the population dynamics reduce to a one-dimensional deterministic recursion.

Model. The state and action spaces are $\mathcal{X} = \{0, 1\}$ and $\mathcal{A} = \{0, 1\}$. State $x = 1$ means that the individual is a customer, and action $a = 1$ means that an advertisement is displayed. Writing $p = \mu(1)$ for the customer proportion, the transition kernel is $P(1 | x, a, \mu) = \min\{p + \kappa_{\text{ad}}a, 1\}$, $P(0 | x, a, \mu) = 1 - P(1 | x, a, \mu)$. Thus advertising increases the probability of being a customer by κ_{ad} , up to saturation at 1, while the dependence on p models positive social influence. The one-step reward is $r(x, a, \mu) = x - c_{\text{ad}}a$, and the terminal reward is zero. The finite-horizon discounted objective is $J_T(\theta; p_0) = \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t (X_t^\theta - c_{\text{ad}}A_t^\theta) \right]$.

Setup. The learned policy is a two-hidden-layer network of width 32 reading (t, μ) , at horizon $T = 5$. MF-REINFORCE uses $\varepsilon = 1$, transport the grid $\lambda \in \{0.1, 0.2, 0.4\}$ at $\eta = 0.85$, and adaptive transport starts from $(\lambda_0, \eta_0) = (0.2, 0.8)$, at the equal budget derived from $B = 200$, $n = 10$. Training randomizes the initial customer share over $\mathcal{U}([0.05, 0.95])$ for 10,000 iterations; validation averages over a fixed grid in the same interval. Appendix B.4 gives the full specification.

Results. Targeted advertising is a plateau-escape problem. REINFORCE converges within a few hundred iterations to 0.9688 ± 0.0000 and never moves again: its learned policy advertises with probability exactly 0 in both states and at every time, so the customer share never leaves its initial value, while the reference policy $\hat{q}$ advertises with probability 1 almost everywhere. The policy that never advertises is a stationary point of the detached objective, and REINFORCE cannot leave it. Every population-aware estimator does leave it, and the ordering of Table 5.1 follows: MF-REINFORCE reaches 0.9946 ± 0.0091 , fixed-scale transport at $\lambda = 0.1$ reaches 0.9795 ± 0.0089 , and adaptive transport 0.9727 ± 0.0078 . Figure 5.3 shows the separation over training.

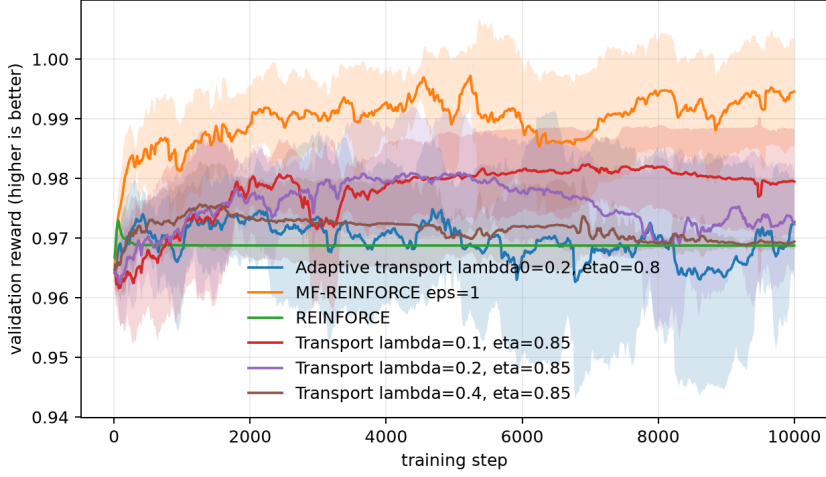


Figure 5.3: Targeted-advertising benchmark at $T = 5$: mean validation reward against training iteration over five seeds, with one standard deviation shaded. REINFORCE is the flat line at 0.9688: it reaches the never-advertise plateau within a few hundred iterations and cannot leave it, while every population-aware estimator escapes.

Neither estimator recovers the reference policy, and they fail in different ways. Table 5.5 gives the learned advertising probability in each of the two states. MF-REINFORCE advertises to everyone with probability 1 for the first two steps and then switches off completely, which brings the customer share from 0.2 to 0.603 and holds it there; the reference keeps advertising until the end. Transport and adaptive transport instead learn a state-targeted policy that advertises to non-customers with probability 1 and to customers with probability 0, at every step. That policy is intuitively reasonable, since advertising to an existing customer costs without converting anyone, and it drives the customer share higher than MF-REINFORCE does, to 0.738; it nonetheless scores lower, because the reference advertises to customers as well and the retention this buys outweighs its cost.

Policy	Advertising probability (non-customer, customer)					Final share
	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	
Reference $\hat{q}$	(1.00, 1.00)	(1.00, 1.00)	(1.00, 1.00)	(1.00, 1.00)	(0.64, 0.64)	—
REINFORCE	(0.00, 0.00)	(0.00, 0.00)	(0.00, 0.00)	(0.00, 0.00)	(0.00, 0.00)	0.200
MF-REINFORCE	(1.00, 1.00)	(1.00, 1.00)	(0.02, 0.01)	(0.00, 0.00)	(0.00, 0.00)	0.603
Transport $\lambda = 0.1$	(1.00, 0.00)	(1.00, 0.00)	(1.00, 0.00)	(1.00, 0.00)	(1.00, 0.00)	0.738
Adaptive transport	(1.00, 0.00)	(1.00, 0.00)	(1.00, 0.00)	(1.00, 0.00)	(1.00, 0.00)	0.738

Table 5.5: Learned advertising probabilities on the targeted-advertising benchmark, by state and time, on the best seed of each estimator, together with the customer share reached at $T = 5$ from the validation law. The reference row is the explicit policy $\hat{q}$ evaluated along the corresponding learned flow. The population-averaged advertising probability plotted in Figure C.7 equals 0.5 for both transport rows, which is an artefact of averaging a policy that is 1 in one state and 0 in the other; the two estimator families in fact learn structurally different policies, one truncated in time and the other targeted in state.

The reward scale compresses all of this: the whole spread between a policy that never advertises and one that nearly matches the reference is under 3% of the objective. The seed dispersion is also bimodal rather than diffuse: at $\lambda = 0.1$ three seeds reach the targeted policy of Table 5.5 and two remain near the REINFORCE plateau with an averaged advertising probability of 0.03, so the mean of 0.31 reported by the policy-error table describes no individual run. Transport escapes the plateau on most seeds but not on every one.

5.3 Continuous-state space benchmarks

Both benchmarks below are exact-chart instances of the construction of Section 3.4, in the sense discussed there: the perturbed population argument is one-dimensional, given by the mean, the chart is the closed moment recursion rather than a fitted Gaussian mixture, and the representation error vanishes by construction. The higher moments are still propagated, but only to evaluate the exact objective and the reference optimum at a frozen policy. They therefore test the perturbation of the population argument and the auxiliary sensitivity recursion in isolation, with no mixture-fitting error superposed on them.

5.3.1 Linear–Quadratic

We first consider a scalar linear–quadratic mean-field control problem. This benchmark is deliberately transparent: the population law is Gaussian, the moment recursion is closed, the perturbed objective and its gradient are available in closed form, and the unperturbed optimizer is given by a Riccati recursion. It therefore provides a continuous-state analogue of the two-state benchmark, where optimization errors can be separated from estimator bias and Monte Carlo variance.

Model. The initial state is $X_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$ and the action space is $\mathbb{R}$. For a population law with mean $\bar{m}_t$, the policy is Gaussian,

$$\pi_t^\theta(\cdot \mid x, m) = \mathcal{N}\left(\theta_t^1 x + \theta_t^2 \bar{m}_t, \tau^2\right), \quad \theta_t = (\theta_t^1, \theta_t^2) \in \mathbb{R}^2.$$

The controlled dynamics are $X_{t+1} = aX_t + bA_t + c\bar{m}_t + \sigma\varepsilon_{t+1}$, $\varepsilon_{t+1} \sim \mathcal{N}(0, 1)$. The implementation uses rewards equal to the negative of the quadratic cost. Equivalently, validation minimizes

$$C_T(\theta) = \mathbb{E} \left[\sum_{t=0}^{T-1} \left(qX_t^2 + rA_t^2 + \gamma\bar{m}_t^2 \right) + q_TX_T^2 + \gamma_T\bar{m}_T^2 \right],$$

and reports the validation reward $-C_T(\theta)$.

Setup. The unperturbed optimizer is given by a Riccati recursion, and the perturbed objective and its gradient are available in closed form. The horizon is $T = 20$. The population argument is perturbed affinely, $M_t^{\lambda, \theta} = (1 + \lambda\zeta_t)\mu_t^\theta + \lambda\beta_t$ with ζ_t, β_t i.i.d. $\mathcal{N}(0, \rho^2)$; the fixed-scale grid is $\lambda \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$ at $\eta = 0.85$ and adaptive transport starts from $(\lambda_0, \eta_0) = (0.2, 0.8)$. The reference budget is $B = 200$ main and $n = 1$ law-gradient trajectories, which transport reallocates to $n_{\text{law}} = 20$ auxiliary trajectories at the same total cost. Both exact moment flow and particle flow are run. Appendix B.5 gives the model, the parameters and the diagnostics protocol.

Results. The Riccati optimizer gives $C(\theta^*) = 7.2238$ in the cost convention. REINFORCE ends at 7.5157 ± 0.0010 and fixed-scale transport at $\lambda = 0.1$ at 7.2952 ± 0.0023 , closing 75.5% of the REINFORCE optimality gap with a seed dispersion two orders of magnitude below the effect. On the full training axis this is invisible, and the comparison even looks favourable to REINFORCE, which comes within 0.05 of its final value at iteration 3,160 against 8,530 for transport, and captures 99.35% of the available improvement against 99.85%. The left panel of Figure 5.4 zooms on the last 6,000 iterations and shows what that hides: the REINFORCE curve is not converged but oscillating between 7.38 and 7.53, while the transport curves descend smoothly and are still improving at the end of the budget. The two estimators do not approach the same limit.

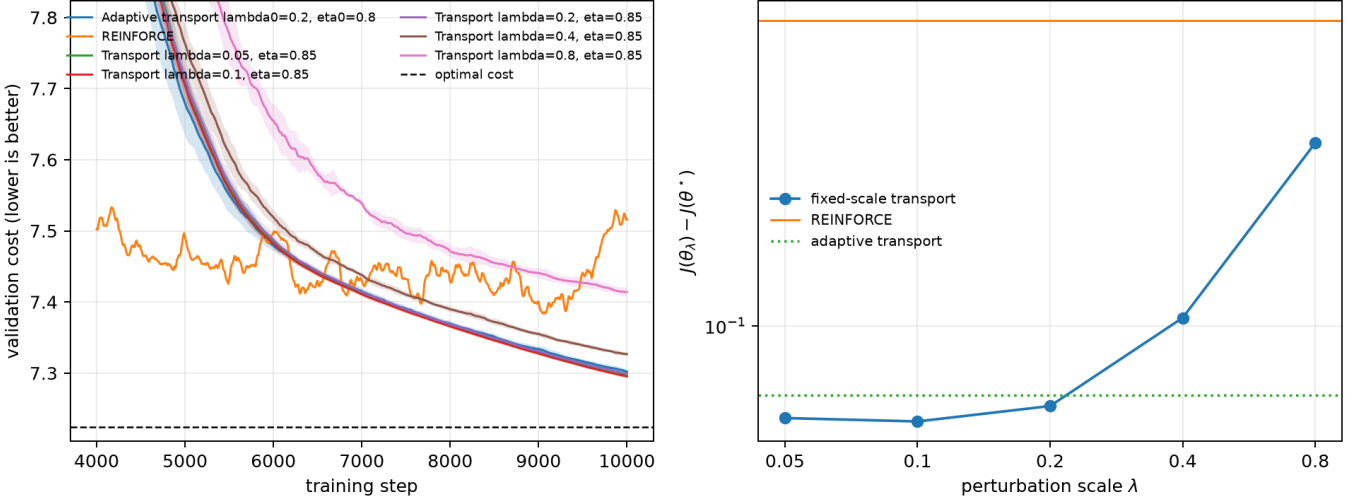


Figure 5.4: Linear-quadratic benchmark at $T = 20$ with exact population flow. Left: validation cost over the last 6,000 training iterations, as mean over five seeds with one standard deviation shaded, against the Riccati optimum (dashed). Right: final optimality gap against the perturbation scale, on logarithmic axes, with REINFORCE and adaptive transport as horizontal references. The gap is minimized at an interior λ and every fixed scale in the grid improves on REINFORCE.

Because the perturbed objective and its optimizer are both available in closed form here, the residual error splits into two interpretable pieces, reported in Table 5.6. The perturbation bias $|J^\lambda(\theta_\lambda^*) - J(\theta^*)|$ is a property of the problem the estimator targets rather than of the run, and grows from 0.275 at $\lambda = 0.05$ to 70.4 at $\lambda = 0.8$; the optimization error $|J^\lambda(\hat{\theta}_\lambda) - J^\lambda(\theta_\lambda^*)|$ measures how far the learned parameter is from the optimizer of that perturbed problem and is the only part a better estimator could remove, staying at 0.073–0.092 for $\lambda \leq 0.2$ before growing to 0.42. The optimality gap in the last column is not their sum: a policy optimized for a heavily biased J^λ is displaced from θ^* by far less than the bias of the objective, which is why $\lambda = 0.8$ carries a perturbation bias of 70.4 and a gap of only 0.19. Together the three columns show a trade-off with an interior minimum at $\lambda = 0.1$, plotted in the right panel of Figure 5.4, whose two mechanisms Section 5.5 measures directly.

λ	$J^\lambda(\theta_\lambda^*)$	Perturbation bias $ J^\lambda(\theta_\lambda^*) - J(\theta^*) $	Optimization error $ J^\lambda(\hat{\theta}_\lambda) - J^\lambda(\theta_\lambda^*) $	Optimality gap $ J(\hat{\theta}_\lambda) - J(\theta^*) $
0.05	7.499	0.275	0.0733 ± 0.0021	0.0723 ± 0.0021
0.1	8.325	1.101	0.0756 ± 0.0025	0.0715 ± 0.0023
0.2	11.628	4.405	0.0916 ± 0.0029	0.0754 ± 0.0021
0.4	24.836	17.613	0.1631 ± 0.0068	0.1028 ± 0.0028
0.8	77.654	70.430	0.4246 ± 0.0211	0.1901 ± 0.0059
Adaptive	—	—	—	0.0783 ± 0.0030
REINFORCE	—	—	—	0.2920 ± 0.0010

Table 5.6: Linear-quadratic benchmark at $T = 20$ with exact population flow, cost convention, lower is better, with $J(\theta^*) = 7.224$. The residual error of a fixed-scale transport run splits into two parts. The perturbation bias is a property of the problem the estimator targets, not of the run: it is the distance between the optimal value of the perturbed objective and the true optimum, and it carries no seed dispersion. The optimization error is how far the learned parameter $\hat{\theta}_\lambda$ is from the optimizer of that perturbed objective, and it is the only part a better estimator could remove. The last column is the quantity that matters, and it is not the sum of the other two, since a policy optimized for J^λ need not be displaced from θ^* by the full bias of the objective. All entries are means and standard deviations over five seeds, computed from closed-form expressions with no Monte Carlo error, so the dispersion is entirely across seeds. The adaptive controller varies λ during training and REINFORCE targets no perturbed problem, so only their optimality gaps are defined.

This is also where the particle approximation is cheapest: replacing the exact moment recursion (B.4) by particle estimates moves the final cost at $\lambda = 0.1$ from 7.29525 to 7.29597, about 1% of the remaining gap, and Figure C.10 shows the two curves indistinguishable over the whole run. The contrast with the two-state benchmark, where the same substitution costs two thirds of the gap, is dimensional: here the population law enters only through its mean, and a mean is what a particle ensemble estimates well.

5.3.2 Mean–Variance Portfolio

We next consider a discrete-time mean–variance portfolio selection problem, of the kind put in mean-field form in [6, 7]. As in the LQ benchmark, the relevant moments satisfy closed recursions and the exact perturbed objective is available, but the state is driven by multiplicative asset returns. This makes the example a useful continuous-state test in which the estimator noise comes from financial returns and the population dependence is carried both by the feedback policy and by an explicit running mean-field penalty.

Model. Let $X_t \in \mathbb{R}$ denote the wealth of an investor at time t and let $\alpha_t \in \mathbb{R}$ denote the monetary amount invested in the risky asset, the remainder being held at the deterministic gross risk-free return $s_t > 0$. Writing R_{t+1} for the excess return of the risky asset over $[t, t+1]$, with mean $\bar{r}_t$ and variance $\sigma_{R,t}^2$, the wealth dynamics are

$$X_{t+1} = s_t X_t + \alpha_t R_{t+1}, \quad t = 0, \dots, T-1.$$

The investor is subject to a precommitment mean–variance criterion on terminal wealth, penalized by a running cost on the aggregate position of the population. With $\mu_t = \mathbb{E}[X_t]$ the population mean, the running and terminal rewards are

$$r(x, a, \mu) = -\gamma \mu^2, \quad g(x, \mu) = x - \chi(x - \mu)^2,$$

where $\chi > 0$ is the risk-aversion coefficient and $\gamma \geq 0$ is a mean-field penalty. The terminal reward yields the usual mean–variance objective $\mathbb{E}[X_T] - \chi \text{Var}(X_T)$ at the consistent population law, while the running term introduces an explicit dependence on the population mean. In the current benchmark we set $\gamma = 2$, so the task contains a genuine population-gradient component in addition to the dependence of the feedback policy on the mean.

Setup. The mean and variance recursions are closed, so the exact objective and the optimal (k_t^*, ℓ_t^*) of (B.6)–(B.7) are available for validation. The horizon is $T = 10$. The population mean is perturbed as in the linear–quadratic benchmark, with the smaller grid $\lambda \in \{0.025, 0.05, 0.1, 0.2, 0.4\}$ at $\eta = 0.85$ to match the scale of the wealth process. The reference budget is $B = 500$, $n = 1$, i.e. 5,110 simulated transitions per update, which transport reallocates to $n_{\text{law}} = 200$ auxiliary trajectories. Appendix B.6 gives the full specification.

Results. The portfolio benchmark produces the largest separation of the two continuous-state examples, and it is the only one on which the adaptive controller is the best method in the comparison. Against the closed-form optimum $J(\theta^*) = -13.133$, REINFORCE ends at -20.535 ± 0.004 , fixed-scale transport between -18.195 and -17.980 , and adaptive transport at -17.193 ± 0.594 , closing respectively 34.5% and 45.1% of the REINFORCE gap. Figure C.8 shows that REINFORCE does not stall but degrades: it drifts below its own initialization from iteration 2,000 onwards and ends more than one unit below where it started, whereas every transport variant climbs.

Two features qualify the result, both about what five seeds establish.

First, the choice of λ barely matters here: across the whole grid the final objective moves only from -18.19 to -17.98 , a range of 0.21, against a distance to REINFORCE of about 2.5. What produces the improvement is using the population-flow derivative at all, not the scale at which it is estimated. This is the opposite of the two-state benchmark, where the same grid spans more than six units of objective and the choice of λ decides whether the run succeeds.

Second, the seed dispersion exceeds that grid range: each fixed scale carries a standard deviation between 1.08 and 1.29 against the 0.21 separating best from worst. Five seeds therefore cannot rank the λ values,

and the apparent ordering inside the transport block of Table 5.1 should not be read as one. The comparison against REINFORCE is a different matter, since a gap of 2.5 against a dispersion of 1.2 is resolved, and in the same direction on every seed.

The same decomposition as on the linear-quadratic benchmark falls out very differently here. Table 5.7 shows a perturbation bias that is small at every scale, from 0.03 at $\lambda = 0.025$ to 7.35 at $\lambda = 0.4$, against an optimization error that never drops below 4.97: the learned policy is far from the optimizer of the perturbed problem it is actually solving, rather than displaced by the bias of that problem. The limitation here is estimator variance, not perturbation bias, consistent with the auxiliary sample size of 200 per coordinate being the largest in the suite and still insufficient.

λ	$J^\lambda(\theta_\lambda^*)$	Perturbation bias $ J^\lambda(\theta_\lambda^*) - J(\theta^*) $	Optimization error $ J^\lambda(\hat{\theta}_\lambda) - J^\lambda(\theta_\lambda^*) $	Optimality gap $ J(\hat{\theta}_\lambda) - J(\theta^*) $
0.025	-13.16	0.03	5.0669 ± 1.2370	5.0617 ± 1.2352
0.05	-13.26	0.12	4.9699 ± 1.2979	4.9494 ± 1.2904
0.1	-13.62	0.49	4.9742 ± 1.3048	4.8891 ± 1.2755
0.2	-15.05	1.92	5.2298 ± 1.3084	4.8533 ± 1.2021
0.4	-20.48	7.35	6.6487 ± 1.4065	4.8472 ± 1.0790
Adaptive	—	—	—	4.0601 ± 0.5937
REINFORCE	—	—	—	7.4021 ± 0.0042

Table 5.7: Portfolio benchmark at $T = 10$ with exact population flow, reward convention, higher is better, with $J(\theta^*) = -13.133$. The residual error of a fixed-scale transport run splits into two parts. The perturbation bias is a property of the problem the estimator targets, not of the run: it is the distance between the optimal value of the perturbed objective and the true optimum, and it carries no seed dispersion. The optimization error is how far the learned parameter $\hat{\theta}_\lambda$ is from the optimizer of that perturbed objective, and it is the only part a better estimator could remove. The last column is the quantity that matters, and it is not the sum of the other two, since a policy optimized for J^λ need not be displaced from θ^* by the full bias of the objective. All entries are means and standard deviations over five seeds, computed from closed-form expressions with no Monte Carlo error, so the dispersion is entirely across seeds. The adaptive controller varies λ during training and REINFORCE targets no perturbed problem, so only their optimality gaps are defined.

The controlled ablation of Section 5.6, which switches the explicit population dependence on and off through γ , attributes the improvement to that dependence rather than to the perturbation acting as a generic exploration device.

5.4 Adaptive perturbation control

The fixed-scale results above are read at the best entry of a grid costing between three and twenty runs per benchmark. The adaptive controller of Algorithm 2 exists to remove that grid, and the question is whether one run from the benchmark-independent initialization $(\lambda_0, \eta_0) = (0.2, 0.8)$ reaches a comparable objective. Table 5.8 reports the scales it selects.

Benchmark	Flow	λ after training	η after training	Best fixed λ	Resolved bias
Two-state	exact	0.039 ± 0.002	0.365 ± 0.023	0.1	66%
Cybersecurity	exact	0.026 ± 0.000	0.214 ± 0.005	0.4	68%
Distribution planning	exact	0.025 ± 0.000	0.201 ± 0.001	0.4	72%
Targeted advertising	exact	0.040 ± 0.004	0.361 ± 0.024	0.1	64%
Linear-quadratic	exact	0.226 ± 0.015	0.558 ± 0.015	0.1	48%
Portfolio	exact	0.597 ± 0.027	0.336 ± 0.040	0.4	47%

Table 5.8: Perturbation scales reached by the adaptive controller of Algorithm 2, started from $\lambda_0 = 0.2$, $\eta_0 = 0.8$ on every benchmark, as mean and standard deviation over five seeds. The last column gives the fraction of checkpoints at which the debiased bias estimate was resolved and the main scale was therefore allowed to move. The controller contracts both scales on every benchmark. On the continuous-state problems the λ it settles on is comparable to the best entry of the hand-tuned grid; on the finite-state ones it contracts below the smallest calibrated scale, which is where it underperforms.

The controller contracts both scales on every benchmark: mildly on the continuous-state problems, where λ settles between 0.20 and 0.60, and strongly on the finite-state ones, where it reaches 0.025–0.040. The auxiliary scale is contracted everywhere, from $\eta_0 = 0.8$ to between 0.20 and 0.68, which is the one systematic disagreement with the fixed-scale evidence, since the calibration grid prefers η near the top of its range. Figure C.12 shows the mechanism on the linear-quadratic benchmark: the balance signals z_λ and z_η are negative at almost every checkpoint, meaning the debiased bias estimate is small relative to the measured variance, and the controller shrinks the scale carrying that variance. For λ this is the correct response and the trajectory is non-monotone, rising to 0.32 before returning to 0.226; for η the same signal drives a monotone contraction that the grid says is unhelpful.

Measured on the objective rather than on the scales, the controller does most of what it is meant to do. It recovers 97% of the benefit of the grid from a single run on the linear-quadratic benchmark, reaching -7.3021 against -7.2952 ; it is the best method outright on the portfolio benchmark; it lands between the fixed grid and MF-REINFORCE on two-state at both horizons; and it escapes the REINFORCE plateau on targeted advertising.

The two failures are cybersecurity, where it is the worst method at -0.1602 , and distribution planning, where it reaches -0.5596 against -0.2616 for REINFORCE and -0.2421 for the best fixed scale. The terminal scale alone does not distinguish them: the controller contracts λ to 0.026 and 0.025 there, but to 0.039 and 0.040 on two-state and advertising where it performs acceptably, and all four lie below $\lambda = 0.05$, the smallest calibrated scale, so it leaves the calibrated range on every finite-state benchmark. What separates the two pairs is the dimension of the population argument, a two-point simplex on two-state and advertising against a four-point and a ten-point one. The perturbation-density score has $N - 1$ coordinates and scales as λ^{-1} , so contracting the main scale inflates a variance that grows with the dimension, and the same terminal λ is harmless on Δ_2 and damaging on Δ_{10} .

The controller is not failing to measure: the bias estimate is resolved on 68% and 72% of checkpoints on these two benchmarks, the highest rates in the suite, against 48% on the linear-quadratic benchmark where the outcome is good. It measures a small bias correctly and draws the wrong conclusion from it, because the balance rule weighs bias against variance without accounting for the λ^{-1} growth of the score.

5.5 Validation of the theoretical results

The benchmark sections above measure optimization outcomes. This section asks a narrower question: do the quantities that Chapter 3 bounds behave, on the saved runs, the way the bounds say they should. Four predictions are testable from the recorded artifacts. Table 5.9 states each one, the quantity that tests it and the outcome; the paragraphs that follow give the measurements. None of these quantities is available to the training algorithms.

Prediction	Measured quantity	Result	Verdict
$d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^\theta) \leq \lambda$	$\max_t d_{\text{TV}}/\lambda$ along the learned flow	holds at 445/445 points, ratio up to 1.00	tight
$ J^\lambda - J \leq C_T \lambda$	bias/ λ^2 over a $16\times$ range of λ	constant to 0.1% (LQ) and 7% (portfolio)	$O(\lambda^2)$
$\mathbb{E}[\widehat{G}] = \nabla_\theta J^\lambda$ under oracle D	χ^2 test, 50 replications	$p = 1.000, 1.000, 0.996$ at $\lambda \leq 0.2$; $p < 10^{-3}$ at $\lambda = 0.8$	$\lambda \leq 0.2$
$\text{MSE} = \text{bias}^2 + \text{dispersion}$	the identity, run by run	closes to the $(R-1)/R$ factor of the variance estimate	holds

Table 5.9: The four predictions of Chapter 3 that the recorded runs can test, and the outcome of each. The first is measured on every finite-state transport run, the others on the linear–quadratic and portfolio benchmarks, the only two exposing a closed-form gradient oracle. On the portfolio benchmark the third test is inconclusive rather than failed, for the reason given below.

The perturbation is no larger than λ . Since $M_t^{\lambda, \theta} = (1 - \lambda)\mu_t^\theta + \lambda q_t$ with $q_t \in \Delta_N$, we have $d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^\theta) = \lambda d_{\text{TV}}(q_t, \mu_t^\theta) \leq \lambda$, the estimate every subsequent bound is built on. Table C.12 records the largest value along the learned flow of every finite-state transport run; the ratio to λ reaches 0.80 on two-state and advertising and 1.00 on distribution planning, so the perturbation is neither degenerate nor larger than the theory allows.

The perturbed optimum converges at the predicted rate. Theorem 2 bounds $|J^\lambda(\theta) - J(\theta)|$ by $C_T \lambda$ uniformly in θ , and its corollary carries that over to the optimizers. The linear–quadratic and portfolio benchmarks give both J^λ and its maximizer in closed form, so the error induced on the optimal value can be computed exactly rather than estimated.

Table 5.10 reports it. The measured error stays well inside the bound, and it shrinks faster than λ : divided by λ the ratio doubles at every step of the grid, while divided by λ^2 it is constant to within 0.1% across a sixteen-fold range of scales. There is a reason for the extra order. Both objectives are quadratic in the population argument, so the linear term of the expansion vanishes at the optimum and what remains is $O(\lambda^2)$. The measurement recovers that exponent, which confirms the derivation rather than contradicting the theorem: the $O(\lambda)$ bound still holds, it is simply not tight on these two problems.

Benchmark	λ	$J^\lambda(\theta_\lambda^*)$	$ J^\lambda(\theta_\lambda^*) - J(\theta^*) $	$/\lambda$	$/\lambda^2$
Linear–quadratic	0.05	7.499	0.2754	5.51	110.17
	0.1	8.325	1.1015	11.01	110.15
	0.2	11.628	4.4047	22.02	110.12
	0.4	24.836	17.6126	44.03	110.08
	0.8	77.654	70.4299	88.04	110.05
Portfolio	0.025	−13.164	0.0308	1.23	49.34
	0.05	−13.256	0.1231	2.46	49.23
	0.1	−13.622	0.4887	4.89	48.87
	0.2	−15.050	1.9171	9.59	47.93
	0.4	−20.479	7.3465	18.37	45.92

Table 5.10: Perturbation bias of the optimal value on the two benchmarks where the perturbed objective and its optimizer are both available in closed form. Theorem 2 bounds this quantity by $C_T \lambda$ in general, so the ratio in the fifth column is bounded and the sixth column would diverge if the rate were exactly linear. It does not: the sixth column is constant to within 0.1% over a sixteen-fold range of λ on the linear–quadratic benchmark, and constant to within 7% on the portfolio benchmark. Both benchmarks are quadratic in the population argument, for which the linear term of the expansion vanishes and the bias is $O(\lambda^2)$; the measurement recovers that rate rather than the generic bound.

The estimator is unbiased for the perturbed gradient. Proposition 3 states that the estimator is exactly unbiased for $\nabla_\theta J^\lambda$ under oracle sensitivities, and Theorem 5 adds the bias the estimated sensitivity introduces, of order $\lambda^{-1} \sqrt{\eta^2 + (n\eta^2)^{-1}}$. The distinction is testable: at a frozen policy, replicate the estimator

and compare its mean with the closed-form $\nabla_{\theta} J^{\lambda}$. Table 5.11 does this on the linear–quadratic benchmark with $R = 50$ replications at 8192 particles. Its fifth column is the p -value of a χ^2 test of the null hypothesis that the estimator is unbiased for $\nabla_{\theta} J^{\lambda}$, that is, the probability of observing a deviation at least as large as the measured one if the true bias were zero; a large p means the observed gap is within replication noise, a small one that it is not.

λ	$\ \nabla J^{\lambda} - \nabla J\ $	$/\lambda^2$	$\ \mathbb{E}[\hat{G}] - \nabla J^{\lambda}\ $	p	Dispersion	$d \cdot \text{MSE}$	cos
0.05	0.0030	1.18	0.0421	1.000	0.5926	0.3500	0.941
0.1	0.0115	1.15	0.0480	1.000	0.6151	0.3763	0.914
0.2	0.0433	1.08	0.0702	0.996	0.7234	0.5224	0.892
0.4	0.1562	0.98	0.1963	0.123	1.2290	1.5232	0.742
0.8	0.5899	0.92	0.9565	0.000	4.7778	23.2859	0.643

Table 5.11: Evaluation-only gradient diagnostics on the linear–quadratic benchmark, at the policies saved at the end of training and averaged over the five seeds, from 50 independent replications of the estimator at 8192 particles. Column two is the perturbation bias of the gradient, $\|\nabla_{\theta} J^{\lambda}(\theta) - \nabla_{\theta} J(\theta)\|$, computed from the closed-form gradient oracle; column three divides it by λ^2 . Column four is the bias of the estimator with respect to the perturbed gradient it targets, and column five the p -value of the χ^2 test that this bias is zero at the available replication count, so a large p means the estimator is not measurably biased for $\nabla_{\theta} J^{\lambda}$. Column six is the dispersion $\|\text{sd}(\hat{G})\|$ of the replications and column seven the mean-square error scaled by the parameter dimension d ; it equals the sum of the squares of columns four and six, up to the factor $(R - 1)/R$ carried by the unbiased dispersion estimate, and the identity is verified run by run to 2%, which is exactly that factor at $R = 50$. The last column is the cosine alignment of the mean estimate with the oracle gradient.

The bias is indistinguishable from zero at $\lambda \leq 0.2$, where $p \geq 0.996$, marginal at $\lambda = 0.4$ with $p = 0.12$, and unambiguous at $\lambda = 0.8$, where p falls below 10^{-3} . The prediction therefore holds throughout the regime the theory targets, and departs exactly where the λ^{-1} prefactor on the auxiliary error is largest relative to the signal.

The portfolio benchmark, Table C.14 in Appendix C.11, is the honest limit of the measurement rather than of the theory: the estimator dispersion is 347 against an exact gradient of norm 1.08, a signal-to-noise ratio of 0.02 per replication, and the measured bias of 50 is indistinguishable from zero at every scale. The test is uninformative there, not passed.

The gradient bias and the mean-square error decompose as stated. The same tables check the two remaining predictions. The gradient perturbation bias, which Theorem 3 bounds by $C_T \lambda$, scales as λ^2 with a coefficient between 0.92 and 1.18 on the linear–quadratic benchmark and between 2.46 and 2.48 on the portfolio benchmark, constant to within 1% across the grid. This is the gradient-level counterpart of the objective-level scaling above, measured on a different quantity by a different route, so the two are independent checks.

The mean-square error of Theorem 5 splits into a squared bias and a dispersion, and the measured quantities satisfy that identity run by run to within the $(R - 1)/R$ factor of the unbiased dispersion estimate. Reading the two contributions against each other makes the trade-off concrete: at $\lambda = 0.05$ the dispersion is 0.59 against a squared bias of 0.0018, so the error is essentially all variance, whereas at $\lambda = 0.8$ the dispersion has grown to 4.78 and the bias to 0.96, and both terms have deteriorated together, with the cosine alignment falling monotonically from 0.94 to 0.64. This is the quantitative form of the interior optimum in λ that Table 5.6 shows on the objective.

The size of the correction, measured on the estimator itself. The same replications give the size of the transport correction relative to the policy-score term, measured on the estimator rather than on an oracle. On the linear–quadratic benchmark it accounts for a fraction 1.27 to 1.58 of the gradient norm at $\lambda \leq 0.2$, falling to 0.24 at $\lambda = 0.8$, and is resolved against its own replication noise at every scale below 0.4. On the portfolio benchmark the measured fraction is 1.00 at every scale with a χ^2 p -value above 0.96, so it is not distinguishable from noise at the saved policies. This does not contradict the training results

but confirms the point of Section 5.6, that the magnitude of the correction is not by itself informative; the evidence that the population channel drives the portfolio improvement comes from the controlled γ ablation.

What is not tested. Three limits are worth recording. The uniformity in θ of Theorem 2 is tested only at the policies the runs actually reach, which concentrate near the optimum, so nothing is said about the constants far from it. The auxiliary error bound of Proposition 4, $C_T(\eta^2 + (n\eta^2)^{-1})$, is the one prediction these artifacts cannot isolate, since separating its two terms would need a sweep in n at a fixed policy; the calibration grid shows only the combined effect through the training outcome. And the replicated diagnostics exist only on the two benchmarks exposing a closed-form oracle and a directly parametrized policy, the finite-state benchmarks having network policies for which an automatic-differentiation oracle for $\nabla_\theta J^\lambda$ is implemented at $\lambda = 0$ only.

5.6 Mean-field correction diagnostic

The benchmarks above do not order the estimators consistently, although the mean-field structure is present in all of them. This section isolates the quantity that separates the regimes: not whether the population-flow term is large, but whether it is collinear with the term REINFORCE already computes.

REINFORCE never differentiates the law supplied to the policy, the reward or the transition, so its target is the partial derivative of the objective at a frozen population flow, whereas the transport estimator targets the total derivative. Both are computable whenever the environment admits a differentiable evaluation path, here the exact population recursion on the finite-state benchmarks and a pathwise rollout under common random numbers on the continuous ones. Writing μ^θ for the population flow, define

$$G_{\text{full}}(\theta) = \nabla_\theta J(\theta, \mu^\theta), \quad G_{\text{det}}(\theta) = \nabla_\theta J(\theta, \mu) \Big|_{\mu=\mu^\theta},$$

where the second expression detaches μ from the computational graph before differentiating, so that the population channel is suppressed. Their difference is exactly the mean-field correction the transport estimator is built to recover, and G_{det} is what REINFORCE estimates; both are evaluation-only and never available during training. The natural summary, the relative size $\|G_{\text{full}} - G_{\text{det}}\|/\|G_{\text{full}}\|$, is not predictive: the policies are optimized with Adam, whose update is invariant to a positive rescaling of the gradient, so if the correction is parallel to G_{det} , then $G_{\text{full}} = (1 + c)G_{\text{det}}$, $c > -1$, the two estimators induce the same update no matter how large c is: a correction can account for half the gradient norm and be worth nothing. What changes the optimization path is the component orthogonal to G_{det} , captured by the alignment

$$\varrho(\theta) = \frac{\langle G_{\text{full}}(\theta), G_{\text{det}}(\theta) \rangle}{\|G_{\text{full}}(\theta)\| \|G_{\text{det}}(\theta)\|}. \quad (5.3)$$

Values close to +1 mean REINFORCE is directionally correct and the correction only rescales the step; values at or below zero mean it ascends a direction that is not an ascent direction for the true objective.

Measured alignment. We evaluate (5.3) at a reference policy obtained by running REINFORCE for 300 iterations from the benchmark initialization, so the diagnostic is read away from the initial point but before any estimator has converged, and repeat over five independent reference policies. Table 5.12 reports the mean and standard deviation together with the ordering observed in training.

Benchmark	$\ G_{\text{full}}\ $	$\ G_{\text{det}}\ $	ϱ	Observed ordering
Two-state ($T = 5$)	8.714	0.872	-1.000 ± 0.000	transport, MF $\gg$ REINFORCE
Cybersecurity	0.003	0.003	$+1.000 \pm 0.000$	REINFORCE $\approx$ MF
Distribution planning	0.029	0.001	$+0.092 \pm 0.075$	MF $\gg$ REINFORCE
Targeted advertising	0.002	0.005	-0.921 ± 0.023	MF, transport $>$ REINFORCE
Linear-quadratic	23.156	4.657	$+0.985 \pm 0.000$	transport $>$ REINFORCE, narrowly
Portfolio ($\gamma = 2$)	1.357	0.083	-0.196 ± 0.007	transport $>$ REINFORCE

Table 5.12: Correction alignment (5.3), as mean and standard deviation over five independent reference policies, together with the observed ordering of the estimators. The alignment separates the benchmarks on which REINFORCE is directionally competitive from those on which it is not; it does not by itself say how much of the residual gap a competitive REINFORCE leaves, as the linear-quadratic row shows.

The alignments fall into two separated groups, with nothing between 0.092 and 0.985. Every benchmark on which REINFORCE fails outright has $\varrho \leq 0.092$: it descends away from its initialization on two-state and portfolio, and freezes at a stationary point of the detached objective on distribution planning and advertising. The two benchmarks with $\varrho \geq 0.985$ are exactly those on which REINFORCE is directionally competitive, which is a weaker statement than being best: on the linear-quadratic benchmark the correction is far from negligible in norm, accounting for 80% of the gradient, yet nearly parallel to the policy-score term and therefore nearly invisible to a scale-invariant optimizer, and transport still ends ahead. It is the distance from $+1$, not the sign of ϱ , that carries the information: distribution planning has $\varrho = +0.092 \pm 0.075$ and still defeats REINFORCE, since a correction nearly orthogonal to the policy score reorients the update as effectively as one that opposes it.

What $\varrho \approx +1$ does and does not certify. The two high-alignment benchmarks do not behave identically, and the difference bounds what the diagnostic can claim. On cybersecurity REINFORCE is the best method outright; on the linear-quadratic benchmark it is not, since transport closes 75.5% of the optimality gap with a seed dispersion two orders of magnitude below the effect. A high alignment therefore certifies that REINFORCE ascends a useful direction, but not that the two estimators share a limit: the alignment is strictly below one, leaving an orthogonal component of size $\sqrt{1 - \varrho^2} = 0.17$, and REINFORCE indeed oscillates around a biased point rather than converging, as the left panel of Figure 5.4 shows. On cybersecurity the correction is negligible rather than merely aligned, $\varrho = +1.000$ to six decimals with a correction fraction of 0.001, so the two gradients have the same zeros to numerical precision. The reading is therefore ordinal: $\varrho \leq 0.1$ predicts that REINFORCE fails, $\varrho \approx +1$ that it is competitive, and the size of the residual gap is governed by the orthogonal component, which the alignment alone does not resolve.

Controlled pairs. The portfolio problem turns the observation above into a controlled comparison, since the market-impact coefficient γ multiplies the only reward term that depends explicitly on the population and can be switched off with the dynamics held fixed.

At $\gamma = 0$ the terminal reward depends on the population only through the centered variable $x - \mu$, so $\partial_\mu \mathbb{E}[-\chi(X - \mu)^2] = 2\chi(\mathbb{E}[X] - \mu) = 0$ at the consistent law, for every risk-aversion level. The measured gradients confirm that the correction vanishes, $\|G_{\text{full}}\| = \|G_{\text{det}}\| = 0.083$ with $\varrho = +1.000$ to three decimals across all five replications, so REINFORCE is unbiased on this instance and transport can only add variance. That is what training shows: at $\gamma = 0$ transport ends below its own initialization while REINFORCE recovers 42% of the available objective range, and at $\gamma = 2$ the ordering reverses. The same pair shows why the magnitude diagnostic misleads, since the correction measured on the estimator accounts for 83% of the gradient at $\gamma = 0$ even though the exact correction is zero; that 83% is entirely Monte Carlo noise from the auxiliary estimate.

Practical use and limitations. The alignment costs two automatic-differentiation passes over one deterministic evaluation and needs no closed-form oracle, so it can be computed on any benchmark with a differentiable evaluation path before committing to a training grid. It is nonetheless local, read at reference policies after a fixed number of REINFORCE iterations, with no guarantee of being constant along a trajectory; group membership never changes across the five replications, but distribution planning ranges

from +0.021 to +0.217 while cybersecurity and advertising are stable to within 0.023. Finally, this is an observation over six benchmarks and one controlled pair, not a theorem: a low alignment indicates that the correction can change the optimization path, not that a model-free estimator of it has low enough variance to exploit it. The portfolio runs at $\gamma = 2$ show that the two requirements are separate: the alignment is -0.196 , so the correction should change the optimization path, and it does, but transport needs the largest auxiliary allocation in the suite, 200 trajectories for a one-dimensional argument, and still leaves an optimization error of at least 4.97.

Chapter 6

Discussion and Conclusion

6.1 Review of the results

The construction of Chapter 3 rests on a structural claim: REINFORCE is biased in mean-field control because it differentiates the objective at a frozen population flow, and the missing term can be recovered model-free by randomizing the population argument. The experiments support that claim, and also show that recovering the missing term is not by itself a reason to pay for it.

The suite does not order the estimators uniformly, and that disagreement is the most informative feature of Table 5.1. Three regimes appear. On the two-state problem at $T = 5$, on the linear-quadratic benchmark and on the portfolio benchmark, an estimator that differentiates the population flow wins by a margin far larger than the seed dispersion, an order of magnitude on two-state. On distribution planning and targeted advertising the population-flow derivative helps, but the logit-perturbed estimator extracts more of it than transport does. On cybersecurity plain REINFORCE is the best method, and the entire spread between best and worst is under 6% of the objective, so that benchmark separates the methods weakly in absolute terms. One entry deserves separate comment: on distribution planning the adaptive controller is worse than every other method, not only than the fixed grid.

Matched simulator budgets, moreover, do not imply matched cost. The transport update spends its budget on an auxiliary sensitivity estimate that is cheap in simulated transitions and expensive in arithmetic, so its wall-clock cost exceeds that of REINFORCE by a factor between 2 and 26. The ratio is a property of the dimension of the population argument rather than of the estimator: the extreme case is distribution planning, whose nine-coordinate sensitivity makes the recursion the dominant cost of the update and which is also the benchmark with the smallest auxiliary sample size relative to that dimension.

The clearest positive evidence is the two-state benchmark at $T = 5$. REINFORCE degrades over iterations: it ends at -31.51 against an initialization of -4.53 and an optimum of -2.64 , while the transport estimator reaches -3.13 and recovers the closed-form optimal policy to within 3.5%. The same pattern, with smaller margins, appears on both continuous-state benchmarks, where transport closes three quarters of the optimality gap on the linear-quadratic problem and beats REINFORCE by more than twice the seed dispersion on the portfolio problem. In all three cases the population channel carries the sign of the useful direction, and an estimator that ignores it optimizes a different function.

The counterexamples are equally informative: on cybersecurity plain REINFORCE is the best method at a fraction of the wall-clock cost, and on distribution planning and targeted advertising the logit-perturbed estimator of [19] extracts more of the population-flow derivative than transport does. Section 5.6 isolates what separates the regimes, and it is not the magnitude of the correction but its alignment ϱ with the policy-score term. The portfolio ablation makes this controlled rather than correlational: switching the market-impact coefficient γ from 2 to 0 leaves the dynamics unchanged, drives ϱ from -0.196 to $+1.000$, and reverses the ordering of the estimators.

Section 5.5 adds the check that the estimators behave as the analysis says. All four testable predictions hold in the regime the theory targets, and the bias-variance trade-off in λ is therefore a measurable feature of the estimator rather than a formal artefact of the bounds, with an interior optimum on every benchmark where a grid was run.

6.2 Strengths and limitations

Strengths. Four properties distinguish the estimator from the finite-state logit perturbation it generalizes. It is stated at the level of probability measures, so the same argument produces a simplex perturbation in finite state spaces and a transport perturbation in continuous ones, the representation entering only through the map Ψ . The simplex version does not require strictly positive population coordinates, which the logit chart does, so it remains defined on the boundary of the simplex where several of the learned flows actually go. The forward recursion for the sensitivity makes the simulator cost linear in the horizon rather than quadratic, and the two-state experiment isolates exactly that effect: at $T = 2$ the logit estimator is the more accurate, and at $T = 5$ the ordering reverses, its policy error growing by a factor 35 against 2.5 for transport at the same budget. Finally, because the correction is an explicit term, its usefulness can be tested before committing to it; the alignment diagnostic costs two automatic-differentiation passes and predicts, without a training grid, whether REINFORCE will be directionally competitive.

Limitations of the method. The estimator is not uniformly better, and the reasons are worth stating plainly rather than as a caveat. Matching the simulator budget does not match the wall-clock cost, since the transport update carries the auxiliary recursion and its automatic-differentiation graph, so where REINFORCE is directionally correct it is the estimator to use. The correction also appears to be worth its variance only when the auxiliary sensitivity can be resolved: distribution planning allocates 1.2 auxiliary trajectories per free coordinate of a nine-dimensional population argument, the worst ratio in the suite, and is where transport loses most heavily to the logit estimator. That association runs the length of the suite, but it remains an association. The reallocation experiment that would test it directly on distribution planning is set out in Section 6.3 and has not been run; the one that has is a negative control on a benchmark whose ratio was already high (Appendix C.8). At the other extreme the portfolio benchmark spends 200 auxiliary trajectories on a one-dimensional argument and still leaves an optimization error of at least 4.97, so a high ratio is not sufficient either. The adaptive controller is the weakest component: it contracts the main scale below the smallest calibrated value on every finite-state benchmark and is the worst method on distribution planning, because at four replications per checkpoint the debiased bias estimate is frequently unresolved, and reading that as an absence of bias drives the controller towards a variance-minimizing corner.

Limitations of the evidence. Other conclusions are bounded by the design of the study rather than by the method. The continuous-state theory is developed only as far as objective-level convergence, so gradient-level convergence, the policy-gradient formula, the sensitivity estimator and the error decomposition remain open in the Gaussian-mixture case, and the continuous-state experiments are run with an estimator whose guarantees are an extrapolation. The replicated gradient diagnostics need an exact oracle and a directly parametrized policy, which only two benchmarks provide, and on the portfolio benchmark the unbiasedness test is inconclusive rather than passed, since at a signal-to-noise ratio of 0.02 per replication fifty replications cannot resolve a bias of the size the theory allows. The auxiliary error bound of Proposition 4 is the one prediction the saved artifacts cannot separate into its two terms. Finally, five seeds resolve the comparisons between estimator families by wide margins but not the ordering within the transport grid on the portfolio benchmark, where the seed dispersion exceeds the spread of the grid.

6.3 Extensions

An actor-critic form of the estimator. The most direct improvement the analysis suggests is variance reduction on the return, along the lines of the actor-critic algorithms developed for mean-field control in [23], where the critic is carried by a network taking moments of the population law as input. In (3.3) the perturbation score is multiplied by the full perturbed return and carries the factor λ^{-1} that the mean-square-error decomposition identifies as the dominant variance term at small scales; a deterministic baseline removes only a constant, whereas an advantage estimated from a learned critic would remove the part of the return uninformative about the current step. The critic is a natural object here, since the perturbed problem is an ordinary Markov decision process on the augmented space in which the randomized population coordinate is part of the state, as the augmented trajectory law (A.2) of Appendix A.3.3 makes explicit, so $V^\theta(t, x, m)$ is well defined. Two questions must be answered first. The critic is evaluated at the perturbed population argument, so its approximation error enters through the same λ^{-1} prefactor as the score, and the

bias it introduces has to be traded against the variance it removes in Theorem 5. The sensitivity recursion also has to be revisited, since the critic depends on θ through the population flow exactly as the reward does.

Beyond exchangeability. The construction rests on exchangeability: a single representative agent, one population law, one perturbation per step. Two relaxations are natural. In non-exchangeable models [17] the agents are coupled through a graph rather than a single mean field and the limit is described by a graphon, so the population argument becomes a function-valued object indexed by a continuum of vertex types. The framework does not obviously break there, since it requires only a finite-dimensional representation of the population argument and a probability-preserving randomization of it, and a graphon admits finite-dimensional approximations of exactly the kind Ψ is designed to accommodate; what must be redone is the sensitivity recursion, which currently propagates a single flow rather than a family indexed by type. Finitely many interacting populations, where the argument is a tuple of laws, is the natural first step and needs little additional theory.

The second relaxation is extended mean-field control, where the coefficients depend on the joint law of the state and the action. This is the more immediate of the two, because the estimator is indifferent to what the population argument represents: it needs a finite-dimensional chart, a randomization within it, and a forward recursion for its sensitivity, and the joint law of (X_t, α_t) has all three whenever the action space is finite or the joint law is summarized by moments. The difficulty is that the policy then influences the population argument twice, directly through the sampled actions and indirectly through the state flow, so the decomposition into a policy score and a population score acquires a third contribution the present derivation does not have.

Completing the continuous-state analysis. The gap identified above is the natural theoretical continuation: gradient-level convergence for the Gaussian-mixture perturbation, the corresponding policy-gradient representation, a model-free estimator of the sensitivity $D_\theta z_{K,t}^\theta$ of the representation coordinates, and the error decomposition. The finite-state proofs give the template, and the additional difficulty is the representation error δ_K^θ , which does not vanish with λ and must be carried through the differentiated estimates. A related and more open question is whether Ψ can be learned rather than fixed, since the construction requires only that the representation be finite-dimensional and that its coordinates be differentiable in θ .

Allocating the auxiliary budget. The equal-budget rule fixes the total number of simulated transitions per update but not their split between the main trajectories and the auxiliary sensitivity estimate, and that split is the leading candidate explanation for the underperformance of transport on the higher-dimensional simplex. The evidence for it is an association across the suite: the auxiliary sample size per free coordinate of the population argument ranges from 1.2 to 200, and distribution planning, the one benchmark where it falls below one sample per coordinate, is also the one finite-state benchmark on which transport loses to MF-REINFORCE by a wide margin (Appendix C.8). The experiment that would turn that association into a result is fully specified and needs no new machinery: on distribution planning, move the split from $(B_{\text{tr}}, n_{\text{tr}}) = (549, 11)$ to $(200, 360)$, which leaves the matched budget $T(B_{\text{tr}} + n_{\text{tr}}) = 2800$ unchanged and raises the auxiliary estimate to 40 trajectories per coordinate. The measured cost is a factor 1.9 in seconds per training step rather than the factor 33 that the increase in auxiliary samples would suggest, since the sensitivity recursion is applied to the batch at once. The control is already in hand: the same reallocation on targeted advertising, where the auxiliary estimate is resolved at the default split, moves the final objective by less than one standard deviation, as the hypothesis predicts. Beyond that single experiment lies the systematic version, the allocation as a function of the dimension of the population argument and of the horizon. Proposition 4 makes it cheaper than a training grid, since the auxiliary sample size enters through its $(n\eta^2)^{-1}$ term, which is measurable at a frozen policy.

Constraining the adaptive controller. Two modifications follow directly from the diagnosis above: a lower bound on λ at the smallest scale for which the perturbation-density score has been validated, and a balance target that accounts for the dimension of the simplex, since the score has $N - 1$ coordinates and scales as λ^{-1} . A third, less local change would be to resolve the bias estimate properly by accumulating it across checkpoints rather than re-estimating it from four replications each time.

A benchmark with a finite-dimensional law statistic. Both continuous-state benchmarks are driven by the population mean, so the representation is one-dimensional. The controlled noisy Kuramoto model [26, 4], in which oscillators on the circle are steered towards synchronization, is a natural next case: although the state is an angle and the law a distribution on the circle, the interaction enters only through the first Fourier mode ($\mathbb{E}[\cos X_t], \mathbb{E}[\sin X_t]$), so the population argument is again finite-dimensional, of dimension two rather than one, and it would test whether the auxiliary recursion degrades when the law statistic gains a coordinate.

6.4 Conclusion

This report set out to determine whether policy gradients for discrete-time mean-field control can be made fully model-free without the finite-state logit representation on which the existing construction depends. The answer developed here is a perturbation framework stated at the level of probability measures: the population argument is replaced by the pushforward of a random transport map, reducing to a probability-preserving mixture on the simplex in finite state spaces and to a random displacement read through a finite-dimensional representation in continuous ones. In both cases the gradient of the perturbed objective splits into the familiar policy score and a population score, both estimable from simulated trajectories without differentiating the transition kernel, the reward, or the population recursion, and a forward recursion for the sensitivity flow makes the simulator cost linear rather than quadratic in the horizon. The two settings are not at the same stage, and Table 1.1 states the difference plainly: on a finite state space the argument is complete, from the perturbation estimate through gradient-level convergence to the mean-square-error decomposition, whereas on a continuous state space it is established at the level of the objective and open at the level of the gradient. The continuous-state estimator is therefore built by analogy with the finite-state one and supported by measurement rather than by proof.

Empirically the picture is more interesting than a uniform improvement would have been. Across six benchmarks at matched simulator budgets no estimator dominates, and the disagreement is itself the main result: whether REINFORCE is directionally competitive is governed by how far the population-flow correction is from collinear with the term it already computes, a condition that a cheap evaluation-only diagnostic detects in advance and that a controlled ablation confirms. Where the correction is strongly non-collinear, population-aware estimators beat REINFORCE outright, by an order of magnitude on the two-state problem and by a wide margin on the portfolio benchmark. Where it vanishes, as on cybersecurity, REINFORCE is the better choice and the perturbation only adds variance. The diagnostic is silent about what happens between those poles, and the linear-quadratic benchmark is exactly that case: the alignment is above 0.98, so REINFORCE is directionally right and captures most of the available improvement, yet a small orthogonal component leaves it a visible residual gap, three quarters of which transport closes.

What remains is set out above: completing the gradient-level analysis in the continuous-state case, reducing the estimator variance through a critic, allocating the auxiliary budget in a principled way, and relaxing exchangeability. The internship continues until the end of October, and these are the directions in which the work is being taken.

Bibliography

- [1] Andrea Angiuli, Jean-Pierre Fouque, and Mathieu Laurière. “Unified reinforcement Q-learning for mean field game and control problems”. In: *Mathematics of Control, Signals, and Systems* 34.2 (2022), pp. 217–271.
- [2] René Carmona and François Delarue. *Probabilistic theory of mean field games with applications. I, volume 83 of Probability Theory and Stochastic Modelling*. 2018.
- [3] René Carmona, Mathieu Laurière, and Zongjun Tan. “Model-free mean-field reinforcement learning: mean-field MDP and mean-field Q-learning”. In: *The Annals of Applied Probability* 33.6B (2023), pp. 5334–5381.
- [4] Li Chen, Yucheng Wang, and Valeriia Zhidkova. “Optimal control for Kuramoto model: from many-particle Liouville equation to diffusive mean-field problem”. In: *arXiv preprint arXiv:2504.09425* (2025).
- [5] Ziheng Cheng et al. “Actor-Critic Learning for Extended Mean Field Control with Deterministic Policies”. In: *arXiv preprint arXiv:2607.11005* (2026).
- [6] Xiangyu Cui, Xun Li, and Duan Li. “Unified framework of mean-field formulations for optimal multi-period mean-variance portfolio selection”. In: *IEEE Transactions on Automatic Control* 59.7 (2014), pp. 1833–1844.
- [7] Markus Fischer and Giulia Livieri. “Continuous time mean-variance portfolio optimization through the mean field approach”. In: *ESAIM: Probability and Statistics* 20 (2016), pp. 30–44.
- [8] Noufel Frikha et al. “Actor-critic learning for mean-field control in continuous time”. In: *Journal of Machine Learning Research* 26.127 (2025), pp. 1–42.
- [9] Evan Greensmith, Peter L Bartlett, and Jonathan Baxter. “Variance reduction techniques for gradient estimates in reinforcement learning”. In: *Journal of Machine Learning Research* 5.Nov (2004), pp. 1471–1530.
- [10] Haotian Gu et al. “Dynamic programming principles for mean-field controls with learning”. In: *Operations Research* 71.4 (2023), pp. 1040–1054.
- [11] Olav Kallenberg. *Foundations of modern probability*. Springer, 1997.
- [12] Diederik P Kingma and Jimmy Ba. “Adam: A method for stochastic optimization”. In: *arXiv preprint arXiv:1412.6980* (2014).
- [13] Vassili N Kolokoltsov and Alain Bensoussan. “Mean-field-game model for botnet defense in cybersecurity”. In: *Applied Mathematics & Optimization* 74.3 (2016), pp. 669–692.
- [14] Pierre L’ecuyer. “A unified view of the IPA, SF, and LR gradient estimation techniques”. In: *Management Science* 36.11 (1990), pp. 1364–1383.
- [15] Mathieu Lauriere. “Numerical methods for mean field games and mean field type control”. In: *arXiv preprint arXiv:2106.06231* (2021).
- [16] David A Levin and Yuval Peres. *Markov chains and mixing times*. American Mathematical Society, 2026.
- [17] Samy Mekkaoui and Huy  n Pham. *Non-Exchangeable Mean Field Markov Decision Processes with common noise : from Bellman equation to quantitative propagation of chaos*. 2026. arXiv: 2603.00009 [math.OC]. URL: <https://arxiv.org/abs/2603.00009>.
- [18] Samy Mekkaoui, Huy  n Pham, and Xavier Warin. “Learning operators on labelled conditional distributions with applications to mean field control of non exchangeable systems”. In: *arXiv preprint arXiv:2603.21683* (2026).
- [19] Matthieu Meunier, Huy  n Pham, and Christoph Reisinger. “Model-free policy gradient for discrete-time mean-field control”. In: *arXiv preprint arXiv:2601.11217* (2026).
- [20] Washim Uddin Mondal et al. “On the approximation of cooperative heterogeneous multi-agent reinforcement learning (MARL) using mean field control (MFC)”. In: *Journal of Machine Learning Research* 23.129 (2022), pp. 1–46.

- [21] Médéric Motte. “Mathematical models for large populations, behavioral economics, and targeted advertising”. PhD thesis. Université Paris Cité, 2021.
- [22] Médéric Motte and Huyên Pham. “Mean-field Markov decision processes with common noise and open-loop controls”. In: *The Annals of Applied Probability* 32.2 (2022), pp. 1421–1458.
- [23] Huyên Pham and Xavier Warin. “Actor-critic learning algorithms for mean-field control with moment neural networks”. In: *Methodology and Computing in Applied Probability* 27.1 (2025), p. 13.
- [24] Huyên Pham and Xiaoli Wei. “Bellman equation and viscosity solutions for mean-field stochastic control problem”. In: *ESAIM: Control, Optimisation and Calculus of Variations* 24.1 (2018), pp. 437–461.
- [25] Martin L Puterman. *Markov decision processes: discrete stochastic dynamic programming*. John Wiley & Sons, 2014.
- [26] Carlo Sinigaglia, Francesco Braghin, and Spring Berman. “Optimal control of velocity and nonlocal interactions in the mean-field Kuramoto model”. In: *2022 American Control Conference (ACC)*. IEEE, 2022, pp. 290–295.
- [27] Richard S Sutton, Andrew G Barto, and Andrew Barto. *Reinforcement learning: An introduction*. Vol. 1. 1. MIT press Cambridge, 1998.
- [28] Richard S Sutton et al. “Policy gradient methods for reinforcement learning with function approximation”. In: *Advances in neural information processing systems* 12 (1999).
- [29] Ronald J Williams. “Simple statistical gradient-following algorithms for connectionist reinforcement learning”. In: *Machine learning* 8.3 (1992), pp. 229–256.
- [30] Yaodong Yang et al. “Mean field multi-agent reinforcement learning”. In: *International conference on machine learning*. PMLR, 2018, pp. 5571–5580.

Appendix A

Technical Proofs

This appendix collects the technical arguments that support the convergence and estimator results used in Chapter 3. They are kept here to preserve the flow of the main text while keeping the derivations available.

A.1 Canonical Construction

We work on a probability space rich enough to support the following independent random variables:

1. an initial variable $\xi \sim \mu_0$;
2. transition noises $(\epsilon_t)_{t=1}^T$ with common law $\rho \in \mathcal{P}(E)$;
3. policy noises $(\tilde{\epsilon}_t)_{t=0}^{T-1}$ with common law $\tilde{\rho} \in \mathcal{P}(\tilde{E})$;
4. perturbation variables $(f_t)_{t=0}^T$ with common law $\nu \in \mathcal{P}(F)$.

By the representation theorem for probability kernels of Kallenberg [11], the transition kernel can be represented by measurable maps $\mathcal{X} \times \mathcal{P}(\mathcal{X}) \times A \times E \ni (x, m, a, e) \mapsto F_t(x, m, a, e)$ such that $F_t(x, m, a, \cdot) \# \rho = P_t(\cdot \mid x, m, a)$. Similarly, each randomized policy can be represented by measurable maps $\mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \tilde{E} \ni (x, m, \tilde{e}) \mapsto \tilde{F}_t^\theta(x, m, \tilde{e})$ such that $\tilde{F}_t^\theta(x, m, \cdot) \# \tilde{\rho} = \pi_t^\theta(\cdot \mid x, m)$.

Remark 1 (Canonical construction of the problem). *The preceding variables can be realized on the canonical product space $\Omega = \mathcal{X} \times E^T \times \tilde{E}^T \times F^{T+1}$, endowed with its Borel σ -algebra and with the product measure*

$$\mathbb{P}(d\omega) = \mu_0(dx_0)\rho^{\otimes T}(d\epsilon_{1:T})\tilde{\rho}^{\otimes T}(d\tilde{\epsilon}_{0:T-1})\nu^{\otimes(T+1)}(df_{0:T}).$$

For a fixed policy π^θ , the unperturbed canonical process is defined by

$$\begin{cases} X_0^\theta(\omega) = x_0, \\ \alpha_t^\theta(\omega) = \tilde{F}_t^\theta(X_t^\theta(\omega), \mathbb{P}_{X_t^\theta}, \tilde{\epsilon}_t(\omega)), \\ X_{t+1}^\theta(\omega) = F_t(X_t^\theta(\omega), \mathbb{P}_{X_t^\theta}, \alpha_t^\theta(\omega), \epsilon_{t+1}(\omega)), \end{cases} \quad t = 0, \dots, T-1.$$

By construction, it follows that

$$\begin{cases} X_0^\theta \sim \mu_0, \\ \mathcal{L}(\alpha_t^\theta \mid X_t^\theta, \mathbb{P}_{X_t^\theta}) = \pi_t^\theta(\cdot \mid X_t^\theta, \mathbb{P}_{X_t^\theta}), \\ \mathcal{L}(X_{t+1}^\theta \mid X_t^\theta, \mathbb{P}_{X_t^\theta}, \alpha_t^\theta) = P_t(\cdot \mid X_t^\theta, \mathbb{P}_{X_t^\theta}, \alpha_t^\theta). \end{cases}$$

A.2 Discrete-State Space Proofs

A.2.1 Perturbation Estimate

We prove Theorem 1. For every ω , $M_t^{\lambda, \theta}(\omega) - \mu_t^\theta = \lambda(q_t(\omega) - \mu_t^\theta)$, hence $\|M_t^{\lambda, \theta}(\omega) - \mu_t^\theta\|_1 = \lambda\|q_t(\omega) - \mu_t^\theta\|_1 \leq 2\lambda$ because both q_t and μ_t^θ belong to Δ_N . The total variation estimate follows from $d_{\text{TV}}(p, p') = \frac{1}{2}\|p - p'\|_1$ on a finite space [16].

For the Wasserstein estimate, let γ be any coupling of $M_t^{\lambda,\theta}$ and μ_t^θ . Since $d_{\mathcal{X}}(i, j) \leq \text{diam}(\mathcal{X})$ and $d_{\mathcal{X}}(i, i) = 0$, we have $\sum_{i,j \in \mathcal{X}} d_{\mathcal{X}}(i, j) \gamma(i, j) \leq \text{diam}(\mathcal{X}) \sum_{i \neq j} \gamma(i, j)$. Taking the infimum over all couplings:

$$W_1(M_t^{\lambda,\theta}(\omega), \mu_t^\theta) \leq \text{diam}(\mathcal{X}) d_{\text{TV}}(M_t^{\lambda,\theta}, \mu_t^\theta).$$

Thus $W_1(M_t^{\lambda,\theta}, \mu_t^\theta) \leq \text{diam}(\mathcal{X})\lambda$. The expectation estimate follows immediately since the previous bounds hold almost surely.

A.2.2 Density and Score Computations

We first justify the density of the logistic perturbation. Work in the coordinates $(p_1, \dots, p_{N-1})$ on the simplex, with $p_N = 1 - \sum_{i=1}^{N-1} p_i$. The inverse of the logistic map is $z_i = \log(p_i/p_N)$. Its Jacobian satisfies

$$\frac{\partial z_i}{\partial p_j} = \frac{\mathbf{1}_{\{i=j\}}}{p_i} + \frac{1}{p_N}, \quad i, j = 1, \dots, N-1.$$

Hence,

$$\begin{aligned} |\det D\varphi^{-1}(p)| &= \det \left(\text{diag} \left(\frac{1}{p_1}, \dots, \frac{1}{p_{N-1}} \right) + \frac{1}{p_N} \mathbf{1}\mathbf{1}^\top \right) \\ &= \prod_{i=1}^{N-1} \frac{1}{p_i} \left(1 + \frac{1}{p_N} \sum_{i=1}^{N-1} p_i \right) = \frac{1}{\prod_{i=1}^N p_i}. \end{aligned}$$

The change-of-variables formula therefore gives

$$f_q(p) = \rho \left(\log \frac{p_1}{p_N}, \dots, \log \frac{p_{N-1}}{p_N} \right) \frac{1}{\prod_{i=1}^N p_i}.$$

For the density of $M_t^{\lambda,\theta}$, the affine inverse is

$$p_i^{\lambda,\theta}(m) = \frac{m_i - (1-\lambda)\mu_t^\theta(i)}{\lambda}, \quad i = 1, \dots, N,$$

with only the first $N-1$ coordinates independent. The Jacobian determinant of this inverse map in simplex coordinates is $\lambda^{-(N-1)}$, so another change of variables gives the density stated in the main text.

It remains to compute the score. Fix m in the interior of the support and write $p_i = p_i^{\lambda,\theta}(m)$, $z_i = \log(p_i/p_N)$, and $a_i(z) = \partial_{z_i} \log \rho(z)$. At fixed m ,

$$\partial_\theta p_i = -\frac{1-\lambda}{\lambda} \partial_\theta \mu_t^\theta(i), \quad i = 1, \dots, N.$$

Since $\sum_i \partial_\theta \mu_t^\theta(i) = 0$, the last coordinate can be eliminated by

$$\partial_\theta \mu_t^\theta(N) = -\sum_{k=1}^{N-1} \partial_\theta \mu_t^\theta(k).$$

Differentiating the logarithm of the density gives

$$\begin{aligned} \partial_\theta \log f_t^{\lambda,\theta}(m) &= \sum_{i=1}^{N-1} a_i(z) \left(\frac{\partial_\theta p_i}{p_i} - \frac{\partial_\theta p_N}{p_N} \right) - \sum_{i=1}^N \frac{\partial_\theta p_i}{p_i} \\ &= -\frac{1-\lambda}{\lambda} \sum_{k=1}^{N-1} \left(\frac{a_k(z)}{p_k} + \frac{\sum_{i=1}^{N-1} a_i(z)}{p_N} - \frac{1}{p_k} + \frac{1}{p_N} \right) \partial_\theta \mu_t^\theta(k). \end{aligned}$$

This is exactly the expression of Corollary 1.

A.2.3 Objective Convergence

We prove the stability of the perturbed state marginal. The estimate is true at $t = 0$ because $\nu_0^{\lambda, \theta} = \mu_0^\theta = \mu_0$. Assume it holds at time t . For any $\psi : \mathcal{X} \rightarrow [-1, 1]$, add and subtract the same kernel evaluated at X_t^θ and $M_t^{\lambda, \theta}$:

$$\begin{aligned} & \left| \sum_j \psi(j) (\nu_{t+1}^{\lambda, \theta}(j) - \mu_{t+1}^\theta(j)) \right| \\ & \leq \left| \sum_i (\nu_t^{\lambda, \theta}(i) - \mu_t^\theta(i)) \sum_j \psi(j) \bar{K}_t^{\lambda, \theta}(j | i) \right| + 2L_K \mathbb{E} [d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^\theta)], \end{aligned}$$

where $\bar{K}_t^{\lambda, \theta}(j | i) = \mathbb{E}[K_t^\theta(j | i, M_t^{\lambda, \theta})]$. Since $\bar{K}_t^{\lambda, \theta}$ is Markov, the first term is bounded by $2d_{\text{TV}}(\nu_t^{\lambda, \theta}, \mu_t^\theta)$. Taking the supremum over ψ yields

$$d_{\text{TV}}(\nu_{t+1}^{\lambda, \theta}, \mu_{t+1}^\theta) \leq d_{\text{TV}}(\nu_t^{\lambda, \theta}, \mu_t^\theta) + L_K \mathbb{E} [d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^\theta)].$$

The perturbation estimate gives $\mathbb{E}[d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^\theta)] \leq \lambda$, so induction gives

$$d_{\text{TV}}(\nu_t^{\lambda, \theta}, \mu_t^\theta) \leq L_K \lambda t.$$

For the objective, write

$$\begin{aligned} J(\theta) &= \sum_{t=0}^{T-1} \sum_i \mu_t^\theta(i) R_t^\theta(i, \mu_t^\theta) + \sum_i \mu_T^\theta(i) g(i, \mu_T^\theta), \\ J^\lambda(\theta) &= \sum_{t=0}^{T-1} \mathbb{E}[R_t^\theta(X_t^{\lambda, \theta}, M_t^{\lambda, \theta})] + \mathbb{E}[g(X_T^{\lambda, \theta}, M_T^{\lambda, \theta})]. \end{aligned}$$

At a running time t , independence of $M_t^{\lambda, \theta}$ and $X_t^{\lambda, \theta}$ gives the decomposition

$$\begin{aligned} & \mathbb{E}[R_t^\theta(X_t^{\lambda, \theta}, M_t^{\lambda, \theta})] - \sum_i \mu_t^\theta(i) R_t^\theta(i, \mu_t^\theta) \\ &= \sum_i (\nu_t^{\lambda, \theta}(i) - \mu_t^\theta(i)) \mathbb{E}[R_t^\theta(i, M_t^{\lambda, \theta})] + \sum_i \mu_t^\theta(i) \mathbb{E}[R_t^\theta(i, M_t^{\lambda, \theta}) - R_t^\theta(i, \mu_t^\theta)]. \end{aligned}$$

The boundedness and Lipschitz assumptions imply

$$\left| \mathbb{E}[R_t^\theta(X_t^{\lambda, \theta}, M_t^{\lambda, \theta})] - \sum_i \mu_t^\theta(i) R_t^\theta(i, \mu_t^\theta) \right| \leq 2\bar{R}L_K \lambda t + L_R \lambda.$$

The terminal reward is handled in the same way:

$$\left| \mathbb{E}[g(X_T^{\lambda, \theta}, M_T^{\lambda, \theta})] - \sum_i \mu_T^\theta(i) g(i, \mu_T^\theta) \right| \leq 2\bar{g}L_K \lambda T + L_g \lambda.$$

Summing over the horizon gives

$$|J^\lambda(\theta) - J(\theta)| \leq \lambda \left(\bar{R}L_K T(T-1) + L_R T + 2\bar{g}L_K T + L_g \right).$$

This proves the announced $C_T \lambda$ bound, uniformly in θ whenever the constants are uniform.

A.2.4 Consistency of Optimizers

Let $\varepsilon_\lambda = \sup_{\theta \in \Theta} |J^\lambda(\theta) - J(\theta)|$. If $\theta^* \in \operatorname{argmax}_{\Theta} J$, the optimality of θ_λ^* for J^λ gives

$$J(\theta^*) \leq J^\lambda(\theta^*) + \varepsilon_\lambda \leq J^\lambda(\theta_\lambda^*) + \varepsilon_\lambda \leq J(\theta_\lambda^*) + 2\varepsilon_\lambda.$$

Along a sequence $\lambda_n \downarrow 0$ such that $\theta_{\lambda_n}^* \rightarrow \bar{\theta}$, continuity of J gives $J(\theta^*) \leq J(\bar{\theta})$. Since θ^* is a maximizer, $\bar{\theta}$ is also a maximizer. If the maximizer is unique, every convergent subsequence has the same limit θ^* , and compactness of Θ gives convergence of the full family.

A.2.5 Gradient-Level Convergence

We first prove the stability of the differentiated state marginal. Since the initial law is independent of θ , $E_0^{\lambda,\theta}[h] = D_0^\theta[h] = 0$. Differentiating the two marginal recursions gives

$$\begin{aligned} D_{t+1}^\theta[h](j) &= \sum_i D_t^\theta[h](i) K_t^\theta(j | i, \mu_t^\theta) + \sum_i \mu_t^\theta(i) \partial_h K_t^\theta(j | i, \mu_t^\theta) \\ &\quad + \sum_i \mu_t^\theta(i) D_m K_t^\theta(j | i, \mu_t^\theta) [D_t^\theta[h]], \\ E_{t+1}^{\lambda,\theta}[h](j) &= \sum_i E_t^{\lambda,\theta}[h](i) \bar{K}_t^{\lambda,\theta}(j | i) + \sum_i \nu_t^{\lambda,\theta}(i) \mathbb{E}[\partial_h K_t^\theta(j | i, M_t^{\lambda,\theta})] \\ &\quad + \sum_i \nu_t^{\lambda,\theta}(i) \mathbb{E}[D_m K_t^\theta(j | i, M_t^{\lambda,\theta})] [(1 - \lambda) D_t^\theta[h]]. \end{aligned}$$

The factor $(1 - \lambda) D_t^\theta[h]$ comes from $\partial_h M_t^{\lambda,\theta} = (1 - \lambda) D_t^\theta[h]$. Add and subtract the unperturbed terms in the second recursion. If $\delta_t = \|E_t^{\lambda,\theta}[h] - D_t^\theta[h]\|_1$, the Markov property of $\bar{K}_t^{\lambda,\theta}$ controls the transported derivative error by δ_t , while the remaining differences are controlled by the state-marginal estimate, the perturbation estimate, and Assumption 2. More explicitly,

$$\begin{aligned} \delta_{t+1} &\leq \delta_t + C \|D_t^\theta[h]\|_1 \mathbb{E}[d_{\text{TV}}(M_t^{\lambda,\theta}, \mu_t^\theta)] + C d_{\text{TV}}(\nu_t^{\lambda,\theta}, \mu_t^\theta) \\ &\quad + C \mathbb{E}[d_{\text{TV}}(M_t^{\lambda,\theta}, \mu_t^\theta)] + C \lambda \|D_t^\theta[h]\|_1 \leq \delta_t + C \lambda. \end{aligned}$$

The horizon is finite and the derivatives $D_t^\theta[h]$ are uniformly bounded, so induction gives

$$\|E_t^{\lambda,\theta}[h] - D_t^\theta[h]\|_1 \leq C_T^D \lambda.$$

It remains to compare the directional derivatives of J^λ and J . They are

$$\begin{aligned} \partial_h J(\theta) &= \sum_{t=0}^{T-1} \sum_i \mu_t^\theta(i) \partial_h R_t^\theta(i, \mu_t^\theta) + \sum_{t=0}^{T-1} \sum_i \mu_t^\theta(i) D_m R_t^\theta(i, \mu_t^\theta) [D_t^\theta[h]] \\ &\quad + \sum_i D_T^\theta[h](i) g(i, \mu_T^\theta) + \sum_i \mu_T^\theta(i) D_m g(i, \mu_T^\theta) [D_T^\theta[h]], \\ \partial_h J^\lambda(\theta) &= \sum_{t=0}^{T-1} \sum_i E_t^{\lambda,\theta}[h](i) \mathbb{E}[R_t^\theta(i, M_t^{\lambda,\theta})] + \sum_{t=0}^{T-1} \sum_i \nu_t^{\lambda,\theta}(i) \mathbb{E}[\partial_h R_t^\theta(i, M_t^{\lambda,\theta})] \\ &\quad + \sum_{t=0}^{T-1} \sum_i \nu_t^{\lambda,\theta}(i) \mathbb{E}[D_m R_t^\theta(i, M_t^{\lambda,\theta})] [(1 - \lambda) D_t^\theta[h]] \\ &\quad + \sum_i E_T^{\lambda,\theta}[h](i) \mathbb{E}[g(i, M_T^{\lambda,\theta})] + \sum_i \nu_T^{\lambda,\theta}(i) \mathbb{E}[D_m g(i, M_T^{\lambda,\theta})] [(1 - \lambda) D_T^\theta[h]]. \end{aligned}$$

Each running contribution differs by $O(\lambda)$: the terms involving $E_t^{\lambda,\theta}[h] - D_t^\theta[h]$ use the differentiated stability bound, the terms involving $\nu_t^{\lambda,\theta} - \mu_t^\theta$ use the state-marginal bound, and the population-argument differences use the Lipschitz regularity together with $\mathbb{E}[d_{\text{TV}}(M_t^{\lambda,\theta}, \mu_t^\theta)] \leq \lambda$. The terminal terms are identical with g in place of R_t^θ . Therefore,

$$|\partial_h J^\lambda(\theta) - \partial_h J(\theta)| \leq C_T^\nabla \lambda.$$

Taking the supremum over all unit directions h gives $\|\nabla_\theta J^\lambda(\theta) - \nabla_\theta J(\theta)\| \leq C_T^\nabla \lambda$.

A.2.6 Policy-Gradient Formula

We prove Theorem 4. Let L_θ^λ be the density of the complete perturbed trajectory Γ^λ . The only direct dependence of L_θ^λ on θ comes from the policy densities $p_t^\theta(a_t | x_t, m_t)$ and from the perturbation densities

$f_t^{\lambda, \theta}(m_t)$; the transition kernels have no direct θ -dependence once the sampled population arguments are fixed. Therefore

$$\nabla_{\theta} \log L_{\theta}^{\lambda}(\gamma) = \sum_{t=0}^{T-1} \nabla_{\theta} \log p_t^{\theta}(a_t \mid x_t, m_t) + \sum_{t=0}^T \nabla_{\theta} \log f_t^{\lambda, \theta}(m_t) = S_{\theta}^{\lambda}(\gamma).$$

Differentiating under the integral sign and using $\nabla_{\theta} L_{\theta}^{\lambda} = L_{\theta}^{\lambda} \nabla_{\theta} \log L_{\theta}^{\lambda}$ yields

$$\begin{aligned} \nabla_{\theta} J^{\lambda}(\theta) &= \nabla_{\theta} \int \mathcal{R}^{\lambda}(\gamma) L_{\theta}^{\lambda}(\gamma) d\gamma = \int \mathcal{R}^{\lambda}(\gamma) \nabla_{\theta} L_{\theta}^{\lambda}(\gamma) d\gamma \\ &= \int \mathcal{R}^{\lambda}(\gamma) S_{\theta}^{\lambda}(\gamma) L_{\theta}^{\lambda}(\gamma) d\gamma = \mathbb{E}[\mathcal{R}_{\theta}^{\lambda} S_{\theta}^{\lambda}]. \end{aligned}$$

For the baseline form of Corollary 3: Since L_{θ}^{λ} is a probability density, differentiating $\int L_{\theta}^{\lambda}(\gamma) d\gamma = 1$ gives $\mathbb{E}[S_{\theta}^{\lambda}] = 0$. Hence, for any deterministic b ,

$$\mathbb{E}[\mathcal{R}_{\theta}^{\lambda} S_{\theta}^{\lambda}] = \mathbb{E}[(\mathcal{R}_{\theta}^{\lambda} - b) S_{\theta}^{\lambda}] + b \mathbb{E}[S_{\theta}^{\lambda}] = \mathbb{E}[(\mathcal{R}_{\theta}^{\lambda} - b) S_{\theta}^{\lambda}].$$

A.2.7 The Discrete Transport Estimator

This section gives in full the construction summarized in Section 3.3.5.

Estimating the population-flow sensitivity. The policy-gradient formula from Theorem 4 still contains the population-flow sensitivity $D_t^{\theta}(k) := \nabla_{\theta} \mu_t^{\theta}(k)$, for $t = 0, \dots, T$ and $k = 1, \dots, N-1$. This quantity is not directly observed from a single trajectory. Moreover, the usual likelihood-ratio identity does not provide an exact model-free estimator in the mean-field setting, because the transition kernel and the policy depend on the whole population law μ_t^{θ} , which itself depends on θ .

We therefore construct an auxiliary estimator of D_t^{θ} using the same perturbation idea as above. Since $\sum_{i=1}^N \mu_t^{\theta}(i) = 1$, we have $\sum_{i=1}^N D_t^{\theta}(i) = 0$ and it is enough to estimate $D_t^{\theta}(1), \dots, D_t^{\theta}(N-1)$ and then recover $D_t^{\theta}(N)$.

Let $\eta \in (0, 1)$ be an auxiliary perturbation parameter, possibly different from the perturbation λ used in the policy-gradient estimator. For $s = 0, \dots, T$, let $\bar{M}_s^{\eta, \theta} := (1 - \eta) \mu_s^{\theta} + \eta \bar{q}_s$, where $\bar{q}_s \in \Delta_N^{\circ}$ is generated by the same logistic parametrization as before. The auxiliary perturbed process is

$$\begin{cases} \bar{X}_0^{\eta, \theta} \sim \mu_0, \\ \bar{\alpha}_s^{\eta, \theta} \sim \pi_s^{\theta}(\cdot \mid \bar{X}_s^{\eta, \theta}, \bar{M}_s^{\eta, \theta}), \\ \bar{X}_{s+1}^{\eta, \theta} \sim P_s(\cdot \mid \bar{X}_s^{\eta, \theta}, \bar{M}_s^{\eta, \theta}, \bar{\alpha}_s^{\eta, \theta}), \end{cases} \quad s = 0, \dots, T-1.$$

The corresponding policy-score term is $\Psi_s^{\eta, \theta} := \nabla_{\theta} \log p_s^{\theta}(\bar{\alpha}_s^{\eta, \theta} \mid \bar{X}_s^{\eta, \theta}, \bar{M}_s^{\eta, \theta})$. The perturbation-density score at time s is

$$-\frac{1 - \eta}{\eta} \sum_{\ell=1}^{N-1} H_{\ell}(\bar{q}_s) D_s^{\theta}(\ell).$$

For fixed $t \in \{1, \dots, T\}$ and $k \in \{1, \dots, N-1\}$, consider the auxiliary objective $\mu_{t,k}^{\eta}(\theta) := \mathbb{E}[\mathbf{1}_{\{\bar{X}_t^{\eta, \theta} = k\}}]$. Applying the perturbed likelihood ratio formula to the terminal reward $\mathbf{1}_{\{\bar{X}_t^{\eta, \theta} = k\}}$ and using causality, gives the formal identity

$$\nabla_{\theta} \mu_{t,k}^{\eta}(\theta) = \mathbb{E} \left[\mathbf{1}_{\{\bar{X}_t^{\eta, \theta} = k\}} \sum_{s=0}^{t-1} \Xi_s^{\eta, \theta} \right].$$

Here $\Xi_s^{\eta, \theta}$ denotes the sum of the policy score and the perturbation-density score at time s . The terms with $s \geq t$ vanish because the event $\{\bar{X}_t^{\eta, \theta} = k\}$ does not depend on future actions or future perturbations.

Since $\bar{M}_s^{\eta,\theta}$ converges to μ_s^θ as $\eta \downarrow 0$, we expect $\mu_{t,k}^\eta(\theta) \xrightarrow{\eta \downarrow 0} \mu_t^\theta(k)$. Under a gradient-level convergence assumption, this yields

$$D_t^\theta(k) = \lim_{\eta \downarrow 0} \mathbb{E} \left[\mathbf{1}_{\{\bar{X}_t^{\eta,\theta}=k\}} \sum_{s=0}^{t-1} \left(\Psi_s^{\eta,\theta} - \frac{1-\eta}{\eta} \sum_{\ell=1}^{N-1} H_\ell(\bar{q}_s) D_s^\theta(\ell) \right) \right].$$

This relation is triangular in time: the sensitivity D_t^θ depends only on $D_0^\theta, \dots, D_{t-1}^\theta$. Since the initial law μ_0 is independent of θ , we have $D_0^\theta(k) = 0$ for $k = 1, \dots, N$. Therefore, the sensitivities can be estimated recursively forward in time.

Let $n \geq 1$ be the number of auxiliary trajectories used to estimate the population-flow sensitivity. For each $r = 1, \dots, n$, simulate an independent auxiliary perturbed trajectory $(\bar{q}_s^{(r)}, \bar{M}_s^{\eta,\theta,(r)}, \bar{X}_s^{\eta,\theta,(r)}, \bar{\alpha}_s^{\eta,\theta,(r)})_{s=0}^T$. Define $\psi_s^{\eta,\theta,(r)} := \nabla_\theta \log p_s^\theta(\bar{\alpha}_s^{\eta,\theta,(r)} \mid \bar{X}_s^{\eta,\theta,(r)}, \bar{M}_s^{\eta,\theta,(r)})$. We initialize $\hat{D}_0^{\eta,n,\theta}(k) = 0$ for $k = 1, \dots, N$. Then, for $t = 1, \dots, T$ and $k = 1, \dots, N-1$, define recursively

$$\hat{D}_t^{\eta,n,\theta}(k) := \frac{1}{n} \sum_{r=1}^n \mathbf{1}_{\{\bar{X}_t^{\eta,\theta,(r)}=k\}} \left[\sum_{s=0}^{t-1} \psi_s^{\eta,\theta,(r)} - \frac{1-\eta}{\eta} \sum_{s=0}^{t-1} \sum_{\ell=1}^{N-1} H_\ell(\bar{q}_s^{(r)}) \hat{D}_s^{\eta,n,\theta}(\ell) \right].$$

Finally, set $\hat{D}_t^{\eta,n,\theta}(N) := -\sum_{k=1}^{N-1} \hat{D}_t^{\eta,n,\theta}(k)$.

Remark 2 (Empirical population law). *In a fully simulation-based implementation, in order to construct $\bar{M}_s^{\eta,\theta} = (1-\eta)\mu_s^\theta + \eta\bar{q}_s$, μ_s^θ may be replaced by an empirical population law*

$$\hat{\mu}_s^\theta(i) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}_{\{X_s^{\theta,(m)}=i\}}, \quad i = 1, \dots, N,$$

computed from a batch of independent unperturbed trajectories. This introduces an additional sampling error, which can be treated separately in the final mean-square error decomposition.

The Monte Carlo estimator. We now combine the perturbed policy-gradient formula with the auxiliary estimator of the population-flow sensitivity to obtain a fully sample-based estimator of $\nabla_\theta J^\lambda(\theta)$. The only non-directly observable term in the policy-gradient formula is $(D_t^\theta(k))_{0 \leq t \leq T, 1 \leq k \leq N-1}$, which we replace by the auxiliary estimator $\hat{D}_t^{\eta,n,\theta}(k)$.

Let $B \geq 1$ be the number of trajectories used to estimate the perturbed policy gradient, and assume that the auxiliary batch used to compute $\hat{D}_t^{\eta,n,\theta}(k)$ is independent of the batch used below. For each $b = 1, \dots, B$, simulate an independent perturbed trajectory $(q_t^{(b)}, M_t^{\lambda,\theta,(b)}, X_t^{\lambda,\theta,(b)}, \alpha_t^{\lambda,\theta,(b)})_{t=0}^T$ according to

$$\begin{cases} X_0^{\lambda,\theta,(b)} & \sim \mu_0, \\ M_t^{\lambda,\theta,(b)} & = (1-\lambda)\mu_t^\theta + \lambda q_t^{(b)}, \\ \alpha_t^{\lambda,\theta,(b)} & \sim \pi_t^\theta(\cdot \mid X_t^{\lambda,\theta,(b)}, M_t^{\lambda,\theta,(b)}), \\ X_{t+1}^{\lambda,\theta,(b)} & \sim P_t(\cdot \mid X_t^{\lambda,\theta,(b)}, M_t^{\lambda,\theta,(b)}, \alpha_t^{\lambda,\theta,(b)}). \end{cases}$$

The corresponding perturbed return is

$$\mathcal{R}_\theta^{\lambda,(b)} := \sum_{t=0}^{T-1} r_t(X_t^{\lambda,\theta,(b)}, M_t^{\lambda,\theta,(b)}, \alpha_t^{\lambda,\theta,(b)}) + g(X_T^{\lambda,\theta,(b)}, M_T^{\lambda,\theta,(b)}).$$

The policy-score term is

$$\mathcal{S}_{\text{pol}}^{\lambda,\theta,(b)} := \sum_{t=0}^{T-1} \nabla_\theta \log p_t^\theta(\alpha_t^{\lambda,\theta,(b)} \mid X_t^{\lambda,\theta,(b)}, M_t^{\lambda,\theta,(b)}).$$

It is useful to isolate the ideal estimator, in which the exact population-flow sensitivity is available. Define

$$\begin{aligned}\mathcal{S}_{\text{pert}}^{\lambda,\theta,(b)} &:= -\frac{1-\lambda}{\lambda} \sum_{t=0}^T \sum_{k=1}^{N-1} H_k(q_t^{(b)}) D_t^\theta(k), \\ \mathcal{S}^{\lambda,\theta,(b)} &:= \mathcal{S}_{\text{pol}}^{\lambda,\theta,(b)} + \mathcal{S}_{\text{pert}}^{\lambda,\theta,(b)}.\end{aligned}$$

The corresponding exact-sensitivity, or oracle, Monte Carlo estimator is

$$G_{\lambda,B}(\theta) := \frac{1}{B} \sum_{b=1}^B \left(\mathcal{R}_\theta^{\lambda,(b)} - b_0 \right) \mathcal{S}^{\lambda,\theta,(b)}.$$

By Theorem 4 and the baseline identity, $G_{\lambda,B}(\theta)$ is exactly unbiased for $\nabla_\theta J^\lambda(\theta)$. This estimator is not model-free, since it assumes access to the exact sensitivities $D_t^\theta(k)$, but it is a useful oracle reference: it isolates the effect of the transport perturbation from the additional error caused by estimating the mean-field derivative. The perturbation-score term is estimated by

$$\hat{\mathcal{S}}_{\text{pert}}^{\lambda,\eta,n,\theta,(b)} := -\frac{1-\lambda}{\lambda} \sum_{t=0}^T \sum_{k=1}^{N-1} H_k(q_t^{(b)}) \hat{D}_t^{\eta,n,\theta}(k).$$

Thus the estimated total score is $\hat{\mathcal{S}}^{\lambda,\eta,n,\theta,(b)} := \mathcal{S}_{\text{pol}}^{\lambda,\theta,(b)} + \hat{\mathcal{S}}_{\text{pert}}^{\lambda,\eta,n,\theta,(b)}$. For a deterministic baseline $b_0 \in \mathbb{R}$, define the Monte Carlo estimator

$$\hat{G}_{\lambda,\eta,B,n}(\theta) := \frac{1}{B} \sum_{b=1}^B \left(\mathcal{R}_\theta^{\lambda,(b)} - b_0 \right) \hat{\mathcal{S}}^{\lambda,\eta,n,\theta,(b)}.$$

The role of the baseline is only variance reduction: when the exact score is used it does not change the expectation, and in the plug-in estimator it only multiplies the same score approximation.

A.2.8 Bias and Mean-Square Error

To state the full decomposition, define the main-batch coefficients

$$A_{t,k}^{\lambda,B}(\theta) := \frac{1}{B} \sum_{b=1}^B \left(\mathcal{R}_\theta^{\lambda,(b)} - b_0 \right) H_k(q_t^{(b)}), \quad a_{t,k}^\lambda(\theta) := \mathbb{E}[A_{t,k}^{\lambda,1}(\theta)].$$

Then the plug-in estimator differs from the exact-sensitivity estimator by

$$\hat{G}_{\lambda,\eta,B,n}(\theta) - G_{\lambda,B}(\theta) = -\frac{1-\lambda}{\lambda} \sum_{t=0}^T \sum_{k=1}^{N-1} A_{t,k}^{\lambda,B}(\theta) \left(\hat{D}_t^{\eta,n,\theta}(k) - D_t^\theta(k) \right).$$

This identity makes explicit where the factor $1/\lambda$ enters the error.

We first prove Proposition 3. The exact unbiasedness follows from Theorem 4 and the baseline identity. The second estimate is the variance of an average of B independent oracle-score samples, bounded by the score moment assumption. The comparison with $\nabla_\theta J(\theta)$ then follows by adding and subtracting $\nabla_\theta J^\lambda(\theta)$ and using the gradient convergence estimate of Theorem 3.

Next, Proposition 4. The auxiliary identity targets the derivative of the η -perturbed marginal, so the deterministic bias is $O(\eta)$ by Theorem 3. For fixed previous sensitivities, the auxiliary estimator is an average of n independent likelihood-ratio variables whose score has second moment of order $1 + \eta^{-2}$, hence variance $O((n\eta^2)^{-1})$. The recursive use of $\hat{D}_s^{\eta,n,\theta}$ for $s < t$ propagates these errors only through a finite triangular system. An induction over t gives the stated bound.

Finally, Theorem 5. Let $\bar{G}_{\lambda,\eta,n}(\theta) := \mathbb{E}[\hat{G}_{\lambda,\eta,B,n}(\theta) \mid \hat{D}^{\eta,n,\theta}]$, where the expectation is only over the main batch. By independence of the main and auxiliary batches,

$$\bar{G}_{\lambda,\eta,n}(\theta) = \nabla_\theta J^\lambda(\theta) - \frac{1-\lambda}{\lambda} \sum_{t=0}^T \sum_{k=1}^{N-1} a_{t,k}^\lambda(\theta) \left(\hat{D}_t^{\eta,n,\theta}(k) - D_t^\theta(k) \right).$$

Combining this identity with Theorem 3 and Proposition 4 gives the bias bound.

For the mean-square error, use the conditional variance decomposition

$$\mathbb{E} \left[\left\| \hat{G}_{\lambda, \eta, B, n}(\theta) - \nabla_{\theta} J(\theta) \right\|^2 \right] = \mathbb{E} \left[\left\| \hat{G}_{\lambda, \eta, B, n}(\theta) - \bar{G}_{\lambda, \eta, n}(\theta) \right\|^2 \right] + \mathbb{E} \left[\left\| \bar{G}_{\lambda, \eta, n}(\theta) - \nabla_{\theta} J(\theta) \right\|^2 \right].$$

The first term is the variance of an average of B independent main-batch samples, and the score moment assumption bounds it by $C_T(1 + \lambda^{-2})/B$. The second term is bounded by the square of the perturbation bias plus the auxiliary plug-in error, which gives $C_T\lambda^2 + C_T\lambda^{-2}(\eta^2 + (n\eta^2)^{-1})$.

A.3 Continuous-State Space Proofs

A.3.1 Representation and Perturbation Estimate

We prove Theorem 6. By the triangle inequality,

$$\begin{aligned} W_1 \left(M_{K,t}^{\lambda, \theta}, \mu_t^{\theta} \right) &\leq W_1 \left(M_{K,t}^{\lambda, \theta}, \bar{\mu}_{K,t}^{\theta} \right) + W_1 \left(\bar{\mu}_{K,t}^{\theta}, \mu_t^{\theta} \right) \\ &= W_1 \left(\Gamma_K(z_{K,t}^{\theta} + \lambda U_t), \Gamma_K(z_{K,t}^{\theta}) \right) + \delta_{K,t}^{\theta}. \end{aligned}$$

Taking expectations and applying Assumption 4 gives

$$\mathbb{E} \left[W_1 \left(M_{K,t}^{\lambda, \theta}, \mu_t^{\theta} \right) \right] \leq C_{\Gamma, K} \lambda + \delta_{K,t}^{\theta}.$$

The second statement follows immediately.

A.3.2 Density and Score of the Randomized Mixture Coordinates

This section states and proves the two results summarized in Section 3.4.2, in the notation fixed there.

For $\lambda > 0$, define

$$\varrho_{K,t}^{\lambda, \theta}(z) := \frac{1}{\lambda^{q_K}} \rho \left(\frac{z - z_{K,t}^{\theta}}{\lambda} \right), \quad z \in \mathbb{R}^{q_K}. \quad (\text{A.1})$$

Proposition 5 (Density of the randomized mixture coordinates). *Let $\lambda > 0$ and suppose that U_t admits density ρ with respect to Lebesgue measure on $\mathbb{R}^{q_K}$. Then $Z_{K,t}^{\lambda, \theta}$ admits density $\varrho_{K,t}^{\lambda, \theta}$ given by (A.1). In particular, $\mathcal{L}(Z_{K,t}^{\lambda, \theta})(dz) = \varrho_{K,t}^{\lambda, \theta}(z)dz$.*

Proof. For fixed (t, K, λ, θ) , consider the affine map

$$\begin{aligned} \Phi_{K,t}^{\lambda, \theta} : \mathbb{R}^{q_K} &\longrightarrow \mathbb{R}^{q_K}, \\ u &\longmapsto z_{K,t}^{\theta} + \lambda u. \end{aligned}$$

Its inverse is

$$\left(\Phi_{K,t}^{\lambda, \theta} \right)^{-1}(z) = \frac{z - z_{K,t}^{\theta}}{\lambda},$$

and

$$\left| \det D \left(\Phi_{K,t}^{\lambda, \theta} \right)^{-1}(z) \right| = \lambda^{-q_K}.$$

The change-of-variables formula therefore gives

$$\begin{aligned} \varrho_{K,t}^{\lambda, \theta}(z) &= \rho \left(\left(\Phi_{K,t}^{\lambda, \theta} \right)^{-1}(z) \right) \left| \det D \left(\Phi_{K,t}^{\lambda, \theta} \right)^{-1}(z) \right| \\ &= \frac{1}{\lambda^{q_K}} \rho \left(\frac{z - z_{K,t}^{\theta}}{\lambda} \right). \end{aligned}$$

□

Let $\theta \in \Theta \subset \mathbb{R}^{d_\theta}$ and suppose $\theta \mapsto z_{K,t}^\theta$ differentiable, with Jacobian $D_\theta z_{K,t}^\theta \in \mathbb{R}^{q_K \times d_\theta}$.

Proposition 6 (Score of the randomized mixture coordinates). *Suppose that ρ is strictly positive and continuously differentiable on $\mathbb{R}^{q_K}$ and that $\theta \mapsto z_{K,t}^\theta$ is differentiable. Then, for every $z \in \mathbb{R}^{q_K}$,*

$$\nabla_\theta \log \varrho_{K,t}^{\lambda,\theta}(z) = -\frac{1}{\lambda} \left(D_\theta z_{K,t}^\theta \right)^\top \nabla \log \rho \left(\frac{z - z_{K,t}^\theta}{\lambda} \right).$$

Consequently, evaluated at the sampled randomized coordinate $Z_{K,t}^{\lambda,\theta} = z_{K,t}^\theta + \lambda U_t$, we obtain

$$\nabla_\theta \log \varrho_{K,t}^{\lambda,\theta} \left(Z_{K,t}^{\lambda,\theta} \right) = -\frac{1}{\lambda} \left(D_\theta z_{K,t}^\theta \right)^\top \nabla \log \rho(U_t).$$

Proof. From Proposition 5,

$$\log \varrho_{K,t}^{\lambda,\theta}(z) = -q_K \log \lambda + \log \rho \left(\frac{z - z_{K,t}^\theta}{\lambda} \right).$$

For fixed z , the first term does not depend on θ . By the chain rule,

$$\begin{aligned} \nabla_\theta \log \varrho_{K,t}^{\lambda,\theta}(z) &= \left[D_\theta \left(\frac{z - z_{K,t}^\theta}{\lambda} \right) \right]^\top \nabla \log \rho \left(\frac{z - z_{K,t}^\theta}{\lambda} \right) \\ &= -\frac{1}{\lambda} \left(D_\theta z_{K,t}^\theta \right)^\top \nabla \log \rho \left(\frac{z - z_{K,t}^\theta}{\lambda} \right). \end{aligned}$$

Evaluating at $z = Z_{K,t}^{\lambda,\theta} = z_{K,t}^\theta + \lambda U_t$ gives (3.5). □

Corollary 6 (Gaussian perturbation score). *Suppose that $U_t \sim \mathcal{N}(0, \Xi)$. Then*

$$\nabla_\theta \log \varrho_{K,t}^{\lambda,\theta} \left(Z_{K,t}^{\lambda,\theta} \right) = \frac{1}{\lambda} \left(D_\theta z_{K,t}^\theta \right)^\top \Xi^{-1} U_t.$$

In particular, for $U_t \sim \mathcal{N}(0, I_{q_K})$, we have

$$\nabla_\theta \log \varrho_{K,t}^{\lambda,\theta} \left(Z_{K,t}^{\lambda,\theta} \right) = \frac{1}{\lambda} \left(D_\theta z_{K,t}^\theta \right)^\top U_t.$$

A.3.3 Likelihood-Ratio Identities and the Augmented Formulation

The usual likelihood-ratio identities hold for the perturbation score. Define $S_{K,t}^{\lambda,\theta} := \nabla_\theta \log \varrho_{K,t}^{\lambda,\theta} \left(Z_{K,t}^{\lambda,\theta} \right)$. Then, under the usual integrability assumptions, $\mathbb{E} \left[S_{K,t}^{\lambda,\theta} \right] = 0$. Indeed,

$$\begin{aligned} \mathbb{E} \left[S_{K,t}^{\lambda,\theta} \right] &= \int_{\mathbb{R}^{q_K}} \nabla_\theta \log \varrho_{K,t}^{\lambda,\theta}(z) \varrho_{K,t}^{\lambda,\theta}(z) dz \\ &= \int_{\mathbb{R}^{q_K}} \nabla_\theta \varrho_{K,t}^{\lambda,\theta}(z) dz \\ &= \nabla_\theta \int_{\mathbb{R}^{q_K}} \varrho_{K,t}^{\lambda,\theta}(z) dz = 0. \end{aligned}$$

More generally, if $H : \mathbb{R}^{q_K} \rightarrow \mathbb{R}$ is sufficiently integrable and does not depend explicitly on θ , then

$$\nabla_\theta \mathbb{E} \left[H \left(Z_{K,t}^{\lambda,\theta} \right) \right] = \mathbb{E} \left[H \left(Z_{K,t}^{\lambda,\theta} \right) S_{K,t}^{\lambda,\theta} \right].$$

This identity is the formal mechanism by which derivatives of the population-dependent reward, policy and transition kernel would be replaced by a model-free score term.

For later use it is convenient to regard the randomized coordinate $Z_{K,t}^{\lambda,\theta}$ as an explicit component of the perturbed trajectory. Define

$$\begin{aligned}\tilde{r}_{K,t}(x, a, z) &:= r_t(x, \Gamma_K(z), a), \\ \tilde{g}_K(x, z) &:= g(x, \Gamma_K(z)), \\ \tilde{\pi}_{K,t}^\theta(da | x, z) &:= \pi_t^\theta(da | x, \Gamma_K(z)), \\ \tilde{P}_{K,t}(dx' | x, a, z) &:= P_t(dx' | x, a, \Gamma_K(z)).\end{aligned}$$

The perturbed objective may therefore be written entirely on the finite-dimensional augmented space $\mathbb{R}^d \times \mathbb{R}^{q_K} \times \mathcal{A}$. The corresponding trajectory law takes the form

$$\begin{aligned}\mathbb{P}_{K,\lambda}^\theta(dx_0, dz_0, da_0, \dots, dx_{T-1}, dz_{T-1}, da_{T-1}, dx_T, dz_T) \\ = \mu_0(dx_0) \prod_{t=0}^{T-1} \left[\varrho_{K,t}^{\lambda,\theta}(z_t) dz_t \tilde{\pi}_{K,t}^\theta(da_t | x_t, z_t) \tilde{P}_{K,t}(dx_{t+1} | x_t, a_t, z_t) \right] \varrho_{K,T}^{\lambda,\theta}(z_T) dz_T.\end{aligned}\quad (\text{A.2})$$

This representation makes the key structural point explicit: after augmenting the trajectory by the randomized finite-dimensional coordinates $(Z_{K,t}^{\lambda,\theta})_{t \in \mathcal{T}}$, the dependence of the trajectory density on θ occurs through only two families of terms, $\tilde{\pi}_{K,t}^\theta$ and $\varrho_{K,t}^{\lambda,\theta}$. In particular, the model transition kernel does not need to be differentiated. This is what suggests a model-free policy-gradient representation of the perturbed continuous-state problem, by analogy with the finite-state derivation of Section A.2.6; establishing it is one of the open points recorded in Table 1.1.

A.3.4 Convergence of the Perturbed Objective

We first record the averaged coefficients on which the whole argument runs.

Averaged controlled kernel and reward. For $\theta \in \Theta$, define the averaged controlled transition kernel

$$\mathcal{K}_t^\theta(dx' | x, m) := \int_{\mathcal{A}} P_t(dx' | x, a, m) \pi_t^\theta(da | x, m),$$

and the averaged running reward

$$\mathcal{R}_t^\theta(x, m) := \int_{\mathcal{A}} r_t(x, a, m) \pi_t^\theta(da | x, m).$$

The law μ_t^θ of the original controlled process therefore satisfies

$$\mu_{t+1}^\theta = \int_{\mathbb{R}^d} \mathcal{K}_t^\theta(\cdot | x, \mu_t^\theta) \mu_t^\theta(dx).$$

We denote by $\nu_{K,t}^{\lambda,\theta} := \mathcal{L}(X_t^{K,\lambda,\theta})$ the law of the state process driven by the randomized population arguments $(M_{K,t}^{\lambda,\theta})_{t \in \mathcal{T}}$.

Since U_t is independent of the state and action randomness up to time t , the randomized measure $M_{K,t}^{\lambda,\theta}$ is independent of $X_t^{K,\lambda,\theta}$. Consequently,

$$\nu_{K,t+1}^{\lambda,\theta} = \mathbb{E} \left[\mathcal{K}_t^\theta(\cdot | X_t^{K,\lambda,\theta}, M_{K,t}^{\lambda,\theta}) \right].$$

When $\lambda = 0$, we similarly denote $\nu_{K,t}^{0,\theta} = \mathcal{L}(X_t^{K,0,\theta})$, where the population argument is the deterministic measure $\bar{\mu}_{K,t}^\theta$.

Notice that, in general, $\nu_{K,t}^{0,\theta} \neq \bar{\mu}_{K,t}^\theta$. The latter is an approximation of the original population law μ_t^θ , whereas the former is the law of the state process obtained after using this approximation as an exogenous population argument.

We first record a general stability estimate that will be used both for the perturbation error and for the Gaussian-mixture representation error.

Let $(m_t)_{t \in \mathcal{T}}$ and $(\tilde{m}_t)_{t \in \mathcal{T}}$ be two possibly random population-argument flows, independent of the corresponding state noises at the current time. Let (Y_t) and $(\tilde{Y}_t)$ be controlled processes driven by the same policy parameter θ and by m_t and $\tilde{m}_t$, respectively. Denote $\eta_t = \mathcal{L}(Y_t)$, $\tilde{\eta}_t = \mathcal{L}(\tilde{Y}_t)$, and set $d_t := \mathbb{E}[W_1(m_t, \tilde{m}_t)]$.

Lemma 3 (Stability of the state marginal). *Under Assumption 5, if $\eta_0 = \tilde{\eta}_0 = \mu_0$, then*

$$W_1(\eta_t, \tilde{\eta}_t) \leq L_{\mathcal{K}, \mu} \sum_{s=0}^{t-1} L_{\mathcal{K}, x}^{t-1-s} d_s, \quad t = 1, \dots, T. \quad (\text{A.3})$$

The proof couples the two state processes optimally at each step and iterates the resulting Wasserstein recursion from $e_0 = 0$; it is given below.

For convenience, define

$$H_t(L) := \begin{cases} 0, & t = 0, \\ \sum_{j=0}^{t-1} L^j, & t \geq 1. \end{cases}$$

Thus, if $d_s \leq d$ uniformly over s , Lemma 3 gives $W_1(\eta_t, \tilde{\eta}_t) \leq L_{\mathcal{K}, \mu} H_t(L_{\mathcal{K}, x}) d$.

Recall that $M_{K,t}^{\lambda, \theta} = \Gamma_K(z_{K,t}^\theta + \lambda U_t)$ and $\bar{\mu}_{K,t}^\theta = \Gamma_K(z_{K,t}^\theta)$. Under Assumption 4,

$$\mathbb{E}[W_1(M_{K,t}^{\lambda, \theta}, \bar{\mu}_{K,t}^\theta)] \leq C_{\Gamma, K} \lambda. \quad (\text{A.4})$$

Applying Lemma 3 with $m_t = M_{K,t}^{\lambda, \theta}$, and $\tilde{m}_t = \bar{\mu}_{K,t}^\theta$, gives

$$W_1(\nu_{K,t}^{\lambda, \theta}, \nu_{K,t}^{0, \theta}) \leq L_{\mathcal{K}, \mu} H_t(L_{\mathcal{K}, x}) C_{\Gamma, K} \lambda. \quad (\text{A.5})$$

We can now control the objective.

Proposition 7 (Perturbation error). *Under Assumptions 4 and 5, for every fixed K , $\theta \in \Theta$, and $\lambda \in [0, \lambda_0]$,*

$$|J_K^\lambda(\theta) - J_K^0(\theta)| \leq C_T C_{\Gamma, K} \lambda, \quad (\text{A.6})$$

where

$$C_T := \sum_{t=0}^{T-1} [L_{\mathcal{R}, x} L_{\mathcal{K}, \mu} H_t(L_{\mathcal{K}, x}) + L_{\mathcal{R}, \mu}] + L_{g, x} L_{\mathcal{K}, \mu} H_T(L_{\mathcal{K}, x}) + L_{g, \mu}.$$

Consequently, for fixed K ,

$$J_K^\lambda(\theta) \xrightarrow{\lambda \downarrow 0} J_K^0(\theta).$$

The proof bounds each running and terminal reward difference by its state and population contributions using Assumption 5, then sums over time.

We now compare J_K^0 with the original objective J . Recall that $\delta_{K,t}^\theta = W_1(\bar{\mu}_{K,t}^\theta, \mu_t^\theta)$, and $\delta_K^\theta = \max_{t \in \mathcal{T}} \delta_{K,t}^\theta$. Applying Lemma 3 with $m_t = \bar{\mu}_{K,t}^\theta$ and $\tilde{m}_t = \mu_t^\theta$, we obtain

$$W_1(\nu_{K,t}^{0, \theta}, \mu_t^\theta) \leq L_{\mathcal{K}, \mu} H_t(L_{\mathcal{K}, x}) \delta_K^\theta.$$

Proposition 8 (Representation error). *Under Assumption 5,*

$$|J_K^0(\theta) - J(\theta)| \leq C_T \delta_K^\theta, \quad (\text{A.7})$$

where C_T is given by (3.9).

The argument is that of Proposition 7 with $M_{K,t}^{\lambda, \theta}$ replaced by $\bar{\mu}_{K,t}^\theta$ and $\bar{\mu}_{K,t}^\theta$ by μ_t^θ .

Proofs. We prove the stability estimate of Lemma 3. We proceed recursively. Since both processes have the same initial law, $W_1(\eta_0, \tilde{\eta}_0) = 0$.

Let $e_t := W_1(\eta_t, \tilde{\eta}_t)$. Choose a coupling $(Y_t, \tilde{Y}_t)$ such that $\mathbb{E}[|Y_t - \tilde{Y}_t|] = e_t$. Conditionally on $(Y_t, \tilde{Y}_t, m_t, \tilde{m}_t)$, couple the next states through an optimal coupling of the two controlled transition kernels. By (3.7),

$$\begin{aligned} e_{t+1} &\leq \mathbb{E} \left[W_1 \left(\mathcal{K}_t^\theta(\cdot \mid Y_t, m_t), \mathcal{K}_t^\theta(\cdot \mid \tilde{Y}_t, \tilde{m}_t) \right) \right] \\ &\leq L_{\mathcal{K},x} \mathbb{E} [|Y_t - \tilde{Y}_t|] + L_{\mathcal{K},\mu} \mathbb{E} [W_1(m_t, \tilde{m}_t)] \\ &= L_{\mathcal{K},x} e_t + L_{\mathcal{K},\mu} d_t. \end{aligned} \tag{A.8}$$

Iterating (A.8) from $e_0 = 0$ gives (A.3).

We now prove the perturbation error of Proposition 7. For each running reward, using (3.8),

$$\begin{aligned} &\left| \mathbb{E} \left[\mathcal{R}_t^\theta \left(X_t^{K,\lambda,\theta}, M_{K,t}^{\lambda,\theta} \right) \right] - \mathbb{E} \left[\mathcal{R}_t^\theta \left(X_t^{K,0,\theta}, \bar{\mu}_{K,t}^\theta \right) \right] \right| \\ &\leq L_{\mathcal{R},x} W_1 \left(\nu_{K,t}^{\lambda,\theta}, \nu_{K,t}^{0,\theta} \right) + L_{\mathcal{R},\mu} \mathbb{E} \left[W_1 \left(M_{K,t}^{\lambda,\theta}, \bar{\mu}_{K,t}^\theta \right) \right]. \end{aligned}$$

Using (A.5) and (A.4), this is bounded by

$$[L_{\mathcal{R},x} L_{\mathcal{K},\mu} H_t(L_{\mathcal{K},x}) + L_{\mathcal{R},\mu}] C_{\Gamma,K} \lambda.$$

Similarly, the terminal reward satisfies

$$\begin{aligned} &\left| \mathbb{E} \left[g \left(X_T^{K,\lambda,\theta}, M_{K,T}^{\lambda,\theta} \right) \right] - \mathbb{E} \left[g \left(X_T^{K,0,\theta}, \bar{\mu}_{K,T}^\theta \right) \right] \right| \\ &\leq [L_{g,x} L_{\mathcal{K},\mu} H_T(L_{\mathcal{K},x}) + L_{g,\mu}] C_{\Gamma,K} \lambda. \end{aligned}$$

Summing over time gives (A.6).

The representation error of Proposition 8 follows the same route. The proof is identical to that of Proposition 7, replacing $M_{K,t}^{\lambda,\theta}$ by $\bar{\mu}_{K,t}^\theta$ and $\bar{\mu}_{K,t}^\theta$ by μ_t^θ .

In particular,

$$\begin{aligned} &\left| \mathbb{E} \left[\mathcal{R}_t^\theta \left(X_t^{K,0,\theta}, \bar{\mu}_{K,t}^\theta \right) \right] - \mathbb{E} \left[\mathcal{R}_t^\theta \left(X_t^\theta, \mu_t^\theta \right) \right] \right| \\ &\leq [L_{\mathcal{R},x} L_{\mathcal{K},\mu} H_t(L_{\mathcal{K},x}) + L_{\mathcal{R},\mu}] \delta_K^\theta. \end{aligned}$$

The analogous bound for the terminal reward and summation over time yield (A.7).

A.3.5 Convergence of Optimal Values and Policies

We prove Corollary 5. The optimal-value estimate follows from the elementary inequality $|\sup_\theta f(\theta) - \sup_\theta g(\theta)| \leq \sup_\theta |f(\theta) - g(\theta)|$.

Let now $\theta_{n_j}^* \rightarrow \bar{\theta}$ be an arbitrary convergent subsequence, which exists by compactness of Θ . For every $\theta \in \Theta$, $J_{K_{n_j}}^{\lambda_{n_j}}(\theta_{n_j}^*) \geq J_{K_{n_j}}^{\lambda_{n_j}}(\theta)$. By uniform convergence, $J(\theta_{n_j}^*) - J_{K_{n_j}}^{\lambda_{n_j}}(\theta_{n_j}^*) \rightarrow 0$ uniformly, and similarly for the right-hand side. Taking $j \rightarrow \infty$ and using continuity of J gives $J(\bar{\theta}) \geq J(\theta)$ for $\theta \in \Theta$. Therefore $\bar{\theta}$ is an optimizer of the original problem. If the optimizer is unique, every convergent subsequence has the same limit θ^* , and compactness therefore implies $\theta_n^* \rightarrow \theta^*$.

Appendix B

Benchmark Specifications

This appendix records the full specification of every benchmark of Chapter 5: the model, the parameter values, the policy parametrization and population recursion, the perturbation choices, and the training and validation protocol. The corresponding subsections of Chapter 5 keep only what is needed to read the results.

B.1 Two-State Two-Action

Parameters. We use the parameter values

$$\lambda_0 = 0.5, \quad \lambda_1 = 0.8, \quad \kappa = 10, \quad p = 0.6.$$

Under these parameters, $\pi^*(\text{ST} \mid 0) = 0.2$ and $\pi^*(\text{ST} \mid 1) = 0.25$. This policy maps every initial population distribution to $\bar{\mu} = (0.6, 0.4)$ after one time step and keeps the population at that distribution thereafter. The main experiment uses the reference horizon $T = 2$, and an additional run with $T = 5$ is included to test the sensitivity of the estimators to longer temporal propagation.

Policy parametrization and population recursion. To isolate the behavior of the gradient estimators, we use the same restricted policy class as in the reference experiment. The policy is stationary and does not depend on the population law. It is parametrized by two Bernoulli logits $\theta = (\theta_0, \theta_1) \in \mathbb{R}^2$:

$$\pi^\theta(\text{MV} \mid x) = \frac{\exp(\theta_x)}{1 + \exp(\theta_x)}, \quad \pi^\theta(\text{ST} \mid x) = \frac{1}{1 + \exp(\theta_x)}, \quad x \in \{0, 1\}.$$

Writing $m_t^\theta = \mu_t^\theta(1)$, the population law can be propagated exactly by the scalar recursion

$$m_{t+1}^\theta = (1 - m_t^\theta)\lambda_0\pi^\theta(\text{MV} \mid 0) + m_t^\theta \left(1 - \lambda_1\pi^\theta(\text{MV} \mid 1)\right). \quad (\text{B.1})$$

Consequently, for a fixed initial mass m_0 , the unperturbed population objective is available without trajectory noise:

$$J(\theta; m_0) = \sum_{t=0}^T \left[m_t^\theta - (m_t^\theta)^2 - \kappa \left| m_t^\theta - (1 - p) \right| \right]. \quad (\text{B.2})$$

This exact recursion is used to provide the population argument and to evaluate frozen policies. It is not differentiated by REINFORCE, MF-REINFORCE, or the transport estimator during training.

Perturbation choices. MF-REINFORCE uses the retained reference value $\varepsilon = 0.2$. For transport, the auxiliary point is sampled as

$$U_t \sim \mathcal{N}(0, 1), \quad q_t = \left(\frac{\exp(U_t)}{1 + \exp(U_t)}, \frac{1}{1 + \exp(U_t)} \right),$$

and this benchmark is used for the broad transport sweep

$$\lambda \in \{0.05, 0.1, 0.2, 0.4, 0.8\}, \quad \eta \in \{0.4, 0.6, 0.85, 0.95\},$$

where λ is the main perturbation scale and η is the auxiliary scale. The adaptive run starts from $\lambda_0 = 0.2$ and $\eta_0 = 0.8$.

Training protocol. At each policy update, the initial population distribution is randomized according to $\mu_0(1) \sim \mathcal{U}([0.1, 0.9])$ and $\mu_0(0) = 1 - \mu_0(1)$. The policy parameters are initialized at $\theta = (0, 0)$ and optimized with Adam for 10,000 training iterations with learning rate 10^{-3} . In the reference-budget configuration, MF-REINFORCE uses $B = 200$ main trajectories and $n = 10$ auxiliary trajectories. REINFORCE, fixed-scale transport, and adaptive transport are assigned the corresponding equal simulator budget according to the rules of Section 5.1.

We run both exact-flow and particle-flow variants. In the exact-flow variant, (B.1) is used to compute $(\mu_t^\theta)_{t=0}^T$. In the particle-flow variant, the population argument is estimated from a finite simulated population before being supplied to the policy and reward. This second setting tests whether the estimator remains stable when the mean-field law is available only through a particle approximation.

Validation and diagnostics. Every ten training iterations, the current policy is frozen and evaluated from the fixed validation distribution $\mu_0^{\text{val}} = (0.2, 0.8)$. Since the exact recursion is available, the validation reward is computed from (B.2) without trajectory noise, even when the policy was trained with a particle-flow approximation.

Policy recovery is measured by the errors

$$e_0(\theta) = \left| \pi^\theta(\text{ST} \mid 0) - 0.2 \right|, \quad e_1(\theta) = \left| \pi^\theta(\text{ST} \mid 1) - 0.25 \right|.$$

We also record the learned population flow from μ_0^{val} and the average tracking error

$$e_{\text{flow}}(\theta) = \frac{1}{T} \sum_{t=1}^T \left| \mu_t^\theta(1) - (1 - p) \right|.$$

Finally, at selected saved policies, the deterministic recursion can be differentiated by automatic differentiation to obtain an evaluation-only oracle for $\nabla_\theta J(\theta; m_0)$. This oracle is used to measure gradient bias, dispersion, mean-square error, and cosine alignment of the model-free estimators under a matched simulator budget. No oracle information is used during training.

B.2 Cybersecurity

Parameters and policy. We use the parameter values from [19]:

$$\begin{array}{llll} \beta_{UU} = 0.3, & \beta_{UD} = 0.4, & \beta_{DU} = 0.3, & \beta_{DD} = 0.4, \\ q_{\text{rec}}^D = 0.5, & q_{\text{rec}}^U = 0.4, & q_{\text{inf}}^D = 0.4, & q_{\text{inf}}^U = 0.3, \\ v_H = 0.6, & \lambda_{\text{sw}} = 0.8, & k_D = 0.3, & k_I = 0.5, \end{array}$$

with $\Delta t = 0.2$ and $\gamma = 0.5$. The training horizon is $T = 3$, while validation is performed over $T_{\text{val}} = 50$.

The policy is a two-layer multilayer perceptron with hidden width 32 and tanh activations. Its input is (t, μ) and its output is a 4×2 matrix of logits. A row-wise softmax gives

$$\pi_t^\theta(a \mid x, \mu) = \frac{\exp(z_{x,a}^\theta(t, \mu))}{\sum_{a' \in \mathcal{A}} \exp(z_{x,a'}^\theta(t, \mu))}.$$

For a fixed policy, the population flow is propagated exactly by

$$\mu_{t+1}^\theta = \mu_t^\theta K_t^\theta(\mu_t^\theta), \quad K_t^\theta(\mu)(x, x') = \sum_{a \in \mathcal{A}} \pi_t^\theta(a \mid x, \mu) P_{\Delta t}^{\mu,a}(x, x').$$

This recursion is used for population propagation and validation, but is not differentiated during model-free training.

Perturbation choices. MF-REINFORCE uses the retained reference value $\varepsilon = 1.0$. For transport, the auxiliary point is sampled as $q_t = \text{softmax}(U_t^{(1)}, U_t^{(2)}, U_t^{(3)}, 0)$ with $U_t \sim \mathcal{N}(0, I_3)$. The fixed-scale transport grid is $\lambda \in \{0.1, 0.2, 0.4\}$ with $\eta = 0.85$, and adaptive transport starts from $\lambda_0 = 0.2$, $\eta_0 = 0.8$.

Training and validation. At each training iteration, the initial law is sampled from $\text{Dirichlet}(1, 1, 1, 1)$. The MF-REINFORCE reference configuration uses $B = 200$ main trajectories and $n = 1$ auxiliary trajectory; REINFORCE and transport are assigned the corresponding equal simulator budget. The policy is trained for 20,000 iterations with Adam and learning rate 10^{-3} .

Every ten iterations, the frozen policy is evaluated from the uniform initial law $\mu_0^{\text{val}} = (1/4, 1/4, 1/4, 1/4)$ over the validation horizon. Since the reward depends only on the state, the validation value is computed deterministically from the population flow. We report the validation reward, the four state probabilities over time, the infected and defended fractions, and the population-averaged switching probability.

B.3 Distribution Planning

Policy parametrization and population recursion. The policy is represented by a two-layer multilayer perceptron with hidden width 256 and tanh activations. Its input is (t, μ) and its output is a 10×3 matrix of logits. A row-wise softmax defines $\pi_t^\theta(a \mid x, \mu)$.

For a fixed policy, the population recursion is exact. If

$$K_t^\theta(\mu)(x, x') = \sum_{a \in \mathcal{A}} \pi_t^\theta(a \mid x, \mu) \mathbf{1}_{\{x' = x+a \pmod{10}\}},$$

then $\mu_{t+1}^\theta = \mu_t^\theta K_t^\theta(\mu_t^\theta)$. Equivalently,

$$\mu_{t+1}^\theta(x) = \sum_{a \in \mathcal{A}} \mu_t^\theta(x - a) \pi_t^\theta(a \mid x - a, \mu_t^\theta),$$

with all indices understood modulo 10. This gives a deterministic validation objective:

$$J_T(\theta; \mu_0) = - \sum_{t=0}^{T-1} \left[\|\mu_t^\theta - \mu_{\text{target}}\|_2^2 + c_{\text{mov}} \sum_{x \in \mathcal{X}} \mu_t^\theta(x) \sum_{a \in \mathcal{A}} |a| \pi_t^\theta(a \mid x, \mu_t^\theta) \right] - \|\mu_T^\theta - \mu_{\text{target}}\|_2^2.$$

This expression is used only for evaluation of frozen policies.

Perturbation choices. MF-REINFORCE uses the retained reference value $\varepsilon = 2.0$. For transport, the auxiliary point is sampled as $q_t = \text{softmax}(U_t^{(1)}, \dots, U_t^{(9)}, 0)$ with $U_t \sim \mathcal{N}(0, I_9)$. The fixed-scale transport grid is $\lambda \in \{0.1, 0.2, 0.4\}$ with $\eta = 0.85$, and the adaptive run starts from $\lambda_0 = 0.2$, $\eta_0 = 0.8$.

Training and validation. At each training iteration, the initial population law is sampled from $\text{Dirichlet}(\mathbf{1}_{10})$, which keeps the logit perturbation well defined almost surely and explores the interior of Δ_{10} . The MF-REINFORCE reference configuration uses $B = 500$ main trajectories and $n = 10$ auxiliary trajectories. The policy is trained with Adam for 30,000 iterations with learning rate 10^{-4} ; the training length is reduced relative to the original reference grid after convergence checks for the retained perturbation settings.

Every ten iterations, the frozen policy is evaluated from the uniform initial law $\mu_0^{\text{val}} = (1/10, \dots, 1/10)$. We report the validation reward, the learned population flow, the final distance $\|\mu_T^\theta - \mu_{\text{target}}\|_2$, the average distance along the trajectory, and the expected cumulative movement. The final distribution is plotted against the initial and target distributions to check whether the learned policy produces coherent transport rather than excessive local oscillation.

B.4 Targeted Advertising

Population recursion and reference policy. For a feedback policy π^θ , define the aggregate advertising probability

$$q_t^\theta = (1 - p_t^\theta) \pi_t^\theta(1 \mid 0, \mu_t^\theta) + p_t^\theta \pi_t^\theta(1 \mid 1, \mu_t^\theta), \quad p_t^\theta = \mu_t^\theta(1).$$

The population recursion is then

$$p_{t+1}^\theta = p_t^\theta + q_t^\theta \min\{\kappa_{\text{ad}}, 1 - p_t^\theta\}, \quad (\text{B.3})$$

and the expected one-step reward is $p_t^\theta - c_{\text{ad}} q_t^\theta$.

We use

$$\gamma = 0.5, \quad \kappa_{\text{ad}} = 0.2, \quad c_{\text{ad}} = 0.15, \quad T = 5.$$

The infinite-horizon source model admits an explicit optimal aggregate advertising probability. For these parameters,

$$\hat{q}(p) = \begin{cases} 1, & p < 0.6, \\ (0.8 - p)/0.2, & 0.6 \leq p < 0.7, \\ 1, & 0.7 \leq p < 0.85, \\ 0, & 0.85 \leq p \leq 1. \end{cases}$$

This reference policy is used as a structural policy-recovery diagnostic for the finite-horizon experiment.

Policy parametrization and perturbation choices. The learned policy is a two-layer multilayer perceptron with hidden width 32 and tanh activations. Its input is (t, μ) and its output is a 2×2 matrix of logits, converted to action probabilities by a row-wise softmax.

MF-REINFORCE uses $\varepsilon = 1.0$. For transport, the auxiliary point is sampled as $q_t = (\text{sigmoid}(U_t), 1 - \text{sigmoid}(U_t))$ with $U_t \sim \mathcal{N}(0, 1)$. The fixed-scale transport grid is again $\lambda \in \{0.1, 0.2, 0.4\}$ with $\eta = 0.85$, and adaptive transport starts from $\lambda_0 = 0.2$, $\eta_0 = 0.8$.

Training and validation. At each training iteration, the initial customer proportion is sampled from $p_0 \sim \mathcal{U}([0.05, 0.95])$, with $\mu_0 = (1 - p_0, p_0)$. The MF-REINFORCE reference configuration uses $B = 200$ main trajectories and $n = 10$ auxiliary trajectories. Policies are trained with Adam for 10,000 iterations with learning rate 10^{-3} .

Every ten iterations, the frozen policy is evaluated on a fixed uniform grid in $[0.05, 0.95]$. The validation value is computed from the exact population recursion (B.3). We report the average validation reward, the learned customer-share trajectories, the mean advertising probability, the discounted advertising expenditure, and the error between the learned aggregate advertising probability and the explicit reference policy $\hat{q}$.

B.5 Linear–Quadratic

Parameters. The baseline configuration is

$$\begin{aligned} T = 20, \quad \mu_0 = 1, \quad \Sigma_0 = 0.25, \quad a = 0.9, \quad b = 0.5, \quad c = 0.05, \\ q = 1, \quad r = 0.1, \quad \gamma = 5, \quad q_T = 1, \quad \gamma_T = 5, \quad \tau = 0.2, \quad \sigma = 0.1, \quad \rho = 1. \end{aligned}$$

Policies are initialized at $\theta = 0$ and optimized for 10,000 Adam iterations with learning rate 10^{-3} .

Moment recursion and oracle policy. Because the dynamics are linear and the policy is affine, the law is summarized by its mean and variance. Under the affine perturbation described below, these moments satisfy

$$\begin{aligned} \mu_{t+1}^{\theta, \lambda} &= (a + b\theta_t^1 + b\theta_t^2 + c) \mu_t^{\theta, \lambda}, \\ \Sigma_{t+1}^{\theta, \lambda} &= (a + b\theta_t^1)^2 \Sigma_t^{\theta, \lambda} + (b\theta_t^2 + c)^2 \lambda^2 \rho^2 ((\mu_t^{\theta, \lambda})^2 + 1) + b^2 \tau^2 + \sigma^2. \end{aligned} \quad (\text{B.4})$$

Substituting (B.4) into the quadratic cost gives an exact $O(T)$ expression for the perturbed objective $C_T^\lambda(\theta)$, and an adjoint recursion gives the exact gradient. These closed-form quantities are used only for validation and estimator diagnostics.

For $\lambda = 0$, the optimal policy belongs to the chosen class. Let $P_T = q_T$ and $R_T = q_T + \gamma_T$, and define backward

$$P_t = q + \frac{ra^2P_{t+1}}{r + b^2P_{t+1}}, \quad R_t = q + \gamma + \frac{r(a+c)^2R_{t+1}}{r + b^2R_{t+1}}.$$

Then the oracle parameters are

$$k_t^* = -\frac{abP_{t+1}}{r + b^2P_{t+1}}, \quad \ell_t^* = -\frac{b(a+c)R_{t+1}}{r + b^2R_{t+1}}, \quad \theta_t^* = (k_t^*, \ell_t^* - k_t^*).$$

Perturbation and training protocol. The population argument is perturbed through the affine map

$$M_t^{\lambda, \theta} = (1 + \lambda\zeta_t)\mu_t^\theta + \lambda\beta_t, \quad \zeta_t, \beta_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \rho^2).$$

The fixed-scale transport grid is

$$\lambda \in \{0.05, 0.1, 0.2, 0.4, 0.8\}, \quad \eta = 0.85,$$

and adaptive transport starts from $\lambda_0 = 0.2$, $\eta_0 = 0.8$. We run both exact-flow and particle-flow transport variants. Exact flow propagates the moment recursion (B.4), whereas particle flow estimates the population moments from simulated particles before applying the same estimator.

The reference continuous-state budget uses $B = 200$ main trajectories and $n = 1$ law-gradient trajectory. REINFORCE is assigned the same simulator budget by increasing its number of trajectories. For transport, the fixed budget is reallocated toward the auxiliary sensitivity estimate, using $n_{\text{law}} = 20$ auxiliary trajectories and the corresponding reduced number of main trajectories.

Validation and diagnostics. Every ten iterations, the frozen policy is evaluated with the exact unperturbed objective. We report the validation reward, the learned mean and variance flows, the distance to the Riccati optimizer, and the optimality gap. Since an exact gradient oracle is available, the saved policies are also used for bias, dispersion, mean-square error, cosine-alignment, and perturbation-bias diagnostics. The LQ derivation gives the expected $O(\lambda^2)$ objective and gradient bias, so this benchmark is also used to check whether the empirical transport bias follows the predicted scaling.

B.6 Mean–Variance Portfolio

Parameters. We use the baseline configuration

$$T = 10, \quad X_0 \sim \mathcal{N}(1, 0.04), \quad s_t = 1, \quad \bar{r}_t = 0.02, \quad \sigma_{R,t} = 0.08, \quad \chi = 10, \quad \tau_t = 0.02, \quad \rho = 1,$$

for $t = 0, \dots, T-1$, together with $\gamma = 2$. The main run uses normally distributed excess returns. The implementation also allows a centered and rescaled Student distribution with five degrees of freedom, which can be used as a robustness check without changing the moment oracle, since the recursions depend only on the first two moments of the returns. The policy parameters are initialized at $k_t = \ell_t = 0$.

Policy parametrization and moment recursions. Since the unconstrained optimizer of the mean–variance problem is affine in the state and in the population mean, we use the direct time-dependent parametrization $\theta = ((k_t, \ell_t))_{t=0}^{T-1}$ with

$$\alpha_t^\theta = k_t(X_t - \mu_t) + \ell_t + \tau_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1),$$

where $\tau_t > 0$ is a fixed exploration scale. Using a neural network is unnecessary here, and the exact optimizer belongs to the chosen class, which isolates the error of the gradient estimator from function-approximation error.

Because the dynamics are linear and the policy is affine, the population law is characterized by its first two moments. Writing $\Sigma_t = \text{Var}(X_t)$ and $h_t = \bar{r}_t^2 + \sigma_{R,t}^2$, the moments propagate as

$$\begin{aligned} \mu_{t+1}^{\theta, \lambda} &= s_t \mu_t^{\theta, \lambda} + \bar{r}_t \ell_t, \\ \Sigma_{t+1}^{\theta, \lambda} &= A_t(k_t) \Sigma_t^{\theta, \lambda} + h_t \tau_t^2 + \sigma_{R,t}^2 \ell_t^2 + h_t k_t^2 \lambda^2 \rho^2 \left((\mu_t^{\theta, \lambda})^2 + 1 \right), \end{aligned}$$

with $A_t(k) = s_t^2 + 2s_t\bar{r}_tk + h_tk^2$. The perturbation is centered, so the mean recursion does not depend on λ , and all perturbation terms enter at order λ^2 . The exact perturbed objective follows:

$$J^\lambda(\theta) = -\gamma \sum_{t=0}^{T-1} \left[(\mu_t^{\theta,\lambda})^2 + \lambda^2 \rho^2 \left((\mu_t^{\theta,\lambda})^2 + 1 \right) \right] + \mu_T^{\theta,\lambda} - \chi \left[\Sigma_T^{\theta,\lambda} + \lambda^2 \rho^2 \left((\mu_T^{\theta,\lambda})^2 + 1 \right) \right]. \quad (\text{B.5})$$

This gives $J^\lambda(\theta)$ in $O(T)$ for every θ and every perturbation level, and an adjoint recursion gives the exact gradient $\nabla_\theta J^\lambda(\theta)$ at the same cost. Both are used only for validation and diagnostics.

The optimal k_t is unaffected by the running penalty, since k_t enters only through the variance recursion and Σ_T is increasing in each A_t :

$$k_t^* = -\frac{s_t\bar{r}_t}{h_t}. \quad (\text{B.6})$$

For $\gamma = 0$ the remaining component is available in closed form,

$$\ell_t^* = \frac{\bar{r}_t}{2\chi\sigma_{R,t}^2} \prod_{j=t+1}^{T-1} \frac{1}{s_j(1-B_j)}, \quad B_j = \frac{\bar{r}_j^2}{h_j}, \quad (\text{B.7})$$

whereas for $\gamma > 0$ the objective remains concave and quadratic in ℓ , so $\nabla_\ell J^0$ is affine and ℓ^* is obtained by an exact linear solve.

Perturbation scheme. The population argument is perturbed by the affine mean perturbation

$$M_t^{\lambda,\theta} = (1 + \lambda\zeta_t)\mu_t^\theta + \lambda\beta_t, \quad \zeta_t, \beta_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \rho^2),$$

independently of the initial wealth, the asset returns, and the policy noises. The fixed-scale grid is

$$\lambda \in \{0.025, 0.05, 0.1, 0.2, 0.4\}, \quad \eta = 0.85,$$

the smaller radii reflecting the scale of the wealth process.

Training protocol. The initial law is fixed at $X_0 \sim \mathcal{N}(1, 0.04)$, so no randomization of the initial condition is applied. Policies are optimized with Adam for 10,000 iterations with learning rate 10^{-2} . The reference budget uses $B = 500$ main trajectories and $n = 1$ law-gradient trajectory, giving a per-update cost of $C = BT + 2nT(T+1)/2 = 5,110$ simulated transitions, and REINFORCE and transport are assigned the same budget.

As in the LQ benchmark, the fixed transport budget is deliberately reallocated toward the auxiliary law-gradient estimate. For the portfolio runs we use $n_{\text{law}} = 200$ auxiliary trajectories and reduce the number of main trajectories accordingly, at the same total transition budget. The fixed-scale runs use exact moment flow, and adaptive transport starts from $\lambda_0 = 0.2$, $\eta_0 = 0.8$.

Validation and diagnostics. Every ten iterations the policy is frozen and evaluated with the exact objective (B.5) at $\lambda = 0$, so the reported validation reward carries no trajectory noise. Policy recovery is measured by the distance to (B.6)–(B.7) and by the optimality gap $J^0(\theta^*) - J^0(\theta)$. We also record the learned mean and variance flows $(\mu_t^\theta, \Sigma_t^\theta)_{t=0}^T$.

Because an exact gradient oracle is available, this benchmark carries the estimator diagnostics of Section 5.1: at frozen policies we compare the Monte Carlo gradient with the oracle and report relative bias, coordinatewise mean-square error, and cosine alignment; we compare $\nabla_\theta J^\lambda$ with $\nabla_\theta J^0$ to exhibit the quadratic perturbation bias against the growth of the estimator variance as $\lambda \downarrow 0$; and we measure the size of the transport correction relative to the pure policy-score term.

Appendix C

Additional Experimental Results

This appendix collects the per-configuration results of Chapter 5. The main text reports, for each benchmark, the best entry of every estimator family at the headline horizon and population-flow mode; the tables below give every configuration that was run. All values are means and standard deviations over the five seeds of the protocol of Section 5.1, and all of them are regenerated from the saved runs by `scripts/plot.py` and `scripts/report_tables.py`.

C.1 Simulator budget and wall-clock cost

Table C.1 gives the realized simulator budget of every estimator at its headline configuration, together with the wall-clock cost of a run. The budgets agree across estimators by construction; the runtimes do not, and the ratio to REINFORCE is the price of the auxiliary sensitivity recursion.

Benchmark	Estimator	Simulator budget	Wall clock (s)	Ratio to REINFORCE
Two-state	REINFORCE	1 300	48	1.0×
	MF-REINFORCE	1 300	252	5.3×
	Transport	1 300	200	4.2×
	Adaptive	1 296	208	4.3×
Cybersecurity	REINFORCE	612	532	1.0×
	MF-REINFORCE	612	1 199	2.3×
	Transport	612	1 122	2.1×
	Adaptive	609	1 026	1.9×
Distribution planning	REINFORCE	2 800	263	1.0×
	MF-REINFORCE	2 800	6 622	25.1×
	Transport	2 800	6 802	25.8×
	Adaptive	2 797	5 975	22.7×
Targeted advertising	REINFORCE	1 300	146	1.0×
	MF-REINFORCE	1 300	463	3.2×
	Transport	1 300	372	2.6×
	Adaptive	1 296	339	2.3×
Linear-quadratic	REINFORCE	4 420	113	1.0×
	Transport	4 420	1 069	9.5×
	Adaptive	4 406	993	8.8×
Portfolio	REINFORCE	5 110	59	1.0×
	Transport	5 110	352	5.9×
	Adaptive	5 108	334	5.6×

Table C.1: Simulator budget and wall-clock cost per run at the headline configuration, averaged over five seeds. The budget is matched by construction on every benchmark in the table. For transport and adaptive transport the slowest entry of the grid is reported. Matching the simulator budget does not match the wall-clock cost: the transport update carries an auxiliary sensitivity estimate that is cheap in simulated transitions and expensive in arithmetic.

C.2 Full objective tables

Estimator	$T = 2$		$T = 5$	
	Exact	Particle	Exact	Particle
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	-3.4293 ± 0.0165	-3.5412 ± 0.0303	-3.3960 ± 0.1141	-3.8235 ± 0.1854
MF-REINFORCE $\varepsilon = 0.2$	-3.4066 ± 0.0219	-3.5583 ± 0.0232	-4.2053 ± 0.3324	-4.3687 ± 0.0859
REINFORCE	-14.1179 ± 0.0000	—	-31.5104 ± 0.0001	—
Transport $\lambda = 0.05, \eta = 0.4$	-3.5685 ± 0.0417	-3.8091 ± 0.0259	-4.3686 ± 0.1455	-4.4935 ± 0.1463
Transport $\lambda = 0.05, \eta = 0.6$	-3.4445 ± 0.0153	-3.6428 ± 0.0241	-3.6856 ± 0.1458	-3.9363 ± 0.1482
Transport $\lambda = 0.05, \eta = 0.85$	-3.4503 ± 0.0182	-3.5845 ± 0.0309	-3.2281 ± 0.0310	-3.5429 ± 0.0981
Transport $\lambda = 0.05, \eta = 0.95$	-3.4516 ± 0.0168	-3.5806 ± 0.0339	-3.1976 ± 0.0115	-3.5182 ± 0.0931
Transport $\lambda = 0.1, \eta = 0.4$	-3.5155 ± 0.0179	-3.6717 ± 0.0213	-4.2866 ± 0.1307	-4.4465 ± 0.1236
Transport $\lambda = 0.1, \eta = 0.6$	-3.5572 ± 0.0249	-3.5401 ± 0.0275	-3.5103 ± 0.0993	-3.7855 ± 0.1090
Transport $\lambda = 0.1, \eta = 0.85$	-3.5645 ± 0.0236	-3.5120 ± 0.0202	-3.1538 ± 0.0419	-3.4901 ± 0.0959
Transport $\lambda = 0.1, \eta = 0.95$	-3.5645 ± 0.0230	-3.5098 ± 0.0208	-3.1332 ± 0.0423	-3.4547 ± 0.0935
Transport $\lambda = 0.2, \eta = 0.4$	-3.7350 ± 0.0240	-3.6509 ± 0.0421	-4.3809 ± 0.1458	-4.5310 ± 0.0973
Transport $\lambda = 0.2, \eta = 0.6$	-3.7828 ± 0.0328	-3.7178 ± 0.0560	-3.8106 ± 0.0521	-4.0614 ± 0.0905
Transport $\lambda = 0.2, \eta = 0.85$	-3.7925 ± 0.0334	-3.7508 ± 0.0474	-3.6026 ± 0.0542	-3.7501 ± 0.0954
Transport $\lambda = 0.2, \eta = 0.95$	-3.7933 ± 0.0335	-3.7506 ± 0.0455	-3.6143 ± 0.0630	-3.7152 ± 0.0827
Transport $\lambda = 0.4, \eta = 0.4$	-4.1495 ± 0.0275	-4.0828 ± 0.0522	-5.3125 ± 0.2710	-5.2589 ± 0.3015
Transport $\lambda = 0.4, \eta = 0.6$	-4.2597 ± 0.0287	-4.2230 ± 0.0512	-4.9786 ± 0.0951	-5.0526 ± 0.0845
Transport $\lambda = 0.4, \eta = 0.85$	-4.2958 ± 0.0373	-4.2713 ± 0.0447	-4.8547 ± 0.0457	-4.8129 ± 0.0602
Transport $\lambda = 0.4, \eta = 0.95$	-4.2985 ± 0.0386	-4.2764 ± 0.0429	-4.8743 ± 0.0533	-4.8215 ± 0.0428
Transport $\lambda = 0.8, \eta = 0.4$	-6.6666 ± 0.2997	-6.6578 ± 0.2880	-6.0035 ± 0.8558	-6.0160 ± 0.9092
Transport $\lambda = 0.8, \eta = 0.6$	-6.7604 ± 0.2124	-6.7625 ± 0.2076	-9.1291 ± 0.4481	-9.1182 ± 0.5054
Transport $\lambda = 0.8, \eta = 0.85$	-6.7145 ± 0.1590	-6.7106 ± 0.1620	-9.8359 ± 0.0746	-9.8331 ± 0.0700
Transport $\lambda = 0.8, \eta = 0.95$	-6.6989 ± 0.1537	-6.6928 ± 0.1493	-9.8770 ± 0.0716	-9.8692 ± 0.0837

Table C.2: Two-state benchmark: final validation objective for every configuration, as mean and standard deviation over five seeds. Higher is better. The exact and particle population flows are shown side by side at each horizon.

Estimator	$T = 3$
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	-0.1602 ± 0.0032
MF-REINFORCE $\varepsilon = 1$	-0.1524 ± 0.0013
REINFORCE	-0.1517 ± 0.0000
Transport $\lambda = 0.1, \eta = 0.85$	-0.1579 ± 0.0038
Transport $\lambda = 0.2, \eta = 0.85$	-0.1552 ± 0.0038
Transport $\lambda = 0.4, \eta = 0.85$	-0.1544 ± 0.0030

Table C.3: Cybersecurity benchmark: final validation objective for every configuration, as mean and standard deviation over five seeds. Higher is better.

Estimator	$T = 5$
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	-0.5596 ± 0.1225
MF-REINFORCE $\varepsilon = 2$	-0.1017 ± 0.0161
REINFORCE	-0.2616 ± 0.0000
Transport $\lambda = 0.1, \eta = 0.85$	-0.2455 ± 0.0375
Transport $\lambda = 0.2, \eta = 0.85$	-0.2715 ± 0.0611
Transport $\lambda = 0.4, \eta = 0.85$	-0.2421 ± 0.0459

Table C.4: Distribution planning benchmark: final validation objective for every configuration, as mean and standard deviation over five seeds. Higher is better.

Estimator	$T = 5$
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	0.9727 ± 0.0078
MF-REINFORCE $\varepsilon = 1$	0.9946 ± 0.0091
REINFORCE	0.9688 ± 0.0000
Transport $\lambda = 0.1, \eta = 0.85$	0.9795 ± 0.0089
Transport $\lambda = 0.2, \eta = 0.85$	0.9723 ± 0.0094
Transport $\lambda = 0.4, \eta = 0.85$	0.9694 ± 0.0015

Table C.5: Targeted advertising benchmark: final validation objective for every configuration, as mean and standard deviation over five seeds. Higher is better.

Estimator	$T = 20$	
	Exact	Particle
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	-7.3021 ± 0.0030	-7.3045 ± 0.0039
REINFORCE	-7.5157 ± 0.0010	—
Transport $\lambda = 0.05, \eta = 0.85$	-7.2961 ± 0.0021	-7.2968 ± 0.0020
Transport $\lambda = 0.1, \eta = 0.85$	-7.2952 ± 0.0023	-7.2960 ± 0.0022
Transport $\lambda = 0.2, \eta = 0.85$	-7.2992 ± 0.0021	-7.2999 ± 0.0020
Transport $\lambda = 0.4, \eta = 0.85$	-7.3266 ± 0.0028	-7.3273 ± 0.0028
Transport $\lambda = 0.8, \eta = 0.85$	-7.4138 ± 0.0059	-7.4145 ± 0.0055

Table C.6: Linear-quadratic benchmark: final validation objective for every configuration, as mean and standard deviation over five seeds. Higher is better. The exact and particle population flows are shown side by side at each horizon.

Estimator	$T = 10$
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	-17.1931 ± 0.5937
REINFORCE	-20.5350 ± 0.0042
Transport $\lambda = 0.025, \eta = 0.85$	-18.1947 ± 1.2352
Transport $\lambda = 0.05, \eta = 0.85$	-18.0824 ± 1.2904
Transport $\lambda = 0.1, \eta = 0.85$	-18.0220 ± 1.2755
Transport $\lambda = 0.2, \eta = 0.85$	-17.9863 ± 1.2021
Transport $\lambda = 0.4, \eta = 0.85$	-17.9802 ± 1.0790

Table C.7: Portfolio benchmark: final validation objective for every configuration, as mean and standard deviation over five seeds. Higher is better.

C.3 Policy recovery on the two-state benchmark

Table C.8 gives the policy-recovery error of every two-state configuration, of which Table 5.2 reports the best entries. Two features of the full table are worth recording. First, the ordering in η that Table C.9 shows on the objective is reproduced on the policy error, so the two measures agree on the calibration. Second, the REINFORCE error is 0.469 at both horizons and to four decimals identical across seeds, which is the signature of a deterministic limit rather than of a noisy one: the estimator converges, it converges to the wrong policy.

Estimator	$T = 2$		$T = 5$	
	Exact	Particle	Exact	Particle
Adaptive $\lambda_0 = 0.2, \eta_0 = 0.8$	0.0100	0.0394	0.1160	0.1754
MF-REINFORCE $\varepsilon = 0.2$	0.0062	0.0422	0.2196	0.2521
REINFORCE	0.4694	—	0.4691	—
Transport $\lambda = 0.05, \eta = 0.4$	0.0452	0.1011	0.2502	0.2647
Transport $\lambda = 0.05, \eta = 0.6$	0.0137	0.0618	0.1646	0.1982
Transport $\lambda = 0.05, \eta = 0.85$	0.0104	0.0487	0.0888	0.1403
Transport $\lambda = 0.05, \eta = 0.95$	0.0104	0.0478	0.0817	0.1367
Transport $\lambda = 0.1, \eta = 0.4$	0.0219	0.0688	0.2350	0.2570
Transport $\lambda = 0.1, \eta = 0.6$	0.0180	0.0397	0.1120	0.1661
Transport $\lambda = 0.1, \eta = 0.85$	0.0178	0.0295	0.0386	0.1014
Transport $\lambda = 0.1, \eta = 0.95$	0.0178	0.0290	0.0347	0.0962
Transport $\lambda = 0.2, \eta = 0.4$	0.0378	0.0580	0.2268	0.2464
Transport $\lambda = 0.2, \eta = 0.6$	0.0365	0.0395	0.0941	0.1448
Transport $\lambda = 0.2, \eta = 0.85$	0.0364	0.0384	0.0435	0.0796
Transport $\lambda = 0.2, \eta = 0.95$	0.0364	0.0384	0.0408	0.0743
Transport $\lambda = 0.4, \eta = 0.4$	0.0846	0.0859	0.2401	0.2458
Transport $\lambda = 0.4, \eta = 0.6$	0.0829	0.0834	0.1476	0.1625
Transport $\lambda = 0.4, \eta = 0.85$	0.0827	0.0834	0.0941	0.1057
Transport $\lambda = 0.4, \eta = 0.95$	0.0826	0.0836	0.0908	0.1017
Transport $\lambda = 0.8, \eta = 0.4$	0.2579	0.2567	0.2802	0.2803
Transport $\lambda = 0.8, \eta = 0.6$	0.2491	0.2485	0.2867	0.2874
Transport $\lambda = 0.8, \eta = 0.85$	0.2419	0.2409	0.2879	0.2881
Transport $\lambda = 0.8, \eta = 0.95$	0.2407	0.2397	0.2877	0.2876

Table C.8: Two-state benchmark: mean absolute deviation of the learned policy from the closed-form optimum, averaged over the two states and over five seeds, for every configuration. The exact and particle population flows are shown side by side at each horizon.

C.4 The two-state calibration grid

Table C.9 is the full (λ, η) sweep on which the auxiliary scale used by every other benchmark was calibrated, and Figure C.1 shows the effect of η at a fixed main scale, with and without a particle approximation of the population flow. Figure C.2 gives the shorter-horizon counterpart at $T = 2$, where the ordering in η reverses, together with the particle-flow validation curve at $T = 5$.

λ	Auxiliary scale η			
	0.4	0.6	0.85	0.95
0.05	-4.369	-3.686	-3.228	-3.198
0.1	-4.287	-3.510	-3.154	-3.133
0.2	-4.381	-3.811	-3.603	-3.614
0.4	-5.312	-4.979	-4.855	-4.874
0.8	-6.003	-9.129	-9.836	-9.877

Table C.9: Two-state calibration grid at $T = 5$ with exact population flow: final validation objective, averaged over five seeds, for every pair of perturbation scales. The optimum is interior in λ , as the bias-variance trade-off of the main perturbation predicts. In η the objective increases over the whole tested range whenever $\lambda \leq 0.1$ and peaks at $\eta = 0.85$ for $\lambda \in \{0.2, 0.4\}$, so the auxiliary estimator wants a radius near the top of the grid: at the auxiliary sample sizes used, the $(n\eta^2)^{-1}$ variance term of Proposition 4 dominates its $O(\eta)$ bias. The ordering reverses only at $\lambda = 0.8$, where the run is already dominated by perturbation bias. This grid fixes $\eta = 0.85$ for the remaining benchmarks.

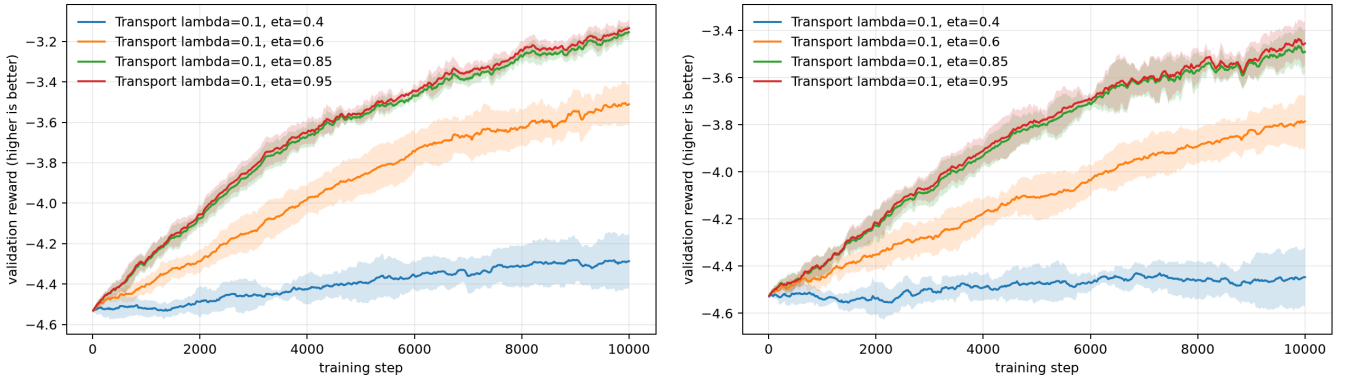


Figure C.1: Two-state benchmark at $T = 5$ and $\lambda = 0.1$: effect of the auxiliary perturbation scale η on the validation reward, as mean over five seeds with one standard deviation shaded, with exact population flow (left) and with a particle approximation of it (right). The ordering in η is unchanged by the substitution; the particle runs are uniformly lower and noticeably noisier across seeds.

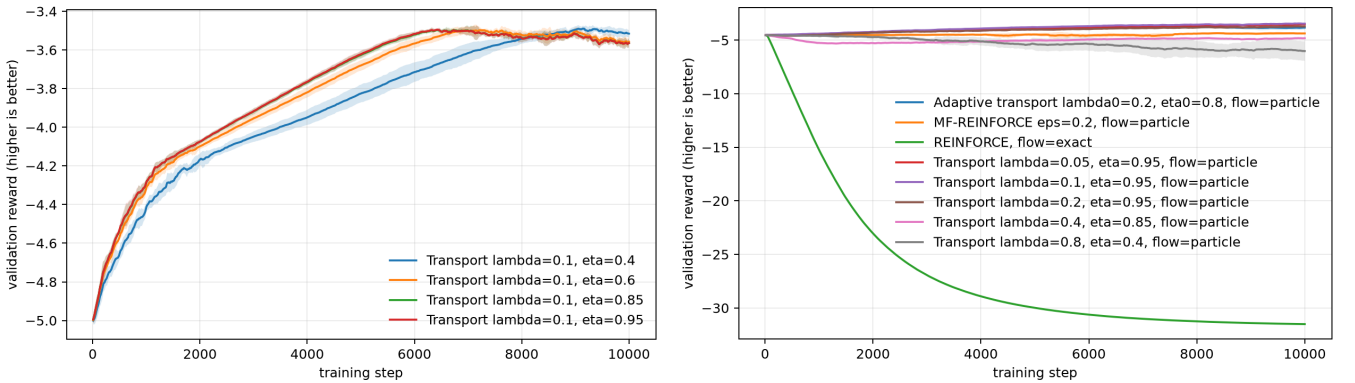


Figure C.2: Two-state benchmark. Left: effect of the auxiliary scale η at $T = 2$ and $\lambda = 0.1$, the shorter-horizon counterpart of Figure C.1. Right: validation reward at $T = 5$ with a particle approximation of the population flow, to be compared with the exact-flow panel of Figure 5.1.

C.5 Learned flows

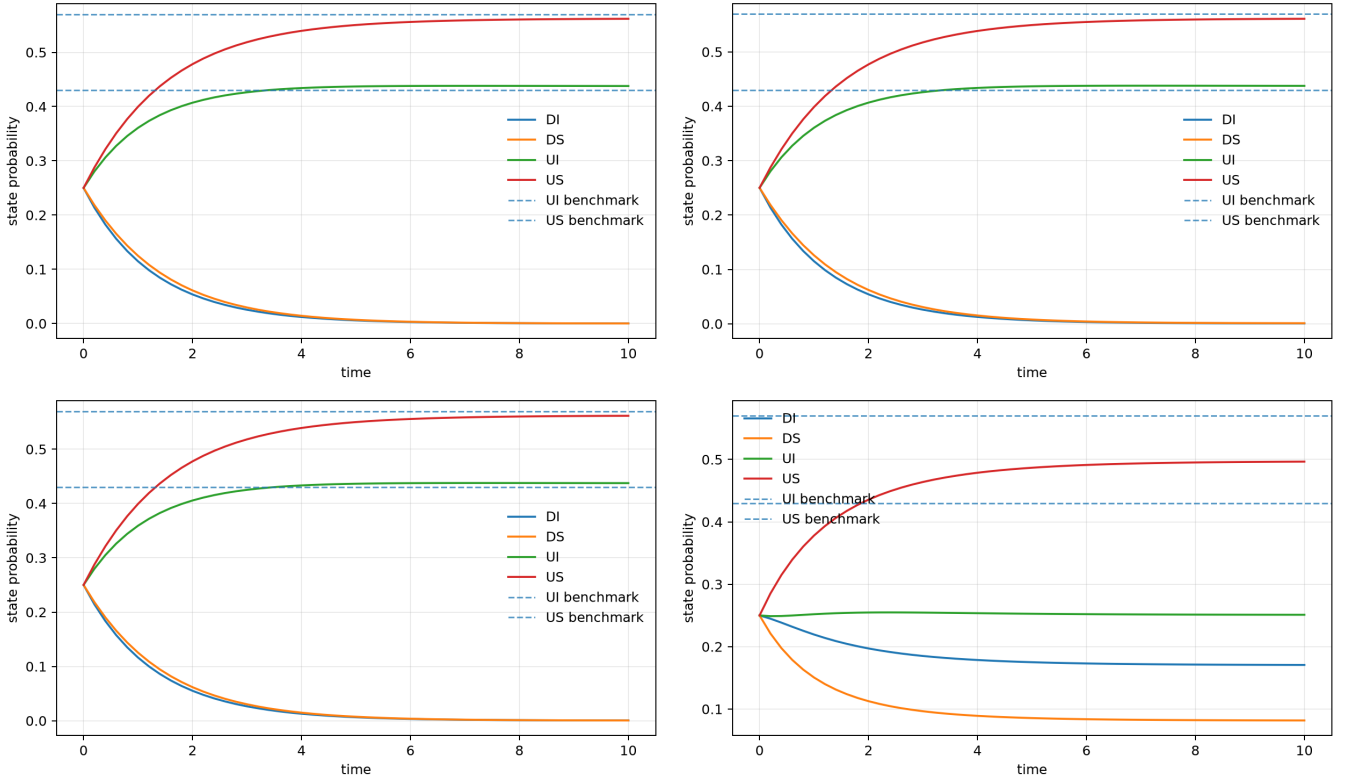


Figure C.3: Learned state distributions on the cybersecurity benchmark, propagated from the uniform validation law over the full validation episode of $T_{\text{val}} = 50$ steps. Top left: REINFORCE. Top right: MF-REINFORCE at $\varepsilon = 1$. Bottom left: fixed-scale transport at $\lambda = 0.4$. Bottom right: adaptive transport. The dashed lines mark the terminal probabilities of the undefended equilibrium. The first three estimators reach it; the adaptive controller keeps most of the population in the two defended states instead.

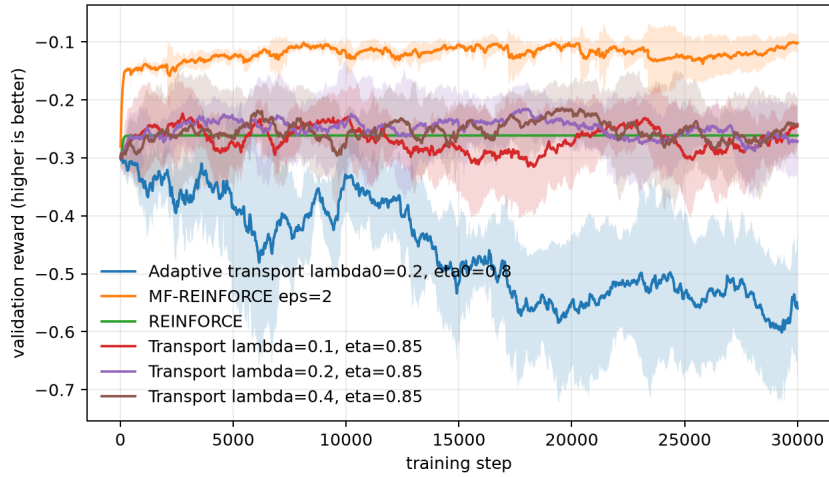


Figure C.4: Distribution-planning benchmark at $T = 5$: validation reward against training iteration, as mean over five seeds with one standard deviation shaded. REINFORCE is the flat line at -0.2616 ; the transport curves oscillate over tens of thousands of iterations without settling.

Figure C.5 compares the population flow learned by all four estimators on the two-state benchmark; the corresponding comparison on cybersecurity, Figure C.3, is kept in the main text because it carries the explanation of that benchmark's ordering. Figure C.9 gives the moment flows on the continuous-state

benchmarks.

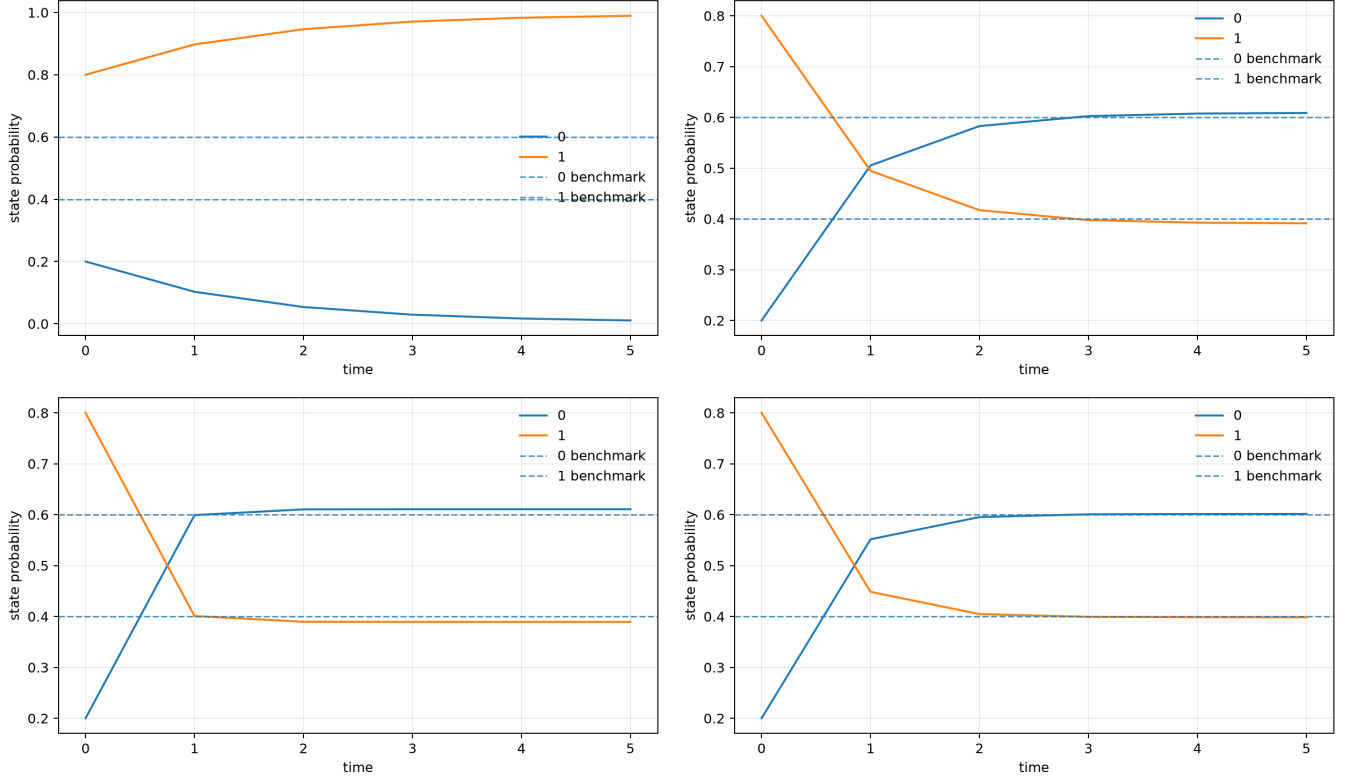


Figure C.5: Learned population flows on the two-state benchmark at $T = 5$ with exact population flow, started from the validation law $\mu_0^{\text{val}} = (0.2, 0.8)$. Top left: REINFORCE. Top right: MF-REINFORCE at $\varepsilon = 0.2$. Bottom left: fixed-scale transport at $\lambda = 0.1$, $\eta = 0.95$. Bottom right: adaptive transport. The dashed lines mark the target proportions $(0.6, 0.4)$ that the closed-form optimal policy sustains.

The remaining validation curves follow.

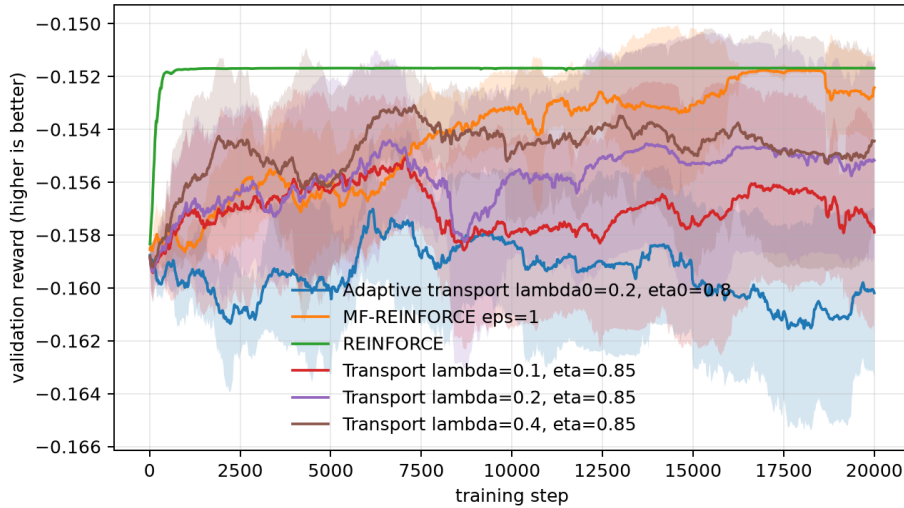


Figure C.6: Cybersecurity benchmark at $T = 3$: validation reward against training iteration, as mean over five seeds with one standard deviation shaded. REINFORCE converges within a few hundred iterations to the best value in the comparison, and the spread between the best and the worst estimator is under 6% of the objective.

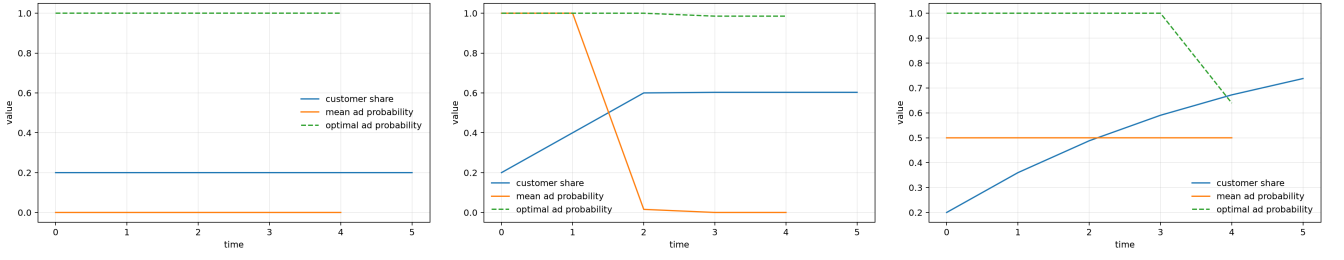


Figure C.7: Targeted-advertising benchmark at $T = 5$: customer share and population-averaged advertising probability of the learned policy against the reference $\hat{q}$, for REINFORCE (left), MF-REINFORCE at $\varepsilon = 1$ (centre) and fixed-scale transport at $\lambda = 0.1$ (right), on the best seed of each. The averaged advertising probability hides a structural difference between the last two, which Table 5.5 resolves.

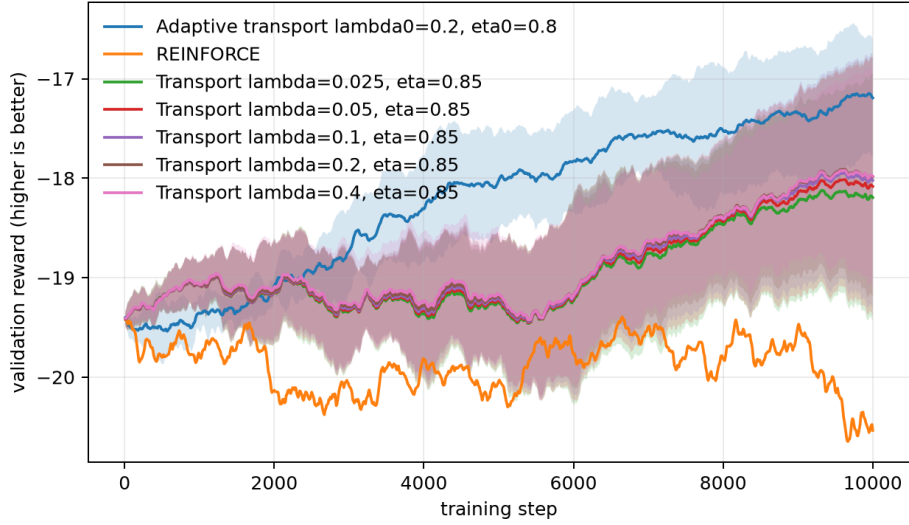


Figure C.8: Mean-variance portfolio benchmark at $T = 10$: validation reward against training iteration, as mean over five seeds with one standard deviation shaded. REINFORCE degrades below its initialization; the transport variants improve throughout, and the adaptive controller improves fastest.

The figure below gives the moment flows on the continuous-state benchmarks. These are qualitative checks rather than measurements: they verify that the learned policies produce coherent population trajectories rather than the oscillation or collapse that a badly scaled sensitivity estimate would cause.

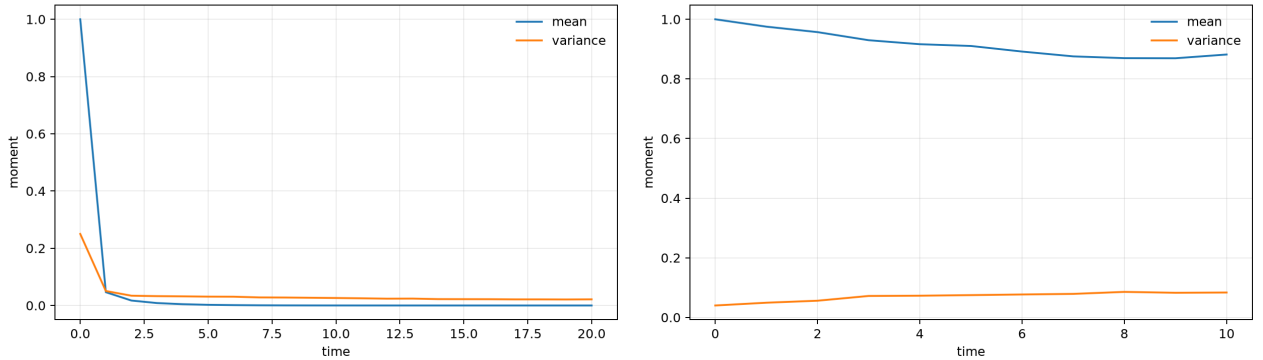


Figure C.9: Learned moment flows on the continuous-state benchmarks, for the best fixed-scale transport configuration of each: linear-quadratic at $T = 20$ and $\lambda = 0.1$ (left) and portfolio at $T = 10$ and $\lambda = 0.4$ (right).

C.6 Population-flow approximation

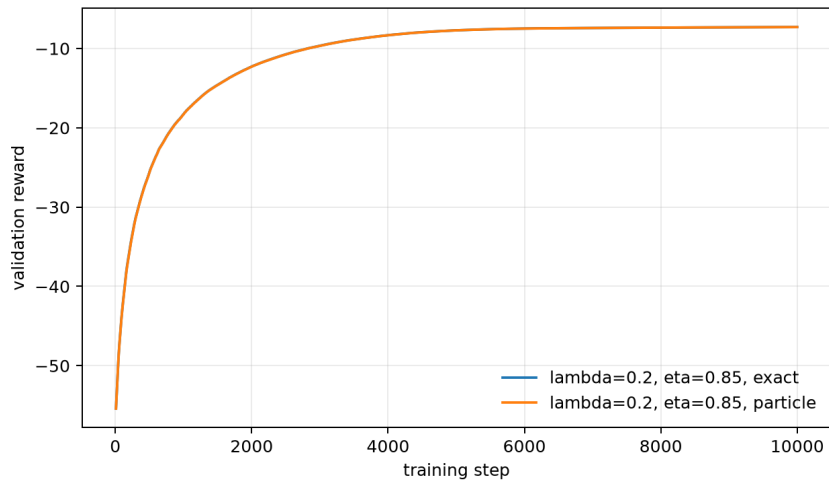


Figure C.10: Linear-quadratic benchmark at $T = 20$ and $\lambda = 0.2$: exact moment recursion against particle estimation of the population flow, as mean over five seeds. The two curves coincide to within the line width.

Figure C.10 compares exact and particle population flow on the linear-quadratic benchmark at $\lambda = 0.2$, and Figure C.11 repeats the comparison at the two ends of the perturbation grid. The two flow modes remain indistinguishable at every scale, which is the expected behaviour when the population law enters the dynamics only through its mean.

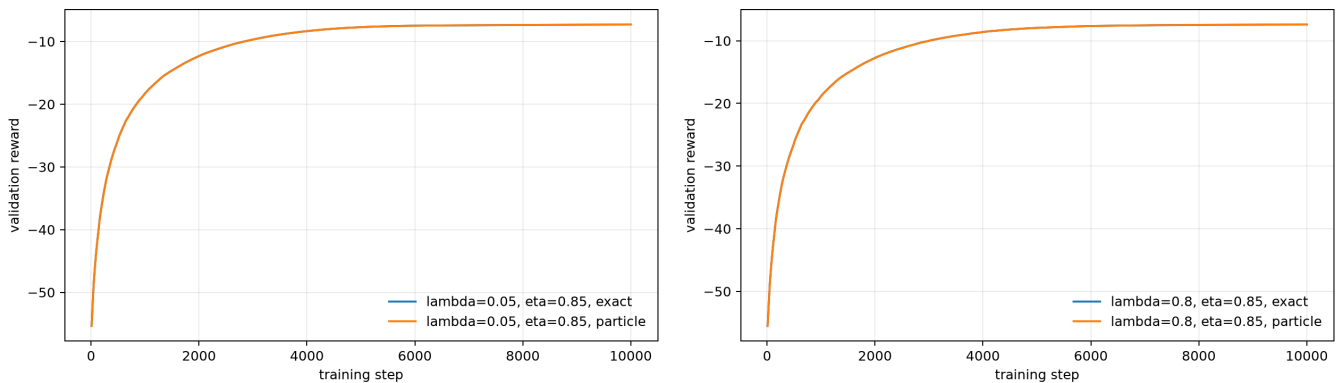


Figure C.11: Linear-quadratic benchmark at $T = 20$: exact against particle population flow at $\lambda = 0.05$ (left) and $\lambda = 0.8$ (right), as mean over five seeds.

C.7 Adaptive schedules

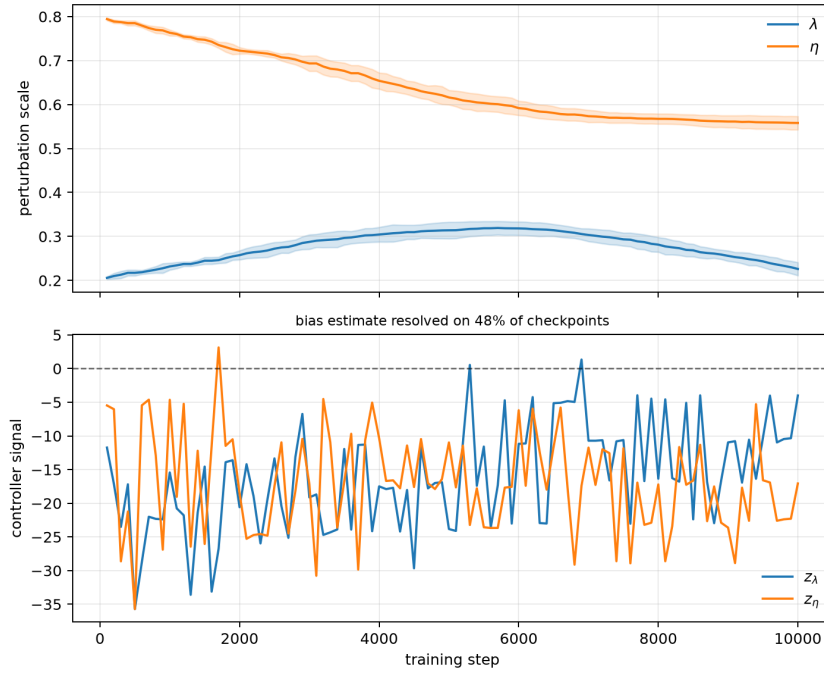


Figure C.12: Adaptive controller on the linear-quadratic benchmark at $T = 20$ with exact population flow. Top: perturbation scales against training iteration, as mean over five seeds with one standard deviation shaded. Bottom: the logarithmic balance signals z_λ and z_η , negative almost everywhere, indicating a variance-dominated regime.

Figure C.12 shows the controller trajectory on the linear-quadratic benchmark. Figure C.13 gives the two benchmarks on which the controller underperforms the fixed grid. Both contract the main scale to the bottom of its range, and the contrast with the linear-quadratic trajectory, which is non-monotone in λ and settles an order of magnitude higher, is the observation discussed in Section 5.4.

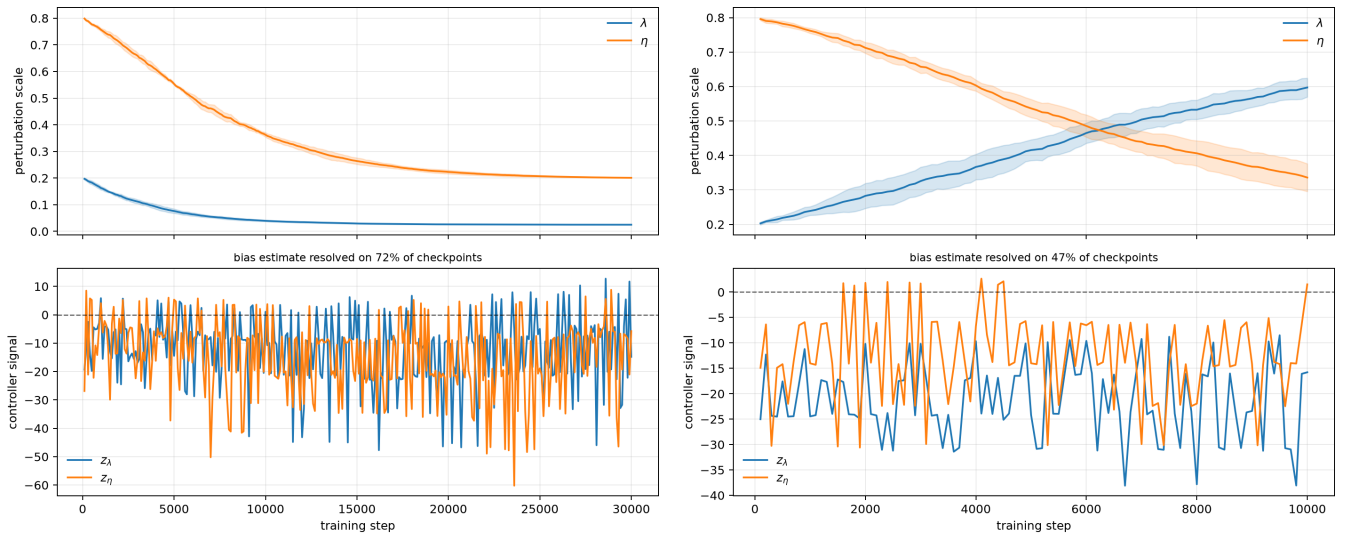


Figure C.13: Adaptive controller on distribution planning at $T = 5$ (left) and on the portfolio benchmark at $T = 10$ (right). Top panels: perturbation scales against training iteration, as mean over five seeds with one standard deviation shaded. Bottom panels: the logarithmic balance signals z_λ and z_η .

C.8 Allocating the auxiliary budget

The equal-budget rule of Section 5.1 fixes the total number of simulated transitions per update, $T(B_{\text{tr}} + n_{\text{tr}})$, but not its split between the main trajectories and the auxiliary sensitivity estimate. Table C.10 records that split on every benchmark, together with the number d of free coordinates of the population argument that the auxiliary estimate has to resolve.

Benchmark	Argument	d	B_{tr}	n_{tr}	n_{tr}/d	Budget
Two-state	Δ_2	1	248	12	12.0	1 300
Cybersecurity	Δ_4	3	184	20	6.7	612
Distribution planning	Δ_{10}	9	549	11	1.2	2 800
Targeted advertising	Δ_2	1	248	12	12.0	1 300
Linear-quadratic	mean	1	201	20	20.0	4 420
Portfolio	mean	1	311	200	200.0	5 110

Table C.10: Split of the matched simulator budget between the main trajectories B_{tr} and the auxiliary sensitivity trajectories n_{tr} of the fixed-scale transport estimator, with d the number of free coordinates of the population argument. The equal-budget rule constrains $T(B_{\text{tr}} + n_{\text{tr}})$ but not the split, which was reallocated towards the auxiliary estimate on the continuous-state benchmarks and left at the default on the finite-state ones. The resulting auxiliary sample size per coordinate spans more than two orders of magnitude, and distribution planning is the one benchmark where it falls close to one.

The ratio n_{tr}/d spans more than two orders of magnitude across the suite, and distribution planning is the extreme case. Its transport runs use $B_{\text{tr}} = 549$ main and $n_{\text{tr}} = 11$ auxiliary trajectories against a ten-point simplex with $d = 9$, that is 1.2 auxiliary trajectories per coordinate, against 6.7 on cybersecurity, 12 on the two-state and advertising problems, 20 on the linear-quadratic benchmark and 200 on the portfolio benchmark. It is the only benchmark on which the auxiliary estimate is asked to resolve more coordinates than it has samples, and the only finite-state benchmark on which transport loses to MF-REINFORCE by a wide margin. Whether that association is causal is the question taken up as an extension in Section 6.3. This section records the two measurements bearing on it: what a reallocation costs, and what it does on a benchmark where the hypothesis predicts nothing.

The cost of reallocating. The split can be changed without changing the budget, since the rule constrains only the sum: on distribution planning, taking $B_{\text{tr}} = 200$ and $n_{\text{tr}} = 360$ leaves $T(B_{\text{tr}} + n_{\text{tr}}) = 2\,800$ unchanged and raises the auxiliary sample size to 40 per coordinate. It is also affordable in arithmetic. One might expect the cost of the update to grow linearly in n_{tr} , but the sensitivity recursion is applied to the whole auxiliary batch at once, and a direct measurement gives a factor 1.9 in seconds per training step when n_{tr} goes from 11 to 360, a 33-fold increase in auxiliary samples. The dominant cost of the update on this benchmark is the exact population recursion over the ten-point simplex, which the reallocation does not touch. Applied to a base cost of 6 802 seconds per transport seed over 30 000 iterations (Table C.1), that factor nonetheless puts a five-seed reallocated arm outside the compute budget of the internship so far, which is why the experiment is proposed in Section 6.3 rather than reported here.

A negative control on targeted advertising. Table C.11 reports the same reallocation on a benchmark where the hypothesis predicts no effect. The advertising population argument is a two-point simplex with a single free coordinate, and its transport runs already use $n_{\text{tr}} = 12$ auxiliary trajectories, the same ratio as the two-state benchmark on which transport is the best estimator at $T = 5$. Moving the split from $(B_{\text{tr}}, n_{\text{tr}}) = (248, 12)$ to $(100, 160)$ at fixed budget takes the final objective from 0.9795 ± 0.0089 to 0.9734 ± 0.0079 , a difference well inside one standard deviation. Where the auxiliary estimate is already resolved, moving budget towards it does not help, which is what the hypothesis predicts; Section 5.2.4 identifies a different mechanism as the limitation there. The control establishes the absence of an effect where the ratio is already high; the positive arm is what the extension of Section 6.3 would supply.

Benchmark	λ	B_{tr}	n_{tr}	n_{tr}/d	Iterations	Objective	Seeds
Targeted advertising	0.1	248	12	12.0	10 000	0.9795 ± 0.0089	5
		100	160	160.0		0.9734 ± 0.0079	5

Table C.11: Recorded check of moving the matched simulator budget from the main trajectories towards the auxiliary sensitivity estimate. Both rows use the same estimator, perturbation scales, simulator budget $T(B_{\text{tr}} + n_{\text{tr}})$ and number of training iterations; only the split differs. The table compares the two allocations at the iteration count given in the sixth column, reading the default arm from its recorded validation history at the same point, and reports the mean and standard deviation over the seeds indicated. Higher is better.

C.9 Perturbation bound along the learned flows

Table C.12 is the first of the measurement tables behind Table 5.9 of Section 5.5, recording the largest value of $d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^{\theta})/\lambda$ along the learned flow of every finite-state transport run. The other two, the objective bias against the perturbation scale and the replicated gradient diagnostics, are reported in the body of that section.

Benchmark	Configurations	Measured points	$\max_t d_{\text{TV}}/\lambda$	Satisfies $d_{\text{TV}} \leq \lambda$
Two-state	80	400	0.800	400/400
Cybersecurity	3	15	0.843	15/15
Distribution planning	3	15	1.000	15/15
Targeted advertising	3	15	0.800	15/15
Total	89	445		445/445

Table C.12: Perturbation bound on the finite-state benchmarks. For every transport run the largest value of $d_{\text{TV}}(M_t^{\lambda, \theta}, \mu_t^{\theta})$ along the learned population flow is compared with λ . The bound holds at every measured point, and is attained to within a factor $d_{\text{TV}}(q_t, \mu_t^{\theta}) \leq 1$ of λ , so the perturbation is neither degenerate nor larger than the theory allows.

C.10 The population-dependence ablation

Table C.13 is the controlled pair discussed in Section 5.6.

Benchmark	Setting	$\ G_{\text{full}}\ $	$\ G_{\text{det}}\ $	ϱ	Observed ordering
Portfolio	$\gamma = 0$	0.083	0.083	$+1.000 \pm 0.000$	REINFORCE > transport
Portfolio	$\gamma = 2$	1.357	0.083	-0.196 ± 0.007	transport > REINFORCE

Table C.13: Switching the explicit population dependence on reverses the sign of ϱ with the dynamics held fixed, and the ordering of the estimators reverses with it. At $\gamma = 0$ the terminal reward depends on the population only through the centred variable $x - \mu$, the correction vanishes identically, and REINFORCE is exact; at $\gamma = 2$ the correction is large and weakly aligned, and the transport estimator wins.

C.11 Gradient diagnostics on the portfolio benchmark

Table C.14 is the portfolio counterpart of Table 5.11. It is reported here rather than in Section 5.5 because, as discussed there, the estimator dispersion on this benchmark is two orders of magnitude above the norm of the exact gradient, so the unbiasedness test it carries is inconclusive rather than passed.

λ	$\ \nabla J^\lambda - \nabla J\ $	$/\lambda^2$	$\ \mathbb{E}[\hat{G}] - \nabla J^\lambda\ $	p	Dispersion	$d \cdot \text{MSE}$	cos
0.025	0.0016	2.48	49.6401	0.494	346.7352	122179.3137	0.124
0.05	0.0062	2.47	50.3068	0.463	347.3983	122572.3083	0.136
0.1	0.0246	2.46	51.1811	0.444	351.4799	125337.4476	0.140
0.2	0.0987	2.47	53.9354	0.429	368.9108	137827.8275	0.138
0.4	0.3934	2.46	65.2150	0.400	442.0454	197315.5292	0.128

Table C.14: Evaluation-only gradient diagnostics on the portfolio benchmark, at the policies saved at the end of training and averaged over the five seeds, from 50 independent replications of the estimator at 8192 particles. Column two is the perturbation bias of the gradient, $\|\nabla_\theta J^\lambda(\theta) - \nabla_\theta J(\theta)\|$, computed from the closed-form gradient oracle; column three divides it by λ^2 . Column four is the bias of the estimator with respect to the perturbed gradient it targets, and column five the p -value of the χ^2 test that this bias is zero at the available replication count, so a large p means the estimator is not measurably biased for $\nabla_\theta J^\lambda$. Column six is the dispersion $\|\text{sd}(\hat{G})\|$ of the replications and column seven the mean-square error scaled by the parameter dimension d ; it equals the sum of the squares of columns four and six, up to the factor $(R-1)/R$ carried by the unbiased dispersion estimate, and the identity is verified run by run to 2%, which is exactly that factor at $R = 50$. The last column is the cosine alignment of the mean estimate with the oracle gradient.